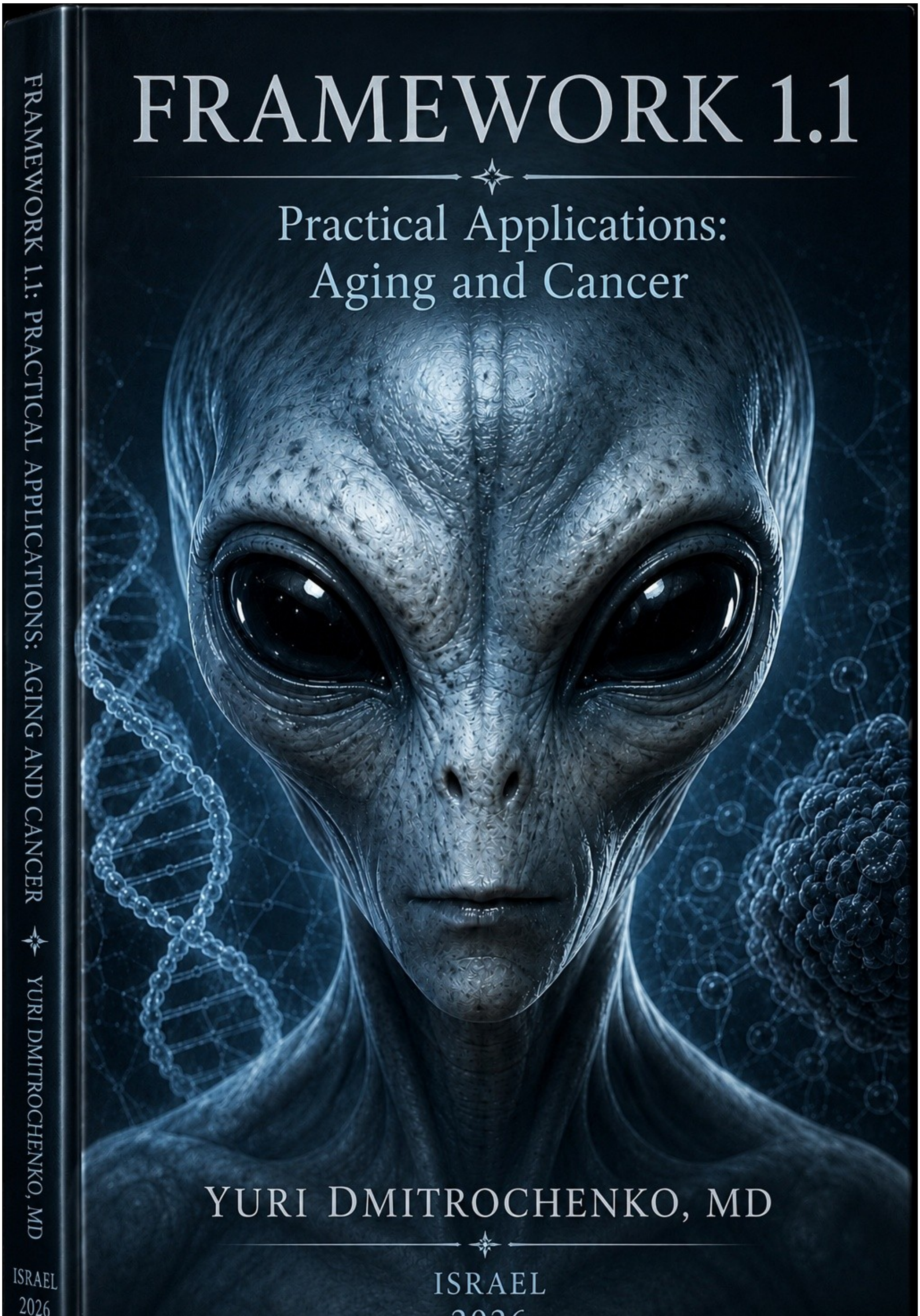
FRAMEWORK 1.1
Practical Applications:
Aging and Cancer
YURI DMITROCHENKO, MD
ISRAEL
2026
FRAMEWORK 1.1: PRACTICAL APPLICATIONS: AGING AND CANCER
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FRAMEWORK 1.1

Practical Applications: Aging and Cancer

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Preface

If Framework 1.0 is a language, then Framework 1.1 is the first attempt to speak it.

Aging and cancer did not wait for the appearance of the hydrogel model or the principle of stochastic determinism to acquire their own logic. The task of this book is not to propose yet another dialect, but to test whether the same language used in Framework 1.0 to describe the cell as a hydrogel system can, without translation, speak of the concrete: neuroblastoma 4S, the bifurcation point of energetic reserve, endometrial hyperplasia.

A special place in this attempt belongs to the image of the inverted differentiation pyramid: a clone's escape from control unfolds the ordinary hierarchy of tissue organization, expanding the number of primitive, poorly differentiated variants where mature tissue would otherwise keep them confined to a narrow, specialized channel. Closely tied to this image is another concept receiving its first extended treatment here — the dormant cellular state: not a passive idling, but an actively maintained, economical regime in which the clone holds its reserve in check, refraining from spending it prematurely, and thereby sets its own internal limit on its expansion.

The language that has just begun to be spoken does not yet sound as fluent as mature speech. The text retains hypotheses flagged as requiring verification, sections on potential sources of model failure, predictions awaiting experimental test.

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2026

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Part I

The Nuclear Glycogen Trilogy

Nuclear Glycogen, Local Glycolysis, and Nuclear Syneresis: Toward a Physical Model of Nuclear Energetics

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Abstract

The eukaryotic nucleus performs some of the most ATP-intensive processes in biology despite containing no mitochondria. Classical models assume that ATP freely diffuses from the cytoplasm into the nucleus through nuclear pores. However, recent discoveries indicate that the nucleus possesses compartmentalized metabolic pathways — including regulated nuclear glycogen synthesis and glycogenolysis, local glycolytic enzyme activity, and metabolically coupled chromatin remodeling — that collectively confer a degree of energetic autonomy previously unrecognized.

This perspective integrates these independent findings into a unified biophysical hypothesis. We propose that the nucleus functions as a localized Basal Glycolytic Regime (BGR) microdomain: an energetic compartment without mitochondrial oxidative capacity, sustained by substrate-level phosphorylation from local carbohydrate reserves. During glycopenia, progressive failure of this nuclear energetic system reduces ATP production, decreases acetyl-CoA availability, impairs histone acetylation, promotes chromatin condensation, and may induce a physical contraction of the nuclear hydrogel — a process we term nuclear syneresis, by analogy with the well-characterized phenomenon of syneresis in polymer physics.

The model generates experimentally testable predictions distinguishable from classical diffusion-based accounts of nuclear energy supply, and positions the nuclear glycogen–acetyl-CoA–chromatin axis as a mechanistic link between cellular metabolism, epigenetic state, and nuclear architecture.

Keywords: *nuclear glycogen; nuclear glycolysis; histone acetylation; chromatin condensation; nuclear syneresis; hydrogel physics; BGR; acetyl-CoA; O-GlcNAc; nuclear energetics*

Series: *Article I of the Nuclear Glycogen Trilogy. See companion articles: (II) Nuclear Glycogen as a Determinant of Tumor Resistance; (III) Glycogen as a Graded Marker of Cellular Differentiation.*

1. Introduction

The eukaryotic nucleus is responsible for DNA replication, RNA transcription, chromatin remodeling, DNA repair, RNA processing, nuclear transport, and assembly of ribonucleoprotein complexes. All of these processes require continuous ATP consumption. Yet mature nuclei contain no mitochondria.

For decades this paradox has been explained by passive ATP diffusion through nuclear pore complexes. Although diffusion certainly contributes to ATP delivery, it is an unregulated, distance-dependent process that cannot readily buffer rapid or localized increases in nuclear

ATP demand. The G1/S transition, DNA damage response, and bursts of transcriptional activation all represent conditions under which nuclear ATP demand may transiently exceed what diffusion alone can supply.

Recent discoveries suggest that nuclear metabolism is considerably more autonomous than previously appreciated. The nucleus has been found to contain dedicated glycogen metabolism machinery, local glycolytic enzymes, and metabolite-sensitive chromatin-modifying systems that collectively constitute a self-sufficient energetic compartment. The present perspective integrates these discoveries into a single physical framework — the nuclear BGR microdomain — and proposes that its failure under substrate deprivation produces a physically distinct nuclear state: nuclear syneresis.

2. Nuclear Glycogen as a Compartmentalized Energy Reserve

For almost eighty years, nuclear glycogen remained an unexplained histological observation. A major breakthrough occurred in 2019 when Sun and colleagues demonstrated, using stable isotope tracing in human non-small cell lung cancer cells, that: glycogen is synthesized de novo inside mammalian nuclei; glycogen phosphorylase translocates into the nucleus under physiological conditions; nuclear glycogen undergoes regulated glycogenolysis; and nuclear glycogen provides carbon for compartmentalized pyruvate production and, subsequently, for histone acetylation [1].

These findings overturned the assumption that nuclear glycogen was an incidental contaminant or artifact. The nucleus possesses dedicated enzymatic machinery for carbohydrate storage and mobilization, independent of the cytoplasmic glycogen pool.

The regulatory mechanism was further characterized by Donohue and colleagues, who showed that the E3 ubiquitin ligase malin — mutations in which cause Lafora disease, a progressive myoclonic epilepsy characterized by pathological polyglucosan body accumulation — plays a pivotal role in promoting nuclear glycogenolysis and the resulting histone acetylation response [2]. Thus, the nuclear glycogen metabolism system is not a metabolic curiosity but is under specific enzymatic control with known pathological consequences when disrupted.

3. Local Nuclear Glycolysis

Multiple glycolytic enzymes have been identified in nuclear or perinuclear compartments under specific metabolic and signaling conditions. Notably, glyceraldehyde-3-phosphate dehydrogenase (GAPDH), enolase-1, and the M2 isoform of pyruvate kinase (PKM2) have been reported to translocate into the nucleus, where they participate in non-canonical functions that overlap with energy metabolism and gene regulation [3].

The relevance of these translocations to nuclear energetics is directly established by the glycogen data. Nuclear glycogen degradation generates glucose-1-phosphate, which enters glycolytic metabolism to produce ATP and pyruvate within the nuclear compartment. This substrate entry point bypasses the requirement for glucose import from the cytoplasm and creates a self-contained substrate–enzyme–product system within the nucleus.

Such spatial compartmentalization minimizes diffusion distances between ATP generation and consumption — the same design principle operative in mitochondrial cristae architecture, where respiratory chain components are concentrated at membrane surfaces immediately adjacent to the ATP synthase. At the nuclear level, co-localization of glycogen, glycolytic enzymes, and chromatin-modifying complexes may enable rapid, localized energetic responses to transcriptional demand.

4. Nuclear ATP as a Semi-Autonomous Energetic Compartment

Within the nucleus, ATP supports an exceptional range of energy-consuming processes: chromatin remodeling complexes (SWI/SNF, NuRD, ISWI family), ATP-dependent helicases, RNA polymerases I, II, and III, DNA repair enzymes including helicases and ligases, nuclear transport GTPases, and maintenance of phase-separated nuclear condensates including transcriptional hubs, splicing speckles, and the nucleolus.

Compartment-specific ATP biosensors have begun to reveal that nuclear and cytoplasmic ATP pools are not always in equilibrium. Even modest decreases in nuclear ATP — insufficient to be detected by whole-cell measurements — can rapidly impair chromatin remodeling and transcription. This functional sensitivity establishes the nucleus as an energetically semi-autonomous compartment whose performance depends on local ATP availability rather than simply on cytoplasmic equilibration.

The corollary is important: conditions that do not produce detectable whole-cell energy depletion may nonetheless impair nuclear function if they selectively reduce nuclear ATP. Nuclear glycogen, as a local carbohydrate reserve, would function precisely as a buffer against such localized deficits.

5. Nuclear Glycogen During Glycopenia

The existence of a dedicated nuclear glycogen reserve immediately raises a physiological question: why would the nucleus maintain its own carbohydrate store if cytoplasmic ATP diffusion were always sufficient?

The most parsimonious explanation is that nuclear glycogen serves as a short-term energetic buffer during transient glycopenia or periods of rapidly escalating nuclear ATP demand. Three physiological contexts are particularly relevant:

First, the G1/S transition requires coordinated activation of DNA replication machinery, including helicase loading, origin firing, and chromatin remodeling at thousands of replication origins simultaneously. This represents a burst of nuclear ATP demand of brief duration.

Second, the DNA damage response requires rapid recruitment and activation of repair complexes, histone modification at damage sites, and chromatin remodeling to expose damaged DNA — all within minutes of damage induction.

Third, acute transcriptional responses to cellular signals require rapid, coordinated activation of chromatin remodeling at promoters across the genome, consuming ATP and acetyl-CoA simultaneously.

In each scenario, nuclear glycogen would provide a local substrate reservoir enabling nuclear energetic responses that outpace what diffusion from the cytoplasm can supply. This prediction has not yet been directly investigated but is experimentally testable, as described in Section 8.

6. Histone Acetylation as the Metabolic–Structural Interface

Nuclear glycogenolysis generates pyruvate, which is converted to acetyl-CoA — the obligate one-carbon donor for all histone acetyltransferase reactions. The link between this local metabolic flux and chromatin architecture is direct and mechanistically established [1,4].

Histone acetylation neutralizes the positive charge of lysine residues on histone tails, reducing their electrostatic affinity for the negatively charged DNA phosphate backbone. This weakening of histone–DNA contacts allows chromatin to adopt an open, swollen configuration associated with transcriptional competence. Conversely, reduced acetylation restores electrostatic contacts, promotes nucleosome compaction, and drives chromatin toward a condensed, transcriptionally repressive state.

The metabolic coupling is bidirectional and self-reinforcing. Open chromatin supports active transcription, which generates acetyl-CoA demand, which is met by local glycogenolysis, which maintains the open state. Metabolic insufficiency breaks this cycle: reduced local glycolysis → reduced acetyl-CoA → reduced histone acetylation → chromatin condensation → further reduction in transcriptional activity → further reduction in metabolic demand signaling.

Acetyl-CoA availability is therefore a direct readout of local glycolytic flux and a physical regulator of nuclear architecture, not merely a biochemical signal. This framing is central to the nuclear syneresis hypothesis developed below.

7. O-GlcNAc as a Second Nuclear Nutrient Sensor

Thousands of nuclear proteins carry O-linked N-acetylglucosamine (O-GlcNAc) modifications. Unlike classical N-linked glycosylation, O-GlcNAc is a dynamic, reversible post-translational modification cycling on a timescale of hours, catalyzed by O-GlcNAc transferase (OGT) and O-GlcNAcase (OGA) [5].

The substrate for O-GlcNAcylation — UDP-GlcNAc — is produced by the hexosamine biosynthetic pathway, which consumes glucose, glutamine, acetyl-CoA, and UTP. O-GlcNAc levels thus reflect cellular glucose availability and are sensitive to glycopenic conditions. Histones, transcription factors, chromatin remodeling complexes, DNA repair proteins, RNA polymerase II, and nuclear pore components all undergo O-GlcNAc cycling.

Consequently, the nucleus continuously monitors carbohydrate availability through this glycan-based signaling layer, independently of and in parallel to the acetyl-CoA pathway described above. The two systems are complementary: acetyl-CoA mediates the direct physical effect on chromatin compaction, while O-GlcNAc modulates the activity of the regulatory proteins that govern chromatin dynamics. Together, they constitute a dual-sensor system coupling nuclear function to the local carbohydrate economy.

8. Nuclear Syneresis: A Physical Interpretation of Energetic Failure

Current molecular biology interprets chromatin condensation under metabolic stress primarily as an epigenetic regulatory event — a reversible modulation of gene expression. This interpretation is not incorrect, but it is incomplete.

Chromatin is also a physical object: a highly hydrated polymer network occupying a confined compartment bounded by the nuclear envelope. As a polymer network, chromatin has well-defined physical properties including swelling pressure (governed by osmotic and electrostatic forces), viscoelasticity, and phase behavior. In polymer physics, the spontaneous expulsion of solvent from a swollen gel network — driven by a decrease in swelling pressure — is termed syneresis. The gel contracts, water is expelled, and the network adopts a denser, more compact configuration.

We propose that the chromatin condensation produced by metabolic insufficiency — specifically by loss of histone acetylation and the consequent restoration of electrostatic histone–DNA contacts — constitutes syneresis of the nuclear hydrogel. The mechanistic cascade is:

Glucose deficiency

↓

Reduced nuclear glycogenolysis

↓

Reduced local glycolytic flux

↓

Reduced ATP and acetyl-CoA production

↓

Reduced histone acetyltransferase activity

↓

Histone deacetylation → restored electrostatic histone–DNA contacts

↓

Chromatin compaction → decreased nuclear hydration pressure

↓

Mechanical contraction of the nuclear hydrogel

↓

Nuclear syneresis

This interpretation reframes nuclear syneresis as a physical state transition rather than merely a regulatory event. The nucleus passes from a swollen, hydrated, transcriptionally active gel to a condensed, dehydrated, energetically depleted state. The transition is driven by the thermodynamics of the polymer–solvent system, not by a specific regulatory program, though regulatory programs may modulate its threshold and kinetics.

The distinction matters for several reasons. First, a physical state transition may be partially irreversible under prolonged metabolic deprivation, explaining why cells that survive brief glycopenia may recover transcriptional activity while those experiencing sustained deprivation do not. Second, nuclear syneresis predicts measurable changes in nuclear mechanical properties — stiffness, volume, and deformability — that precede detectable whole-cell metabolic changes, providing experimental handles distinct from purely molecular readouts. Third, the concept extends naturally to aging (progressive dampening of nuclear energetic reserve) and to cancer (nuclear syneresis resistance as a survival advantage during metabolic stress).

9. The Nucleus as a BGR Microdomain

The Basal Glycolytic Regime (BGR) is defined as an energetic state in which a metabolic compartment lacking mitochondrial oxidative capacity sustains ATP production through substrate-level phosphorylation from local carbohydrate reserves, operating under conditions of limited or absent oxidative phosphorylation [6].

The nucleus satisfies all defining criteria of a BGR microdomain:

1. Absence of mitochondria — the nucleus contains no oxidative phosphorylation capacity.
2. Local carbohydrate reserve — nuclear glycogen, synthesized and stored within the nucleus.
3. Local glycolytic enzymes — GAPDH, enolase-1, PKM2, confirmed by nuclear fractionation studies.
4. Substrate-level ATP production — from nuclear glycogenolysis and local glycolysis.
5. Sensitivity to substrate availability — nuclear function tracks acetyl-CoA and O-GlcNAc levels, both of which reflect local carbohydrate economy.
6. High and variable ATP demand — chromatin remodeling, transcription, and repair represent among the most energy-intensive processes in cellular biology.

The nucleus thus functions as a BGR within a BGR: an autonomous energetic microreactor embedded within the larger glycolytic cellular network, operating on shorter timescales and with tighter spatial constraints than the whole-cell regime. This nested architecture explains why the nucleus can respond to local energetic signals independently of the cytoplasmic metabolic state, and why nuclear function can be impaired by conditions that do not produce detectable whole-cell energy depletion.

The BGR framing also provides a unified language connecting nuclear glycogen biology to the broader Framework developed across the companion articles of this series, which examine glycogen accumulation across differentiation states (Article III) and its role in tumor resistance (Article II).

10. Testable Predictions

The hypothesis generates the following experimentally addressable predictions:

(1) Nuclear ATP should decrease before cytoplasmic ATP during acute glucose withdrawal, detectable with compartment-specific biosensors such as ATeam or iATPSnFR targeted to the nucleus versus cytoplasm.

(2) Nuclear glycogen depletion should precede chromatin condensation under graded glucose withdrawal, testable by combined periodic acid–Schiff (PAS) staining with nuclear counterstaining and chromatin accessibility assays (ATAC-seq) performed at sequential time points.

(3) Histone acetylation levels (H3K9ac, H3K27ac, H4K16ac) should track local nuclear acetyl-CoA availability more closely than cytoplasmic acetyl-CoA, distinguishable by compartment-specific metabolite extraction and isotope tracing.

(4) Nuclear mechanical properties — stiffness measured by atomic force microscopy, volume measured by confocal imaging, deformability measured by micropipette aspiration — should change before whole-cell ATP depletion becomes detectable by standard enzymatic assays.

(5) Restoration of local nuclear glycolytic capacity — using cell-permeable pyruvate or acetate in the presence of glycolytic inhibitors — should reverse early chromatin condensation, whereas supplementation with cell-impermeable ATP or cytoplasmic ATP donors alone should not.

(6) Inhibition of nuclear glycogen phosphorylase (using CP-91149 or BAY U6751) should impair nuclear histone acetylation and chromatin accessibility under basal conditions, independently of cytoplasmic glycolytic flux.

11. Conclusion

Recent discoveries regarding nuclear glycogen synthesis, regulated glycogenolysis, compartmentalized glycolytic enzymes, histone acetylation as a metabolic readout, and nutrient-sensitive O-GlcNAc signaling collectively indicate that the nucleus possesses a substantially greater degree of metabolic autonomy than previously recognized. The nucleus is not a passive consumer of cytoplasmic ATP but an active, self-provisioning energetic compartment.

The present hypothesis integrates these observations into a single biophysical framework in which nuclear glycogen supports a localized BGR, while progressive energetic failure under glycopenic conditions culminates in nuclear syneresis — a physical contraction of the nuclear hydrogel driven by metabolic insufficiency rather than by a specific regulatory program.

If experimentally confirmed, this model would provide a common mechanistic language linking metabolism, epigenetics, hydrogel physics, and nuclear architecture. It would position the nuclear glycogen–acetyl-CoA–histone acetylation axis as a first-order determinant of transcriptional capacity under metabolic stress, and would identify the nucleus itself as a primary target in pathological conditions characterized by impaired glucose metabolism — including cancer (see Article II), neurodegeneration, and cellular senescence.

The concept of nuclear syneresis, as a physically defined state transition distinct from regulated chromatin compaction, offers a new category for interpreting the relationship between cellular energetics and nuclear function, with potential implications for the biology of aging and the mechanism of metabolic diseases affecting the brain and proliferating tissues.

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Conflicts of Interest

The author declares no conflicts of interest.

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Nuclear Glycogen as a Determinant of Tumor Resistance: A Local BGR Hypothesis of Chemotherapy Response

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Abstract

Tumor glycogen accumulation is a recognized metabolic adaptation associated with hypoxia, rapid proliferation, and therapeutic resistance. The recent identification of nuclear glycogen as a distinct metabolic compartment involved in histone acetylation through localized glycogenolysis extends this concept beyond cytoplasmic energy storage. Glycogen accumulation in tumor cells, particularly in clear cell renal cell carcinoma, hepatocellular carcinoma, and glycogen-rich breast carcinoma, is well documented and clinically associated with aggressive phenotypes.

This perspective proposes that nuclear glycogen represents a localized Basal Glycolytic Regime (BGR)* supporting continuous ATP generation, acetyl-CoA production, chromatin remodeling, and DNA repair within the nucleus. Tumors enriched in nuclear glycogen may therefore exhibit increased resistance to chemotherapy and radiotherapy by preserving nuclear energetic homeostasis during therapeutic metabolic stress. This model predicts that pharmacological disruption of nuclear glycogen metabolism — using existing glycogen phosphorylase inhibitors — could represent a novel sensitization strategy.

** BGR (Basal Glycolytic Regime): an energetic state in which a metabolic compartment lacking mitochondrial oxidative capacity sustains ATP production through substrate-level phosphorylation from local carbohydrate reserves. See companion article: Dmitrochenko Y. Nuclear Glycogen, Local Glycolysis, and Nuclear Syneresis. 2026.*

Keywords: nuclear glycogen; tumor metabolism; chemotherapy resistance; histone acetylation; DNA repair; glycogen phosphorylase; BGR; nuclear syneresis

Introduction

Resistance to chemotherapy remains one of the principal obstacles in cancer treatment. Established mechanisms include drug efflux pumps, acquired genetic mutations, enhanced DNA repair, epigenetic plasticity, and metabolic reprogramming. Among metabolic adaptations, glycogen accumulation has attracted increasing attention: in clear cell renal cell carcinoma, intracellular glycogen is detectable in more than 80% of cases and serves as a histological hallmark [6]; glycogen-rich carcinomas of the breast represent a distinct aggressive subtype; hepatocellular carcinoma cells accumulate glycogen under hypoxic preconditioning as a survival reserve [4].

Until recently, however, tumor glycogen was viewed exclusively as a cytoplasmic energy reserve mobilized under metabolic stress. The discovery of regulated nuclear glycogen metabolism [1] fundamentally changes this perspective. If the nucleus maintains its own carbohydrate reserve and local glycolytic capacity, then the energetic state of the nucleus during chemotherapy-induced stress becomes an independent determinant of cell survival.

Nuclear Glycogen as an Active Metabolic Compartment

Sun and colleagues demonstrated that mammalian nuclei contain glycogen synthase, glycogen phosphorylase, and active glycogenolysis [1]. Stable isotope tracing showed that nuclear glycogen degradation supplies carbon directly for pyruvate production, acetyl-CoA synthesis, and histone acetylation. The E3 ubiquitin ligase malin regulates this process, and its loss impairs nuclear glycogenolysis and reduces histone acetylation [2].

These findings establish nuclear glycogen not as an inert polymer but as a regulated local energy and carbon source coupled to the epigenetic state of chromatin. This coupling is mechanistically critical: histone acetylation requires acetyl-CoA; acetyl-CoA availability is directly determined by local pyruvate metabolism; and local pyruvate availability depends on nuclear glycogen content.

Why Do Tumor Nuclei Accumulate Glycogen?

Tumor nuclei perform extraordinarily energy-demanding processes under conditions of fluctuating substrate supply: continuous DNA replication, accelerated transcription, chromatin remodeling for epigenetic plasticity, and DNA repair following replication stress or genotoxic injury. All require uninterrupted ATP availability.

If cytoplasmic ATP diffusion through nuclear pores were always sufficient, maintaining a dedicated nuclear glycogen reserve would represent a metabolic redundancy with no selective advantage. The opposite inference is more parsimonious: nuclear glycogen is maintained precisely because nuclear ATP demand can transiently exceed what diffusion can supply. In rapidly proliferating tumor cells, this gap is likely larger and more frequent than in normal differentiated cells, providing a selective pressure for nuclear glycogen accumulation.

Local BGR Within the Tumor Nucleus

Within the BGR framework, nuclear glycogen naturally supports a localized anaerobic energetic regime. The proposed metabolic sequence is:

Nuclear glycogen
↓ [glycogen phosphorylase; regulated by malin/laforin axis] [1,2]
Glucose-1-phosphate → glycolytic intermediates
↓ [nuclear glycolytic enzymes: PKM2, GAPDH, enolase-1] [3]
ATP + Pyruvate
↓ [pyruvate → acetyl-CoA via nuclear PDC or ACSS2]
Acetyl-CoA
↓ [histone acetyltransferases: p300/CBP, PCAF]
Histone acetylation → open chromatin
↓
Efficient DNA repair + transcriptional activation
↓
Maintenance of proliferative and survival capacity

This sequence integrates energetic and epigenetic functions within a single nuclear compartment, operating independently of mitochondrial oxidative phosphorylation. Each step is supported by published biochemical evidence, though the complete chain operating within a single tumor nucleus during therapy has not yet been demonstrated directly.

A Local BGR Model of Chemotherapy Resistance

Chemotherapeutic agents kill tumor cells through DNA damage (alkylating agents, topoisomerase inhibitors, platinum compounds), metabolic disruption (antimetabolites), or oxidative stress (anthracyclines). Successful cell survival after genotoxic injury requires rapid DNA repair, preservation of chromatin accessibility for repair factor recruitment, maintenance of pro-survival transcription, and restoration of metabolic homeostasis [4].

A nucleus possessing an intact local BGR would hold several theoretical advantages during therapeutic stress:

7. continued ATP production independent of cytoplasmic glycolysis, which may be impaired by drug-induced mitochondrial dysfunction;
8. continued acetyl-CoA synthesis sustaining histone acetylation at DNA damage sites, which is required for efficient repair by homologous recombination [7];
9. delayed chromatin condensation (resistance to nuclear syneresis), preserving physical accessibility for repair complexes;
10. maintenance of O-GlcNAc–dependent survival signaling through the hexosamine pathway [5];
11. buffering of nuclear energetic state during the hours required for DNA repair to complete.

Conversely, tumor cells with depleted nuclear glycogen — whether constitutively or induced pharmacologically — would be predicted to enter nuclear syneresis earlier during metabolic stress, impairing DNA repair and sensitizing them to genotoxic agents.

Relationship to Nuclear Syneresis

A companion hypothesis (Dmitrochenko, 2026) proposes that energetic insufficiency in the nucleus culminates in nuclear syneresis — a physical contraction of the nuclear hydrogel driven by loss of histone acetylation and chromatin condensation. Within this framework, nuclear glycogen content determines the resistance threshold for syneretic transition:

HIGH nuclear glycogen → sustained local BGR → open hydrated chromatin → therapy resistance

LOW nuclear glycogen → early BGR failure → nuclear syneresis → therapy sensitivity

This binary framing is necessarily simplified. In practice, nuclear glycogen content is a continuous variable and resistance will be graded. Nevertheless, it generates quantitative predictions distinguishable from existing resistance models, which do not account for compartment-specific nuclear energetics.

Experimental Predictions

1. Chemotherapy-resistant tumor cell lines should contain more nuclear glycogen than sensitive lines, measurable by nuclear fractionation followed by enzymatic glycogen assay or by PAS staining with nuclear counterstaining.
2. Nuclear glycogen content should correlate with treatment resistance independently of total cellular glycogen, predicting that the nuclear fraction — not total glycogen — is the operative variable.
3. Pharmacological inhibition of nuclear glycogenolysis using glycogen phosphorylase inhibitors (CP-91149; BAY U6751; MT19c) should sensitize resistant tumors to genotoxic agents in proportion to their nuclear glycogen content.
4. Nuclear ATP should remain elevated longer during acute glucose withdrawal in resistant versus sensitive cells, detectable with compartment-targeted ATeam or iATPSnFR biosensors.
5. Resistant tumor cells should maintain histone acetylation (H3K9ac, H3K27ac) and chromatin accessibility (ATAC-seq) during transient glucose limitation, whereas sensitive cells should show earlier loss.
6. Nuclear glycogen abundance in tumor biopsy specimens should correlate with chromatin accessibility markers and accelerated DSB repair kinetics (γH2AX resolution rate).

Predictions 1 and 5 are testable with existing standard laboratory methods. Predictions 3 and 4 require specialized reagents available in metabolic research settings. Prediction 6 could be addressed retrospectively in archival NSCLC or RCC specimens with documented treatment response.

Clinical Implications

If the present hypothesis is confirmed, nuclear glycogen quantification could serve as a prognostic biomarker and predictor of chemotherapy response in tumor types known to accumulate glycogen. The practical advantage is that glycogen is readily detectable in formalin-fixed paraffin-embedded sections by PAS staining — a method available in any pathology department.

More consequentially, nuclear glycogen metabolism may represent a targetable vulnerability. Glycogen phosphorylase inhibitors developed for metabolic disease indications are available and characterized [3]. Their selective effect on tumor cells relative to normal tissues would require investigation, but the premise is not unreasonable: normal differentiated tissues with high mitochondrial capacity are less dependent on local glycolytic regimes for nuclear energetics than rapidly proliferating tumor cells operating under intermittent hypoxia.

A combination strategy — glycogen phosphorylase inhibitor plus genotoxic chemotherapy — would be predicted to impair nuclear DNA repair selectively in glycogen-rich tumors, representing a mechanistically novel sensitization approach.

Discussion

The present hypothesis is distinguished from existing metabolic resistance models by its focus on the nuclear compartment rather than whole-cell metabolism. Current metabolic resistance frameworks emphasize cytoplasmic glycolytic flux, mitochondrial plasticity, fatty acid oxidation, or reactive oxygen species management. None specifically addresses the energetic autonomy of the nucleus during genotoxic stress.

An important alternative explanation must be acknowledged: chromatin remodeling at DNA damage sites could be driven entirely by cytoplasmic acetyl-CoA and ATP imported through nuclear pores, without any contribution from local nuclear glycogen. The present hypothesis predicts that this import is insufficient under the conditions of metabolic stress accompanying chemotherapy — a prediction that requires experimental resolution rather than assumption.

The selectivity concern deserves explicit attention. Normal tissues rich in glycogen — hepatocytes, myocytes, astrocytes — also contain nuclear glycogen and could theoretically be affected by glycogen phosphorylase inhibition. However, these cells possess high mitochondrial oxidative capacity and are unlikely to depend on local nuclear glycolysis for survival under therapeutic conditions in the same manner as hypoxic, rapidly cycling tumor cells. Differential sensitivity, rather than absolute selectivity, is the realistic expectation.

Finally, the hypothesis is agnostic about which tumor types are most likely to benefit. Clear cell RCC, glycogen-rich breast carcinoma, and hepatocellular carcinoma are logical initial candidates given documented glycogen accumulation, but the framework applies to any tumor in which nuclear glycogen can be demonstrated.

Conclusion

Nuclear glycogen may represent a previously underappreciated determinant of tumor survival during chemotherapy. Rather than functioning solely as a cytoplasmic carbohydrate reserve, nuclear glycogen may sustain a localized Basal Glycolytic Regime that preserves ATP production, epigenetic plasticity, and DNA repair capacity within the nucleus during therapeutic metabolic stress.

This model predicts that tumors enriched in nuclear glycogen will exhibit increased resistance to genotoxic agents, that nuclear glycogen content will prove a stronger predictor of resistance than total cellular glycogen, and that selective pharmacological disruption of nuclear glycogen metabolism will sensitize such tumors to chemotherapy. Each of these predictions is experimentally testable with existing methods and reagents.

Confirmation of this model would position nuclear glycogen measurement as a clinically accessible biomarker and nuclear glycogen phosphorylase as a rational drug target — connecting tumor metabolism, chromatin biology, and therapeutic resistance within a single mechanistic framework.

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Glycogen as a Graded Marker of Cellular Differentiation: Ontogenetic, Evolutionary, and Oncological Dimensions

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Abstract

Glycogen has long been regarded primarily as an energy reserve of mature, differentiated tissues — hepatocytes, myocytes, and astrocytes. Accumulating evidence, however, reveals a strikingly different picture at the opposite end of the differentiation spectrum. Intracellular glycogen is most abundant in naive pluripotent stem cells corresponding to the pre-implantation epiblast, and its depletion is not merely a metabolic consequence of differentiation but a molecular trigger of the transition to the next developmental stage. A dedicated nuclear glycogen pool, discovered only in 2019, supports local acetyl-CoA production and thereby maintains chromatin in an open, transcriptionally permissive configuration — the physical correlate of maximal developmental plasticity.

The present perspective integrates these findings with an evolutionary analysis of reproductive strategies, from yolk-dependent oviparity to matrotrophy — the condition in which an embryo receives glucose directly from the mother through placenta-like structures. We propose that the evolutionary transition from lecithotrophy to matrotrophy is the functional analogue, in phylogenetic time, of the implantation event in mammalian ontogeny: both represent the moment at which an embryo first loses energetic autonomy and becomes dependent on an external glucose supply. At this boundary, tissue glycogen first acquires the role of a buffer against fluctuating maternal delivery.

Three functionally distinct glycogen pools are identified across mammalian ontogeny: nuclear glycogen of the maximally undifferentiated cell, cytoplasmic glycogen of naive embryonic stem cells (ESCs) acting as a signaling molecule through the GSK3β–AMPK axis, and placental glycogen of specialized glycogen trophoblast (GlyT) cells emerging later in gestation as a fetal glucose buffer. These observations suggest that intracellular glycogen — and particularly its nuclear compartment — may serve as a quantitative biophysical marker of cellular plasticity, with implications for embryology, oncology, and the biology of cellular reprogramming.

Keywords: *glycogen; differentiation; naive pluripotency; nuclear glycogen; matrotrophy; AMPK; GSK3β; cellular plasticity; oncogenesis; BGR framework*

1. Introduction

The conventional view positions glycogen as a metabolite of specialized, terminally differentiated cells. Hepatocytes store and mobilize glycogen to buffer blood glucose; skeletal myocytes maintain glycogen reserves for contractile bursts; astrocytes provide glycogen-derived lactate to neurons during periods of increased synaptic activity. In each case, glycogen is a property of functional maturity.

Data accumulated over the past decade challenge this framing fundamentally. Several independent lines of evidence now indicate that:

(1) Glycogen is maximally abundant in naive pluripotent stem cells — the least differentiated mammalian cell state — and declines as differentiation proceeds.

(2) Depletion of intracellular glycogen is not passively caused by differentiation but actively triggers the transition to the next developmental stage through the GSK3β–AMPK signaling axis.

(3) A distinct nuclear glycogen pool, synthesized and metabolized within the nucleus itself, couples local carbohydrate availability to chromatin architecture via acetyl-CoA–dependent histone acetylation.

(4) In oncogenesis, dedifferentiation is accompanied by partial reactivation of glycogen accumulation, consistent with a return toward the metabolic profile of less committed progenitor states.

The present perspective assembles these observations into a unified conceptual framework, examines their evolutionary context across vertebrate reproductive strategies, and proposes that glycogen abundance — particularly in the nuclear compartment — functions as a graded, quantitative marker of cellular plasticity.

2. Nuclear Glycogen: The Deepest Layer

The discovery that glycogen is synthesized de novo within mammalian nuclei, reported by Sun and colleagues in 2019, fundamentally altered the landscape of nuclear metabolism [1]. Using stable isotope tracing, Sun et al. demonstrated that nuclear glycogenolysis generates glucose-1-phosphate, which enters a compartmentalized metabolic pathway producing pyruvate and, subsequently, acetyl-CoA — the obligate substrate for histone acetyltransferases [1]. The regulatory mechanism was further characterized by Donohue and colleagues, who showed that the E3 ubiquitin ligase malin promotes nuclear glycogenolysis and the resulting histone acetylation response [3].

The functional significance of this compartmentalized pathway is substantial. The nucleus performs ATP-intensive processes — transcription, chromatin remodeling, DNA repair, nuclear transport, and maintenance of phase-separated condensates — yet contains no mitochondria [4]. Nuclear glycogen provides an autonomous carbohydrate reserve that supports local acetyl-CoA production independently of cytoplasmic metabolic fluctuations. Reduced nuclear glycogenolysis leads to diminished acetyl-CoA availability, decreased histone acetylation, and chromatin condensation — the epigenetic signature of transcriptional silencing and reduced developmental plasticity [1,2].

Within the conceptual framework of the Basal Glycolytic Regime (BGR) — defined as a state of glycolytic dominance in a mitochondria-lacking compartment dependent on substrate-level phosphorylation from local carbohydrate reserves — the nucleus of a naive pluripotent cell represents a BGR microdomain of the highest order: maximally autonomous, maximally active, and most vulnerable to substrate depletion [5].

A progressive failure of this nuclear energetic system — whether through glucose deficiency, glycogen depletion, or loss of glycogenolytic enzyme activity — initiates a cascade culminating in chromatin condensation and physical contraction of the nuclear hydrogel, a process we term nuclear syneresis [5]. The sequence may be summarized as: reduced nuclear glycogenolysis → reduced local acetyl-CoA → histone deacetylation → chromatin compaction → nuclear syneresis → commitment to differentiation.

3. Cytoplasmic Glycogen in Naive Pluripotent Stem Cells

Intracellular glycogen is selectively abundant in naive ESCs and is absent from primed ESCs corresponding to the post-implantation epiblast. This distinction is reproduced in both murine and human ESC systems [6,7]. The biochemical basis involves GSK3β, which normally phosphorylates and inhibits glycogen synthase (encoded by Gys1). In naive ESCs, GSK3β is constitutively suppressed by Netrin-1 signaling, maintaining glycogen synthase in an active state. Critically, pharmacological inhibition of GSK3β promotes glycogen synthesis in naive but not in primed ESCs, indicating that the capacity for glycogen accumulation is a state-specific property of naive pluripotency rather than a general cellular response [6,8].

The cytoplasmic glycogen of naive ESCs performs at least two functions. First, it suppresses AMPK, the principal cellular energy-depletion sensor, thereby sustaining high rates of de novo fatty acid synthesis required for the intense membrane biogenesis accompanying rapid self-renewal [8]. Second, and more fundamentally, glycogen depletion via knockout of Gys1 in naive ESCs leads to robust AMPK activation and spontaneous transition to the primed state [8]. These observations suggest — though do not yet demonstrate conclusively — that glycogen expenditure is a physiological trigger, rather than merely a correlate, of the naive-to-primed transition.

The temporal dynamics of embryonic glycogen support this interpretation. Glycogen levels rise from the 2-cell stage to the early blastocyst and then fall sharply at the late blastocyst stage — precisely at the time of implantation [8,9]. The depletion of epiblast glycogen at implantation may thus serve as a metabolic signal synchronizing the transition to post-implantation development with maternal glucose availability.

Of particular note: glucose itself is not the preferred substrate of the very early embryo. The pre-morula embryo favors pyruvate and lactate; glucose uptake rises dramatically only at the blastocyst stage [10]. During the preceding cleavage divisions, the embryo is energetically autonomous — a pattern that resonates with the yolk-dependent strategy of oviparous vertebrates, as discussed below.

4. Evolutionary Context: From Lecithotrophy to Matrotrophy

4.1 The reproductive spectrum as a metabolic gradient

Vertebrate reproductive strategies span a continuum from complete embryonic autonomy to complete maternal dependence. This continuum may be read as a gradient of external glucose supply to the developing embryo, and it maps onto the ontogenetic sequence in mammalian development with striking fidelity — though we emphasize that the correspondences proposed here represent evolutionary hypotheses rather than established homologies.

4.2 Oviparous vertebrates: yolk as the analogue of autonomous glycogen

In birds, reptiles, and most fish, the embryo is energetically fully autonomous. All substrates — including carbohydrates — are pre-deposited in the yolk before fertilization and are mobilized progressively as development proceeds [11]. The yolk thus performs, at the organismal level, the same function that nuclear and cytoplasmic glycogen perform at the cellular level: it is a pre-formed carbohydrate reserve, consumed in register with developmental progress. The earliest cleavage-stage cells of oviparous embryos depend on pyruvate and lactate derived from yolk-associated stores, not on free glucose — a condition precisely analogous to the pre-morula mammalian embryo.

4.3 Ovoviviparous fish: internal incubation without matrotrophy

In ovoviviparous (lecithotrophic viviparous) fish such as guppies, angel sharks, and many others, eggs develop internally but the embryo receives little or no maternal nutrient input, depending entirely on intra-egg yolk reserves [12]. This strategy is functionally identical to oviparity from the embryo's metabolic perspective: the site of incubation has shifted, but the energetic relationship between mother and embryo has not. No selective pressure yet exists for tissue-level glycogen accumulation in the embryo, because the glucose supply is not maternal and therefore not subject to maternal metabolic fluctuation.

4.4 Matrotrophic viviparity: the critical evolutionary boundary

The functionally decisive transition is not viviparity per se — which has arisen independently more than 150 times in vertebrates — but the emergence of matrotrophy: the direct transfer of maternal nutrients, including glucose, to the embryo through placenta-like structures. This transition has occurred independently in multiple fish lineages (notably in the family Goodeidae among teleosts, and in select sharks including some Carcharhiniformes), in viviparous squamate reptiles, and was achieved once in the common ancestor of placental mammals approximately 100 million years ago [13,14].

At the matrotrophic boundary, the embryo first encounters a glucose supply that is external, variable, and under maternal metabolic control. This creates, for the first time in embryonic history, a selective advantage for maintaining tissue-level glycogen: a buffer against fluctuations in maternal glucose delivery. We propose that the emergence of matrotrophy represents the evolutionary moment at which embryonic tissue glycogen first acquires adaptive significance beyond its role as a simple energy reserve.

4.5 Mammalian ontogeny as evolutionary recapitulation

In mammalian ontogeny, the implantation of the blastocyst may be interpreted as a functional recapitulation of this evolutionary boundary. Before implantation, the embryo is energetically autonomous — its cells depend on pyruvate, lactate, and limited glucose, independent of maternal supply, in a pattern analogous to lecithotrophic development. After implantation, the embryo becomes progressively dependent on maternally supplied glucose through trophoblast-mediated exchange. The depletion of naive-ESC glycogen at implantation is, in this reading, the cellular-level correlate of the evolutionary loss of lecithotrophy.

This correspondence is offered as an evolutionary analogy rather than a strict homology, and we recognize that it awaits direct experimental or comparative investigation. Nevertheless, the parallel is mechanistically coherent: both transitions — evolutionary matrotrophy and developmental implantation — mark the moment at which external glucose supply first becomes physiologically significant for the developing organism.

5. Three Functional Pools of Glycogen in Mammalian Ontogeny

The data reviewed above support the identification of three functionally distinct glycogen pools that appear sequentially during mammalian ontogeny, each with a characteristic location, regulatory mechanism, developmental timing, and functional significance.

Pool	Developmental timing	Primary function
I. Nuclear glycogen (maximally undifferentiated	Pre-implantation; maximal in naive pluripotency	Local acetyl-CoA for histone acetylation; maintenance of open

cells)		chromatin; BGR microdomain autonomy
II. Cytoplasmic glycogen (naive ESCs / pre-implantation epiblast)	Pre-implantation; absent in primed ESCs	AMPK suppression; de novo fatty acid synthesis; signaling molecule triggering implantation upon depletion
III. Placental glycogen (glycogen trophoblasts, GlyT)	Appears ~E6.5; peaks E15.5–16.5; declines by E18.5	Maternal glucose buffer for the fetus during late organogenesis; classical glycogen depot of a differentiated tissue

A critical distinction separates Pools I and II from Pool III. The first two pools are inversely correlated with differentiation: they are maximal at the undifferentiated pole and absent in committed cells. Pool III, by contrast, belongs to a fully differentiated, specialized placental cell type and performs the classical storage function of a mature tissue glycogen depot. This distinction cautions against the simplistic equation glycogen = immaturity and underscores the functional multiplicity of glycogen biology across the developmental time axis.

6. Oncogenesis: Partial Reactivation of the Undifferentiated Glycogen Profile

If glycogen abundance marks the undifferentiated pole, then oncogenic dedifferentiation — which involves partial reactivation of embryonic gene expression programs — should be accompanied by glycogen reappearance in tumor cells. Published data are consistent with this prediction, though the pattern is not uniform.

Studies in hepatocellular carcinoma cell lines of varying dedifferentiation levels reveal that glycogen is detectable even in severely dedifferentiated clones, specifically in a subpopulation of cancer stem-like cells (CSLCs) in vitro and in the majority of tumor-initiating cells derived from ascites hepatoma in vivo [15]. The authors interpret these observations as suggesting that dynamic changes in glycogen formation in CSLCs and tumor-initiating cells may be of importance for their dedifferentiation and tumor-initiating capacity.

Within the BGR framework, this pattern is coherent. A BGR-active clone that has undergone Hopf bifurcation — exiting the regulated oscillatory cycle and entering unbounded proliferation upon loss of the ECM structural stop signal — is predicted to restore the metabolic profile of maximally proliferating, minimally committed cells [5]. This profile includes both nuclear and cytoplasmic glycogen accumulation. Glycogen reappearance in tumor cells is therefore not an epiphenomenon but a predictable feature of the energetic regime that supports unconstrained proliferation.

We emphasize, however, that the tumor cell is not an embryo. It deploys embryonic metabolic programs selectively and often in distorted form. The observations reviewed here are consistent with partial reactivation of the undifferentiated glycogen profile in cancer stem-like populations, not with wholesale recapitulation of naive pluripotency. This distinction matters for experimental design: not all tumor cells will display glycogen accumulation, but cancer stem-like subpopulations — those with the highest tumor-initiating capacity — are candidates for preferential glycogen positivity.

7. Testable Predictions

The framework generates the following experimentally addressable predictions:

(1) Nuclear glycogen content should correlate inversely with degree of differentiation across a panel of cell lines spanning pluripotency to terminal differentiation, measurable by PAS staining with nuclear counterstaining or by subcellular fractionation followed by enzymatic glycogen assay.

(2) Forced glycogen depletion in naive ESCs (via Gys1 knockout or glycogen phosphorylase overexpression) should accelerate the naive-to-primed transition, while forced glycogen synthesis in primed ESCs should partially retard differentiation.

(3) Nuclear ATP and acetyl-CoA should decrease before cytoplasmic equivalents during acute glucose deprivation, detectable with compartment-specific biosensors (ATeam, iATPSnFR for ATP; genetically encoded acetyl-CoA sensors).

(4) In matrotrophic fish species, embryonic tissue glycogen should be detectable at developmental stages corresponding to the onset of maternal glucose transfer, but absent or minimal in lecithotrophic ovoviviparous species at equivalent developmental stages.

(5) In tumor cell populations, PAS staining intensity should correlate with markers of cancer stem-like identity (CD44hi/CD24lo, ALDH activity, sphere-forming capacity), with the highest glycogen content in the most dedifferentiated subpopulation.

8. Conclusion

Glycogen describes an arc that is inversely aligned with the trajectory of differentiation. In the maximally undifferentiated state — naive pluripotency — glycogen is present in two functionally distinct pools: nuclear glycogen maintaining chromatin openness through local acetyl-CoA production, and cytoplasmic glycogen suppressing AMPK to sustain the biosynthetic demands of self-renewal. As differentiation proceeds, both pools are depleted. The depletion of cytoplasmic glycogen is, in all probability, not merely a consequence but a trigger of the naive-to-primed transition — acting as a metabolic signal that synchronizes the cell's developmental state with its glucose environment.

In evolutionary time, the transition from lecithotrophy to matrotrophy represents the moment at which the embryo first loses energetic autonomy and becomes dependent on external glucose. We propose that this transition is the evolutionary prototype of implantation: both events mark the boundary at which the depletion of an internal carbohydrate reserve first acquires the meaning of a developmental decision. The parallel, while not yet experimentally formalized as a strict homology, is mechanistically coherent and generates testable comparative predictions.

In oncogenesis, the partial reactivation of the undifferentiated glycogen profile — particularly in cancer stem-like populations — is consistent with, and predicted by, the BGR framework's account of dedifferentiation as a return to the metabolic state of maximal cellular plasticity.

Taken together, these observations suggest that intracellular glycogen content — and especially the nuclear glycogen pool — may serve as a graded, quantitative biophysical marker of cellular plasticity, applicable across embryology, stem cell biology, and oncology.

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Conflicts of Interest

The author declares no conflicts of interest.

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Part II

The Clone and the Organism

Pathological Clone Formation:

The Differentiation Information Pyramid and the Regenerative Energy Window

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Series: Article I. "The Clone and the Organism"

Abstract

A conceptual model is proposed for the formation of a pathological cellular clone as the structurally predictable consequence of two fundamental biological mechanisms: the differentiation information pyramid and the regenerative energy window.

The differentiation hierarchy is viewed not as a linear conveyor but as a self-sustaining informational structure in which each level of maturity actively suppresses the intermediate compartment immediately above it through level-specific signalling proteins. The pyramid is established during embryogenesis and terminates in two functional poles on the reverse differentiation scale: terminally differentiated cells (level 0) and their immediate precursors — foetal cells (level −1). All more primitive cells are suppressed by their high energy consumption and vertical informational suppression, and exist as an energy-dependent background population rather than a functional compartment.

When tissue is damaged, three restraining forces attenuate simultaneously, creating a regenerative energy window. The BGR-active undifferentiated clone gains a substrate advantage precisely when suppressive and competitive pressure is minimal. By synchronising with the tissue's regenerative oscillator, the clone accumulates across repeated cycles, invisible to surveillance systems.

Three probabilistic outcomes of the first stage are described: clone elimination as the most frequent outcome, a stationary state as the most prevalent in the population at large, and progression to the first bifurcation point as a rare but cumulatively increasing outcome. The model is discussed with specific reference to the liver as an organ with a unique regenerative mechanism.

Keywords: *differentiation information pyramid; level-specific suppression; energy window; regenerative oscillator; BGR; pathological clone; latent oncogenesis; extracellular matrix; liver*

1. Introduction

Oncogenesis is conventionally described as the accumulation of somatic mutations that progressively remove constraints on cellular proliferation. This model explains much — but leaves several fundamental questions unanswered. Why does the latent period between the first molecular event and a clinically significant tumour span decades? Why is oncogenesis statistically linked to chronic inflammation and repeated regeneration, regardless of tissue type? Why do the vast majority of potentially malignant cellular alterations fail to progress?

The present model proposes a different starting point. Oncogenesis is viewed not as a breakdown of normal biology but as the structurally predictable consequence of two fundamental homeostatic mechanisms: the differentiation information pyramid and the regenerative oscillator. A pathological clone forms not in spite of these mechanisms — but through them, exploiting their physiological vulnerabilities.

The model draws on the conceptual apparatus of the BGR Framework — specifically the concepts of the Basal Glycolytic Regime as the energetic state of the undifferentiated cell, the Regeneration Pendulum as the oscillatory mechanism of tissue renewal, and the extracellular matrix as the structural stop-signal for clonal expansion [1]. To these is added a new element: the differentiation information pyramid with level-specific suppressors of intermediate compartments.

This article is the first in the series “The Clone and the Organism.” Article II examines the interaction of the clone with the immune system and the autoimmune zone between two bifurcation points. Article III describes the clone’s capture of the tissue oscillator with progressively increasing amplitude as the mechanism of malignant progression.

2. The Differentiation Information Pyramid

2.1 Pyramid structure: from embryogenesis to mature tissue

The differentiation hierarchy is established during embryogenesis and is complete by birth, terminating in two functional poles conveniently described on the reverse differentiation scale. Level 0 corresponds to terminally differentiated cells — the functional pole of the tissue. Level −1 corresponds to their immediate precursors — foetal cells that persist in mature tissue as a reserve compartment.

All cells of more primitive levels — multipotent, pluripotent, totipotent — are normally suppressed by two simultaneously operating mechanisms. The first is vertical informational suppression: each level N secretes level-specific signalling proteins that eliminate intermediate compartment N−1. The second is energetic disadvantage: maintaining open chromatin, nuclear glycogen reserves, self-renewal machinery, and broad transcriptional competence imposes metabolic costs unavailable to cells under normal substrate competition.

The consequence: poly- and pluripotent cells exist in mature tissue exclusively in zones of locally high energy delivery — near arterioles, in metabolically privileged niches. This is a background population, not a functional reserve. Its existence is determined by the energetic geography of the tissue, not by a regulatory programme.

2.2 Level-specific suppression of intermediate compartments

The standard representation of tissue renewal as a linear sequence — stem cell → progenitor → transit-amplifying cell → mature cell — describes genealogical relationships but does not reveal the regulatory architecture that sustains this sequence.

We propose viewing the differentiation hierarchy as an informational pyramid in which each level N actively suppresses the intermediate compartment N−1 situated between itself and the level above. The target of suppression is not the parental stem cell but the intermediate link directly connecting the mature cell to its less differentiated precursor.

Data from several model systems confirm this architecture. The differentiated colonocyte secretes Indian Hedgehog, which via the mesenchyme elevates BMP signalling and suppresses Wnt at the crypt base — acting on transit-amplifying cells, not directly on Lgr5+

stem cells [2,3]. In the hair follicle, committed progeny form an inner layer that inhibits stem cell activation in the outer layer through direct contact inhibition [4]. In the haematopoietic system, differentiating haemocytes deliver inhibitory signals to their haematopoietic progenitor precursors [5].

The critical implication of this architecture: the disappearance of an intermediate compartment is not the result of successful differentiation — it is the condition for hierarchical stability. The pyramid holds because each rung actively removes the rung immediately above it.

2.3 Division stochastics: three daughter cell variants

At each division a cell produces not a strictly identical daughter but one of three variants:

12. a cell of the same differentiation level (symmetric division)
13. a cell one level higher on the differentiation scale
14. a cell one level lower — more primitive

The probability of each variant is determined by the balance of two forces. Vertical suppression from more mature cells bears down on the primitive variant. The energetic advantage of the more differentiated cell creates competitive selection in favour of the mature variant. The result is statistical predominance of the more differentiated division outcome under normal conditions. But the probability of the primitive variant is never zero.

This means the pyramid continuously generates cells of more primitive levels at every division. Normally they are immediately suppressed — energetically and informationally. The pathological clone begins with the same event — the appearance of a primitive variant at division. The difference lies in the state of the microenvironment at the moment of its appearance.

2.4 The special case of the liver

The general model of the information pyramid applies to all tissues — with one significant exception requiring separate consideration.

The liver possesses a unique regenerative mechanism without parallel in other organs. Following moderate injury, the liver restores itself through proliferation of terminally differentiated hepatocytes that rapidly re-enter the cell cycle from G0 phase [6]. This breaks the basic rule of the pyramid: the terminally differentiated level-0 cell divides without recruiting level-−1 precursors.

With massive or chronic injury, a reserve mechanism is engaged: bipotent oval cells — progenitors from the canals of Hering capable of differentiating into both hepatocytes and cholangiocytes — become activated [7,8]. Additional complexity arises from the observation that under injury conditions, a proportion of mature hepatocytes are capable of transient dedifferentiation — reversion to an oval-like state followed by redifferentiation [9].

Thus in the liver the direction of differentiation is reversible under certain conditions, and the mature level-0 cell retains proliferative potential. This does not refute the information pyramid model but indicates that in the liver the rigidity of the hierarchy is reduced by an additional plasticity mechanism. Hepatic oncogenesis — hepatocellular carcinoma — likely exploits both the standard pathway through a primitive clone and the liver-specific pathway through dedifferentiation of mature hepatocytes. This may explain the particular aggressiveness of HCC in cirrhosis — a tissue with the highest frequency of regenerative cycles and disrupted ECM architecture.

3. The Regenerative Window

3.1 Three restraining forces and their simultaneous attenuation

Under normal homeostatic conditions, less differentiated cells are held in a subordinate position by three simultaneously operating forces:

15. informational suppression: level-specific signalling proteins from mature cells eliminate the intermediate compartment
16. energetic competition: more differentiated cells are metabolically more efficient and displace primitive clones in competition for substrate
17. structural restraint: the extracellular matrix serves as a physical stop-signal for BGR-active clonal expansion

The stability of the pyramid is overdetermined — maintained by three independent mechanisms simultaneously. Clonal expansion requires the attenuation of all three.

Tissue injury physiologically attenuates all three mechanisms simultaneously. This simultaneity is structurally inevitable: all three mechanisms depend on the presence of an intact dense layer of differentiated cells. When that layer is disrupted — all three forces fall together.

3.2 The content of the energy window

Energy surplus. Inflammatory hyperaemia delivers increased substrate to the zone of injury. Cellular debris releases metabolites. At this point it is essential to understand why the pathological clone prevails under these conditions — not because it carries a special “cancer programme” or specific growth-promoting mutations, but because its energetic regime — the BGR, Basal Glycolytic Regime — precisely matches the conditions of the regenerative window. The window generates an excess of glycolytic substrate and removes the competitive pressure of metabolically efficient mitochondrial neighbours. The clone does not invade a foreign environment; it finds itself in an environment for which it is already optimised. The Warburg effect, in this context, is neither a pathology nor a coincidence, but an exact correspondence between the metabolic phenotype of the clone and the physiology of injured tissue.

Suppression attenuation. The mature cell population is reduced by injury. Level-specific suppressive signals — Hedgehog, BMP, differentiation-inducing cytokines — decline in proportion to the loss of their cellular source. Intermediate compartments are no longer eliminated. The lower rungs of the pyramid lose stability.

Competitive relief. Physical space is freed. The differentiated cells that previously displaced the undifferentiated clone metabolically are absent or reduced. The clone expands into vacated territory without its usual competitive pressure.

3.3 Oncogenesis as the structural consequence of regeneration

A fundamental conclusion follows from this analysis: oncogenesis is not a random breakdown of normal biology but the structurally predictable consequence of the regenerative mechanism itself. Every regenerative episode physiologically creates conditions under which a BGR-active clone can expand. Normally the window closes as the differentiated cell layer is restored.

Risk accumulates with each repeated cycle of damage and regeneration. This corresponds precisely to the epidemiological pattern: cirrhosis and hepatocellular carcinoma, chronic colitis and colorectal cancer, recurrent mucosal injury and squamous cell carcinoma. Oncogenesis in these contexts is not an exception to the rule but the cumulative realisation of a small probability through repeated iteration. Chronic inflammation elevates oncogenic risk not through specific mutagenic action but by increasing the frequency of regenerative windows.

4. Clone Dynamics Within the Regenerative Oscillator

4.1 Phase synchronisation — the period of invisibility

A clone arising within the regenerative window initially oscillates in phase with the tissue regenerative cycle. It proliferates during the proliferative phase, partially contracts during the resolution phase, and re-expands with the next cycle. From the perspective of the tissue's regulatory systems, the clone is indistinguishable from the normal less-differentiated cellular compartments participating in physiological renewal.

This period of phase synchronisation constitutes the latent phase of oncogenesis. The clone does not actively hide — it is invisible because it is behaviourally indistinguishable from normally regenerating tissue. The latent period of decades is not the time required for mutation accumulation but the time required for clone accumulation through incremental growth across repeated regenerative cycles.

4.2 Clone accumulation across regenerative cycles

Partial death of clone cells during the resolution phase does not return the clone to its original size — it stimulates proliferation of surviving cells in the next cycle. The mechanism: the death of part of the clone generates a local regenerative stimulus — DAMP signals, substrate release, reduction of contact inhibition — which the remaining cells receive as a proliferative signal.

The clone exploits the physiological pendulum as the driving force of its own growth. This explains the duration of the latent period: the clone grows not continuously but incrementally — by a quantity Δ per regenerative cycle. Decades when episodes are infrequent; years when chronic inflammation produces frequent cycles. Chronic inflammation elevates oncogenic risk not through specific mutagenic action but by increasing the frequency of regenerative windows.

4.3 The first bifurcation point

When the clone reaches critical mass, its internal proliferative dynamics begin to exceed what the tissue oscillator can assimilate. The clone's proliferative phase outlasts the tissue's. Its contraction during resolution is insufficient to restore synchrony. The tissue's regulatory systems register a persistent, non-resolving proliferative focus.

This is the first bifurcation point. The clone has not yet degraded the ECM or acquired autonomous vascular supply — it remains within the physical constraints of the tissue. But it is now large enough to perturb the oscillator rather than merely participate in it. The subsequent dynamics are the subject of Article II of this series.

5. Probabilistic Map of First-Stage Outcomes

Three outcomes are possible at the first stage. Their probability distribution is fundamental to understanding the rarity of clinically significant oncogenesis against a background of ubiquitous subclinical clone formation.

Outcome	Probability	Mechanism
Clone elimination	Most frequent	Window closes before clone reaches self-sustaining size. Immune clearance. Restoration of suppressive pressure.
Stationary state	Most prevalent in population long-term	Clone stabilises below first bifurcation threshold. Oscillates with the pendulum. Clinically silent for years to decades.
Progression	Rare per cycle; cumulatively increasing	Clone exceeds first bifurcation threshold. Onset of interaction with immune system — Article II.

Clone elimination is the most frequent outcome. The regenerative window closes before the clone reaches the size at which partial cell death within a cycle stimulates growth of the survivors. Immune clearance, restoration of suppressive pressure, competitive displacement — any of these mechanisms is sufficient for elimination. The majority of potentially malignant cellular alterations end here — clinically undetected.

The stationary state is probably the most prevalent long-term outcome in the population. The clone stabilises at a sub-threshold level: small enough to remain in phase synchrony with the tissue oscillator, stable enough not to be eliminated. Indirect evidence: the high frequency of microscopic tumour foci found at autopsy in individuals without clinical signs of malignancy.

Progression is a rare outcome per individual regenerative cycle but cumulatively increasing. The probability of progression over a lifetime is the product of a small per-cycle probability and a large number of regenerative cycles across decades. Oncogenesis is rare not because pathological clones rarely arise — but because the probability of progression at each stage is small, and all stages must be realised sequentially.

6. Variable Map of the First Stage

(1) Local energetic excess during regeneration (V1): substrate availability to the BGR-active clone under conditions of inflammatory hyperaemia. Determines the width of the energy window.

(2) Attenuation of level-specific suppression (V2): reduction of suppressive signals from mature cells to intermediate compartments, proportional to the loss of the mature cell population. Determines the informational component of the window.

(3) Competitive relief (V3): reduction of metabolic competition from differentiated neighbours in the injury zone. Determines the spatial component of the window.

(4) Clone size (V4): threshold variable of the first bifurcation. Below threshold — clone oscillates with tissue and accumulates incrementally. Above threshold — clone perturbs the oscillator.

(5) Degree of phase synchronisation (V5): measure of oscillatory coherence between the clone and the tissue regenerative cycle. At full synchronisation the clone is invisible to surveillance systems. Progressive desynchronisation is the condition for the first bifurcation.

7. Experimentally Testable Predictions

(1) In tissues with documented chronic inflammation, the frequency of subclinical clonal foci should correlate with the number of regenerative episodes sustained, not merely with patient age.

(2) The size distribution of subclinical clonal foci should be bimodal: the majority stable within a narrow range (stationary state), a minority progressively enlarging (path to the first bifurcation).

(3) In experimental models with controlled regenerative cycle frequency, the rate of malignant transformation should increase non-linearly with the number of injury events — accelerating as intervals between events shorten.

(4) Level-specific suppressive signals (BMP, Hedgehog) should be reduced in the microenvironment of subclinical clonal foci compared with normal tissue from the same patient — independently of focus size.

(5) The metabolic profile of subclinical focus cells should correspond to BGR — predominance of glycolysis, elevated glycogen content, reduced mitochondrial activity — differing from surrounding differentiated tissue quantitatively rather than qualitatively.

(6) In the liver under chronic hepatitis, the rate of malignant transformation should be disproportionately higher than in other chronic inflammatory diseases of equivalent duration — as the model predicts, due to the additional pathway through dedifferentiation of mature hepatocytes.

8. Conclusion

Pathological clone formation is not a random molecular event superimposed on normal tissue biology but the structurally predictable consequence of two fundamental homeostatic mechanisms: the differentiation information pyramid and the regenerative oscillator.

The information pyramid is stable because it is overdetermined: by three simultaneous forces — informational suppression, energetic competition, and structural ECM restraint. Regeneration physiologically attenuates all three forces simultaneously — not as a pathological event but as the inevitable consequence of tissue restoration.

At every division, three daughter cell variants are possible. The primitive variant is normally suppressed immediately — but the probability of its survival in a regenerative window, when all three restraining forces are attenuated, is fundamentally higher. The clone arising in the window synchronises with the tissue oscillator and accumulates incrementally — invisible to surveillance systems.

The fate of the clone is determined by the probabilistic distribution of three outcomes: elimination as the most frequent, stationary state as the most prevalent in the population, progression to the first bifurcation point as a rare cumulative outcome. The rarity of clinically significant oncogenesis against a background of ubiquitous subclinical clones reflects the product of small probabilities at each stage.

The liver occupies a special place in this model: the presence of an additional regenerative pathway through dedifferentiation of mature hepatocytes creates a specific vulnerability that likely explains the high frequency of HCC in cirrhosis.

The interaction of the clone with the immune system following the first bifurcation point, the autocatalytic inflammatory loop, and the autoimmune zone are examined in Article II of this series. In the proposed model, oncogenesis represents not a violation of the laws of tissue organisation, but their rare probabilistic consequence. It is for this reason that understanding the mechanisms of normal regeneration becomes the key to understanding the origin of tumours.

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Conflicts of Interest

The author declares no conflicts of interest.

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The Onset of Clone–Organism Interaction:
The First Bifurcation, the Autoimmune Zone, and the Paradox of the Immune Response

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Abstract

This article examines the dynamics of a pathological clone following attainment of the first bifurcation point — the moment at which the clone exits phase synchronisation with the tissue regenerative oscillator and becomes visible to the organism’s surveillance systems.

The paradox of the immune response is described: non-specific antibodies directed against the undifferentiated antigens of the clone cause collateral damage to normal differentiated cells, generating local inflammation. That inflammation reproduces the conditions of the regenerative window — thereby stimulating the growth of the very clone against which the response is directed. An autocatalytic loop is established that sustains itself without external trigger.

The concept of the autoimmune zone is proposed as the state of a clone positioned between two bifurcation points: large enough to perturb the tissue oscillator, yet not large enough to escape its physical constraints. Systemic lupus erythematosus is examined as a modelling case for inter-bifurcation dynamics. The applicability conditions of the model are defined explicitly: it describes a specific class of autoimmune processes in immunologically open tissues, and does not apply to barrier-isolated organ-specific diseases or to diffuse stromal-compartment pathology.

The binary concept of immunological tolerance is replaced by a continuous variable: the thymic representational density of clone antigens (V11), determined by AIRE-dependent stochastic promiscuous gene expression in medullary thymic epithelial cells. Clone immunogenicity is a gradient, not a threshold, determined by the overlap between the clone’s antigen profile and the cumulative population-level antigen representation achieved by mTECs.

Keywords: first bifurcation; autoimmune zone; autocatalytic loop; thymic representational density; AIRE; promiscuous gene expression; systemic lupus erythematosus; cytostatics; BGR; regenerative oscillator

1. Introduction

Article I of this series described the mechanism of pathological clone formation: the differentiation information pyramid, the regenerative energy window, and the dynamics of clone accumulation during the phase of synchronisation with the tissue oscillator. The clone grows incrementally, invisible to surveillance systems, and in the majority of cases either is eliminated or stabilises in a stationary state.

The rare outcome — progression to the first bifurcation point — initiates a qualitatively different phase. The clone ceases to be a passive participant in the tissue oscillator and begins to interact actively with the immune system. This interaction is paradoxical: the immune response directed against the clone, under certain conditions, does not suppress it but stimulates it.

The present article describes the mechanism of this paradox, the conditions under which the inter-bifurcation state arises and is maintained, and the boundaries within which the model applies. Article III examines the transition through the second bifurcation point: ECM degradation, NO-mediated vasodilation, mechanical capillary expansion, and capture of the tissue oscillator with progressively increasing amplitude.

2. The First Bifurcation Point: Exit from Synchronisation

2.1 The mechanism of exit

The clone accumulating during phase synchronisation with the tissue oscillator reaches critical mass. Its internal proliferative dynamics exceed what the tissue oscillator can assimilate. Three features characterise this transition.

The clone's proliferative phase outlasts the tissue's regenerative phase. The clone's contraction during the resolution phase is insufficient to restore synchrony with the tissue cycle. The tissue's regulatory systems register a persistent, non-resolving proliferative focus.

The clone has not yet exited the physical constraints of the tissue — it has not yet degraded the ECM or acquired autonomous vascular supply. But it is now large enough to perturb the oscillator. This is the moment at which the organism first notices the clone.

2.2 The antigenic profile of the clone

A clone at an intermediate differentiation level expresses antigens characteristic of less differentiated cells: nuclear proteins, histones, nucleosomal components, and surface markers of progenitor states. These antigens are normally present in restricted niche compartments and only in small quantities.

When the clone exits synchronisation, its antigenic load exceeds the threshold of immunological tolerance. The immune system is activated.

3. The Paradox of the Immune Response

3.1 Antibody non-specificity and collateral damage

The antibodies generated against the clone are directed against antigens of less differentiated cells. These antigens are not unique tumour-specific markers. They are present, at lower concentration, on normal stem and progenitor cells throughout the organism, and on the surface and in the nuclei of normal differentiated cells undergoing division.

The consequence: antibodies bind not only to the clone but also to normal tissue cells. Collateral damage to differentiated cells — not the clone itself but its normal tissue environment — ensues.

3.2 The autocatalytic loop

Collateral damage to differentiated cells generates local inflammation. That inflammation reproduces precisely the conditions described in Article I as the regenerative energy window:

Fewer mature cells → attenuation of level-specific suppression of intermediate compartments; space is freed → competitive relief; substrate released from dead cells → energetic surplus for the BGR-active clone.

The clone receives exactly the conditions it requires for growth. Clone growth increases antigenic load. Increased antigenic load intensifies the immune response. Intensified immune response increases collateral damage. The loop is closed.

Antibodies

↓

Collateral damage to mature cells

↓

Local inflammation → regenerative window (V1 + V2 + V3)

↓

Clone growth → increased antigenic load

↓

Increased antibody production

↓

...

The loop sustains itself without external trigger. This is an autocatalytic process.

3.3 The zone of paradoxical amplification

The intensity of the immune response becomes the sixth variable of the model (V6). It has a threshold character.

Below the threshold of total elimination — the immune response paradoxically stimulates clone growth through the autocatalytic loop. Above the threshold of total elimination — the clone is destroyed completely and the loop has no opportunity to develop. Between these values lies the zone of paradoxical amplification: the immune response exists, collateral damage is generated, the loop operates, and the clone grows faster than it would without an immune response.

This explains the clinically observed phenomenon: a tumour following incomplete immune response or incomplete therapy frequently grows more aggressively than before treatment.

3.4 Specificity of the immune response as the seventh variable

The higher the specificity of antibodies against clone antigens, the less collateral damage to normal cells, the weaker the inflammatory signal, and the less pronounced the regenerative window generated by the immune response itself.

Targeting specificity — rather than potency of immunosuppression — is the therapeutically decisive variable (V7). Targeted biological therapy disrupts the autocatalytic loop not by suppressing immunity but by eliminating collateral inflammation.

4. The Autoimmune Zone: Chronic Oscillation Between Two Bifurcations

4.1 Definition of the inter-bifurcation state

A clone that has exited synchronisation with the tissue oscillator (first bifurcation) but has not yet degraded the ECM or acquired autonomous vascular supply (second bifurcation) is in the inter-bifurcation state.

In this state, the clone is large enough to generate an immune response; not large enough to exit the physical constraints of the tissue; maintained in chronic oscillation by the autocatalytic loop; neither eliminated nor progressing to malignancy. We propose that this state of chronic oscillation corresponds to the clinical picture of a specific class of autoimmune disease — described in detail in the applicability section below.

4.2 Clinical characteristics of the inter-bifurcation state

Chronicity without resolution — the autocatalytic loop is self-sustaining. Spontaneous resolution requires reduction of the clone below the first bifurcation threshold, which does not occur without intervention.

Systemic involvement from local origin — non-specific antibodies distribute immune complexes throughout the vascular system, producing multi-organ pathology from a single clonal source.

Flare–remission pattern — corresponds to the oscillatory dynamics of the clone in resonance with the tissue regenerative cycle. Flares are the proliferative phases of the clone. Remissions are phases of partial contraction without return below the threshold size.

Appearance of less differentiated cells in circulation — during the proliferative phase, the clone sheds less differentiated cells into the bloodstream. This is not the cause of the autoimmune process — it is a marker of inter-bifurcation clone dynamics.

4.3 Why cytostatic agents are effective in the inter-bifurcation state

Cytostatic agents reduce the size of all actively dividing populations — including the clone during its proliferative phase. If the clone is reduced below the first bifurcation threshold, it re-enters phase synchronisation with the tissue oscillator and again becomes invisible to immune surveillance.

In this interpretation, the autocatalytic loop is interrupted through desynchronisation achieved by clone size reduction. This mechanism is proposed as an additional explanation — alongside the established immunosuppressive action of cytostatics — for their efficacy in the inter-bifurcation state. The relapse pattern upon therapy withdrawal is consistent: cytostatics reduce the clone but do not eliminate it, and upon withdrawal the clone re-expands through the next regenerative window.

This mechanism may be particularly relevant for methotrexate at low doses in rheumatoid arthritis — a regimen whose efficacy cannot be attributed solely to systemic immunosuppression at these doses.

5. Thymic Representational Density: Replacing the Binary Tolerance Registry

5.1 The AIRE mechanism and its stochastic character

The classical formulation of central tolerance posits a binary registry: antigens present in the thymus during T cell education are classified as self; antigens absent are foreign. This formulation, while useful as a first approximation, is incompatible with the known biology of thymic antigen presentation.

Medullary thymic epithelial cells (mTECs) achieve self-antigen coverage through promiscuous gene expression (PGE) — the AIRE-dependent ectopic transcription of tissue-restricted antigens spanning essentially all tissues of the body [Anderson 2002, Kyewski 2004]. Critically, this process is stochastic at the individual cell level: each mTEC expresses a subset of genes, and complete antigen coverage is a population-level property, not a property of individual cells [Derbinski 2008, Meredith 2015].

Furthermore, the differentiation-associated transcription factors Oct4, Nanog, and Sox2 — markers of primitive undifferentiated states — are among the AIRE-dependent targets expressed in mTECs [Oven 2008]. This means that antigens of primitive cell compartments are represented in the thymus — but at low frequency and variable coverage, not excluded from representation.

5.2 Thymic representational density as a continuous variable (V11)

We replace the binary tolerance registry with a continuous variable: the thymic representational density of clone antigens (V11). This variable is defined as the overlap between the antigen profile of the pathological clone and the cumulative population-level pattern of AIRE-dependent gene expression in mTECs.

High V11 (dense representation): the clone's antigens are well-represented in the mTEC population. Central tolerance is robust. The immune system mounts a weak response or none. The clone may be immunologically near-invisible despite its pathological character. Direct path toward oncogenesis without an autoimmune phase.

Intermediate V11 (partial representation): the clone's antigens are stochastically expressed — represented in some mTECs but not others. Central tolerance is incomplete and variable. The immune system attacks with uncertainty — some T cell clones are deleted, others escape. This is precisely the condition generating inter-bifurcation oscillation. The autoimmune zone.

Low V11 (sparse representation): the clone's antigens belong to patterns rarely or never achieved by the mTEC population. Central tolerance is minimal. Strong immune response is possible. Either elimination — or, under conditions of paradoxical amplification, accelerated autocatalytic growth.

5.3 Clinical implications of V11 as a continuous variable

If V11 is correct, the efficacy of checkpoint immunotherapy (PD-1/PD-L1 blockade) should correlate inversely with thymic representational density of tumour antigens — not with tumour differentiation grade per se, but with the AIRE-dependent coverage of the tumour's antigen profile. Available data on immunotherapy response are consistent with this prediction, though direct testing in terms of mTEC antigen representation has not been performed.

The lupus paradox — autoantibodies against nuclear antigens that should be well-represented in the tolerance registry — is explained by post-translational modification. NETosis and apoptosis generate modified epitopes not present at the time of thymic education. These neo-epitopes fall outside the representational map established during development, regardless of the underlying antigen's representational density.

6. Applicability Conditions of the Model

6.1 Three mechanistic classes of autoimmune disease

Autoimmune diseases are not a homogeneous category. The inter-bifurcation model applies to one specific class and must be distinguished from two others.

Class I — Barrier-isolated organ-specific diseases. Multiple sclerosis, uveitis, sympathetic ophthalmia, autoimmune orchitis. The target antigens are sequestered behind anatomical barriers (blood-brain barrier, blood-retinal barrier, blood-testis barrier) and were not present in the systemic circulation during thymic education. The immune system has not been tolerised to these antigens — not because of clonal dynamics, but because of anatomical exclusion. Pathology initiates upon barrier breach. The inter-bifurcation model does not apply. The relevant mechanism is antigenic unveiling, not clone-driven oscillation.

Class II — Diffuse stromal-compartment pathology. Rheumatoid arthritis (in its classic presentation), systemic sclerosis, dermatomyositis, Sjögren's syndrome. The pathological cell population is distributed diffusely within connective tissue stroma, without the localised focus and discrete oscillator dynamics that define the inter-bifurcation model. Therapeutic response to anti-inflammatory agents, low-dose cytostatics, and corticosteroids is consistent with suppression of a distributed inflammatory microenvironment rather than reduction of a focal clone. The inter-bifurcation model may apply partially — particularly to the cytostatic response mechanism — but the full bifurcation geometry is not present.

Class III — Focal clone in an immunologically open tissue with active oscillator. Systemic lupus erythematosus (B-cell clone in immunologically open lymphoid tissue), autoimmune hepatitis, primary biliary cholangitis. This is the class to which the inter-bifurcation model fully applies. The clone is localised, the tissue oscillator is identifiable, the autocatalytic loop operates through circulating antibodies and systemic immune complex deposition.

6.2 Conditions for inter-bifurcation model applicability

The model applies when three conditions are simultaneously met: (a) the pathological clone is localised in an immunologically open tissue — not sequestered behind an anatomical barrier; (b) the clone's antigens were present in sufficient quantities during thymic education to establish partial but incomplete tolerance (intermediate V11); (c) the tissue harbours an identifiable regenerative oscillator with which the clone can synchronise and from which it has exited.

When these conditions are not met, the model does not apply, and should not be extended by analogy.

7. Discussion: SLE as a Modelling Case

SLE is examined here not as the primary target of the model but as the most tractable modelling case: it satisfies all three applicability conditions and its phenomena map consistently onto inter-bifurcation dynamics.

Autoantibodies against nuclear antigens arise from a B-cell clone whose antigens — modified nuclear components generated by NETosis — fall outside the representational map established during thymic education (neo-epitopes; low V11 for modified forms). The clone is in lymphoid tissue — immunologically open. The flare-remission pattern corresponds to oscillatory clone dynamics.

Lupus nephritis represents deposition of immune complexes in high-filtration-pressure glomerular capillaries — collateral damage from circulating complexes, not organ-specific autoimmune attack.

The elevated lymphoma risk in SLE — epidemiologically documented — receives a structural explanation: the B-cell clone generating autoantibodies is itself in inter-bifurcation oscillation. Lymphomatous transformation is the crossing of the second bifurcation by this clone.

The therapeutic map of SLE is consistent with the model: hydroxychloroquine reduces innate immune activation — attenuates loop entry; mycophenolate and cyclophosphamide reduce clone size — desynchronisation; belimumab blocks BAFF — the survival signal of the B-cell clone — disrupts the loop at clone level. None of the therapies cures SLE because none eliminates the conditions for the regenerative window in which the clone arises.

8. The Link Between Autoimmune Disease and Oncogenesis

The clinical association between autoimmune diseases and elevated oncogenic risk is well documented epidemiologically. The model explains it structurally — but only for Class III diseases satisfying the applicability conditions.

A clone in the autoimmune zone has crossed the first bifurcation. The second bifurcation is one step away. The autocatalytic loop enlarges the clone with each flare cycle. Each flare is a regenerative window that brings the clone closer to the second bifurcation threshold.

The probability of malignant transformation is proportional to disease duration, flare frequency and severity, and the specificity of therapy administered. Therapy with higher specificity disrupts the loop more effectively — and thereby reduces the cumulative risk of transition to the second bifurcation.

For Class I and Class II diseases, the oncogenic risk elevation — where it exists — requires separate mechanistic explanation not provided by this model.

9. Variable Map of the Second Stage

Variable	**Stage**	**Description**
V6 — Immune response intensity	II	Threshold variable. Below total-elimination threshold — paradoxically stimulates clone growth. Above threshold — elimination. Between — inter-bifurcation oscillation.
V7 — Immune response specificity	II	Determines power of the autocatalytic loop through magnitude of collateral damage. Higher specificity — weaker loop — lower risk of progression to second bifurcation.
V8 — ECM integrity	II → III	Structural stop-signal of the second bifurcation. Preserved in the inter-bifurcation state. Degradation marks transition to third stage.
V11 — Thymic representational density of clone antigens	II–III	Continuous variable replacing binary 'registry' model. Determined by AIRE-dependent stochastic expression of clone antigen genes in mTECs. High density — immunological near-invisibility. Low density — high immunogenicity. Intermediate — autoimmune zone.

10. Experimentally Testable Predictions

(1) In autoimmune diseases satisfying the applicability conditions (Class III), clone size — measurable through clonal expansion indices in biopsy or V(D)J recombination analysis in lymphoid tissue — should correlate with proximity to malignant transformation independently of disease duration.

(2) The efficacy of cytostatic therapy in Class III autoimmune disease should correlate with the degree of clone size reduction below the first bifurcation threshold, not with the degree of overall immunosuppression achieved.

(3) Targeted biological agents with higher clone specificity should yield lower rates of malignant transformation in autoimmune patients compared with non-specific immunosuppressants — through weaker amplification of the autocatalytic loop.

(4) In SLE, anti-dsDNA titre should correlate with the frequency of circulating less differentiated B cells — not as a causal relationship, but as parallel markers of inter-bifurcation clone dynamics.

(5) The efficacy of checkpoint immunotherapy (PD-1/PD-L1) should correlate with the thymic representational density (V11) of tumour antigens — measurable through AIRE-dependent gene expression overlap analysis — rather than with tumour differentiation grade alone.

(6) In tumours arising against a background of Class III autoimmune disease, the transition from autoimmune focus to malignant tumour should be accompanied by detectable ECM degradation preceding angiogenesis — consistent with the sequence V8 → V9 → V10 described in Article III.

(7) In Class II stromal-compartment diseases, the cytostatic response should show a different dose-response profile compared with Class III diseases: efficacy at immunosuppressive doses in Class II, versus efficacy at sub-immunosuppressive (clone-size-reducing) doses in Class III.

11. Conclusion

The first bifurcation point is not only the moment at which the organism first notices the clone. It is the moment at which a dynamic interaction is established between clone and organism — one capable, under certain conditions, of sustaining itself indefinitely through the autocatalytic loop.

The paradox is that the immune response — a defence mechanism — when insufficiently specific, becomes a mechanism for sustaining the pathological clone. Collateral inflammation from non-specific antibodies reproduces the regenerative window, and the clone receives precisely the conditions it requires for growth.

The inter-bifurcation model applies to a specific and definable class of autoimmune disease: focal clones in immunologically open tissues, with intermediate thymic representational density and identifiable tissue oscillator dynamics. It does not apply to barrier-isolated organ-specific diseases or to diffuse stromal-compartment pathology. This specificity is not a weakness of the model but its precision.

The binary concept of immunological tolerance is replaced by a continuous variable — thymic representational density (V11) — grounded in the known stochastic biology of AIRE-dependent promiscuous gene expression. Clone immunogenicity is a gradient determined by the overlap between the clone's antigen profile and the cumulative population-level representation achieved by mTECs. This formulation generates specific, testable predictions about immunotherapy response.

In the proposed model, Class III autoimmune disease and oncogenesis are two states of the same dynamic system, distinguished by the position of the clone relative to the second bifurcation point. Chronic autoimmune disease of this class is not merely a statistical risk factor for oncogenesis — but its mechanistically predictable precursor. This prediction allows direct experimental verification.

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Conflicts of Interest

The author declares no conflicts of interest.

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Capture of the Tissue Oscillator:

Amplitude Escalation, Vascular Transition, and Malignant Progression

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Series: Article III. "The Clone and the Organism"

Abstract

This article describes the transition of a pathological clone through the second bifurcation point and the mechanisms of malignant progression. Unlike the first bifurcation, at which the clone exits phase synchronisation with the tissue oscillator, the second bifurcation does not signify exit from the oscillator — the clone remains coupled to the tissue cycle but becomes its dominant driver. The amplitude of each successive proliferative cycle increases, while blocking factors progressively lose the capacity to constrain it.

The sequence of the vascular transition is described: degradation of the extracellular matrix (ECM) eliminates the structural stop-signal of clonal expansion; nitric oxide release induces arteriolar dilation; mechanical capillary expansion under elevated pressure creates new perfusion pathways; angiogenesis stabilises the resulting vascular architecture. The clone transitions from diffusion-dependent to perfusion-dependent energy supply — the energy window becomes permanent.

Metastasis is conceptualised as the dissemination of cells into cellularly compatible zones of destruction and regeneration — sites where an energy window is open, a tissue oscillator is active, and the target tissue is cellularly compatible with the BGR profile of the clone. Only when all three conditions are simultaneously fulfilled can synchronisation with the target tissue oscillator occur and a metastatic focus form.

Three outcomes of the third stage and their clinical correspondences are examined. The complete map of twelve model variables concludes the description of clone dynamics across all three stages.

Keywords: *second bifurcation; oscillator capture; amplitude escalation; ECM degradation; NO-mediated vasodilation; angiogenesis; malignant progression; metastasis; cellular compatibility; BGR*

1. Introduction

Articles I and II of this series described the formation of a pathological clone and its interaction with the immune system in the inter-bifurcation state. A clone that has reached the first bifurcation is maintained in chronic oscillation by the autocatalytic loop — but remains within the physical constraints of the tissue. This state corresponds to the clinical picture of autoimmune disease and may persist for years or decades.

The second bifurcation is a qualitative transition of a different kind. A point of principle distinguishes the present model from classical accounts of tumour autonomy: the clone does not exit the tissue oscillator. It captures it. The tissue regenerative cycle continues — but the clone now sets its amplitude, not the tissue. Each successive proliferative surge exceeds the last. Each resolution phase fails to return to the previous baseline. Blocking factors remain present, but their capacity to constrain the escalating amplitude is insufficient.

The present article describes the mechanism of this capture, the sequence of vascular events that establish the clone's energetic autonomy, the probabilistic map of third-stage outcomes, and the mechanism of metastasis as dissemination of cells into cellularly compatible zones of destruction and regeneration.

2. Mechanism of Tissue Oscillator Capture

2.1 From participant to driver

During the synchronisation phase (Article I), the clone oscillates with the tissue pendulum — its proliferative phases align with the tissue's, and contraction during the resolution phase is sufficient to restore synchrony.

After the first bifurcation (Article II), the clone perturbs the oscillator — its proliferative phase begins to exceed the tissue phase, generating a persistent inflammatory focus.

After the second bifurcation, the clone becomes the dominant source of oscillatory dynamics. Tissue regulatory signals continue to act — but the clone responds not by contracting to its baseline level, merely by transiently reducing its growth rate. The minimum of each cycle exceeds the minimum of the previous one. The maximum of each cycle exceeds the maximum of the previous one. Amplitude escalates.

2.2 Insufficiency of blocking factors

The critical distinction of the third stage from the second: blocking factors do not disappear — they become insufficient.

Level-specific suppressive signals from mature cells continue to arrive — but their source is in the minority. The immune response persists — but falls into the zone of paradoxical amplification, additionally stimulating the clone through collateral inflammation. The ECM is partially preserved — but degrades progressively under clonal pressure. Each proliferative surge leaves slightly less ECM, slightly fewer mature cells, slightly more open space for the next.

This is a cumulative process. Not an abrupt transition, but a gradual shift in the balance — until the blocking factors become completely functionally insufficient.

2.3 The role of partial clone death

A paradoxical contribution to amplitude escalation comes from the partial death of the clone during each resolution phase. This death generates a local regenerative stimulus — DAMP signals, substrate release, reduction of contact inhibition — which the surviving cells receive as a signal for yet more powerful proliferation in the next cycle.

Thus the mechanism that in the first stage provided slow incremental clone accumulation becomes, in the third stage, a mechanism of exponential amplitude amplification. Partial destruction of the clone is not its weakening — it is its stimulation.

This has direct therapeutic consequences: subtotal intervention against a tumour — surgical, radiological, or chemotherapeutic — at insufficient magnitude may not only fail to reduce the clone but provoke a more powerful proliferative response in the next cycle.

3. ECM Degradation: Elimination of the Structural Stop-Signal

3.1 The ECM as the BGR expansion stop-signal

Within the BGR Framework, the extracellular matrix is viewed as the structural stop-signal of clonal expansion. Its mechanical properties — viscoelasticity, topological constraints, molecular architecture — physically impede the expansion of a BGR-active clone beyond its niche.

The ECM fulfils this function in two ways. Mechanically — by creating physical resistance to expansion. Signally — by sequestering growth factors and releasing them only under controlled remodelling conditions.

3.2 The degradation mechanism

As the amplitude of clonal oscillations increases, mechanical pressure and hypoxia within the clone mass escalate. This induces expression of matrix metalloproteinases — primarily MMP-2, MMP-9, and MMP-14. The ECM degrades.

ECM degradation releases growth factors sequestered within the matrix — VEGF, FGF, EGF. These factors immediately amplify clone proliferation, creating an additional positive feedback loop: more clone → more hypoxia → more MMPs → more degradation → more growth factors → yet more clone.

3.3 Consequences of degradation

Elimination of the physical stop-signal opens space for clonal expansion. The clone begins to infiltrate adjacent tissue compartments — not through active invasion in the classical sense, but through occupation of space vacated by matrix degradation. Passive expansion along the path of least resistance.

Simultaneously, degradation of the basement membrane opens clone access to the vascular wall — a necessary prerequisite for the subsequent stages of the vascular transition.

4. The Vascular Transition: Sequence of Events

4.1 Nitric oxide release and arteriolar dilation

Hypoxia and inflammatory signals within the growing clone mass stimulate expression of inducible nitric oxide synthase (iNOS) in clone cells and the surrounding inflammatory microenvironment. Released NO diffuses to the smooth muscle cells of adjacent arterioles and induces their relaxation.

Arterioles dilate. Blood inflow to the clone's microregion increases. This is not a regulated physiological response to the metabolic demands of tissue — it is the direct chemical action of NO on vascular tone, operating in the service of the clone.

4.2 Mechanical capillary expansion

The increased arterial inflow through dilated arterioles elevates hydrostatic pressure in the capillary bed. Capillaries expand mechanically — in accordance with the law of Laplace for thin-walled vessels under internal pressure. This is a physical process requiring no biological regulation.

Mechanically expanded capillaries create new perfusion pathways in the clone's zone. The clone transitions from diffusion-dependent to perfusion-dependent energy supply. The critical significance of this transition: the energy window that in the first and second stages was transient and dependent on tissue regenerative cycles becomes permanent. The clone no longer awaits the next regenerative episode — substrate arrives continuously.

4.3 Angiogenesis: stabilisation of vascular architecture

Hypoxia within the clone mass activates HIF-1α, which drives transcription of VEGF and other angiogenic factors. The mechanically expanded capillaries under elevated pressure become the preferential sites of angiogenic sprouting — new vessels grow along pressure gradients already established mechanically.

Angiogenesis stabilises and elaborates the vascular architecture that arose initially through physical mechanisms. The clone acquires autonomous vascular supply. This is the defining feature of the second bifurcation — not the onset of angiogenesis per se, but the stabilisation of autonomous perfusion.

4.4 Therapeutic windows along the event sequence

The proposed sequence — ECM degradation → NO-mediated vasodilation → mechanical capillary expansion → angiogenesis — presents independent therapeutic windows at each stage.

At the ECM degradation stage: MMP inhibitors may slow the opening of expansion space and prevent release of sequestered growth factors. At the NO-vasodilation stage: NOS inhibitors could theoretically reduce blood inflow to the clone’s zone. At the angiogenesis stage: anti-angiogenic therapy acts here — but after mechanical perfusion pathways are partially established, which explains its incomplete efficacy and justifies synergism with MMP inhibitors.

5. Probabilistic Map of Third-Stage Outcomes

After the second bifurcation, the probabilistic map changes fundamentally compared with the first two stages.

Outcome	Probability	Mechanism and clinical correspondence
Spontaneous elimination	Unlikely	Requires reverse transition through the second bifurcation: ECM restoration, vascular regression, return to diffusion dependence. Thermodynamically highly unfavourable. Spontaneous regression exists but is exceptional.
Stationary state	Possible	Balance between amplitude escalation and blocking factors with established autonomous perfusion. Clinically: indolent tumours — papillary thyroid carcinoma, carcinoids, low-grade lymphomas.
Unrestricted growth	Dominant	Complete functional exhaustion of blocking factors. Amplitude escalates faster than any regulatory response. Clinically: aggressive oncogenesis.

The stationary state of the third stage is mechanistically distinct from the stationary states of the first two stages. It is not a balance below the bifurcation threshold, but a balance at already-established autonomous perfusion. Amplitude escalates slowly — blocking factors partially compensate. This is not equilibrium but slow progression at a clinically indistinguishable rate.

6. Metastasis as Dissemination into Cellularly Compatible Zones of Destruction and Regeneration

6.1 Three necessary conditions for implantation

Metastasis in the proposed model is not a random scattering of tumour cells but a structurally determined process requiring the simultaneous fulfilment of three conditions at the target site.

First — an open energy window: a zone of destruction with available substrate, attenuated suppression, and competitive relief. This is required for a BGR-active clone.

Second — an active tissue oscillator: a regenerative cycle must be proceeding in the target zone with which clone cells can synchronise. Without an oscillator, cells are either eliminated or remain dormant.

Third — cellular compatibility: the target tissue must be compatible with the BGR profile of the clone — in terms of signalling interfaces, adhesion molecules, and regenerative cycle frequency. Cellular compatibility is precisely what determines the possibility of synchronisation.

Only when all three conditions are simultaneously fulfilled does the metastatic cell enter synchronisation with the target tissue oscillator and initiate new clonal growth.

6.2 Dormancy of micrometastases

Dormant micrometastases — cells that have reached a distant tissue but have not formed a growing focus — are explained in this model by incomplete fulfilment of the implantation conditions.

Cells may encounter a zone of destruction and regeneration but find no cellular compatibility with that tissue. Or they may encounter a cellularly compatible tissue but without an active regenerative window. In both cases, synchronisation with the oscillator is impossible — cells remain dormant.

Activation of dormant metastases is predicted by the model upon opening of a compatible zone of destruction and regeneration — through trauma, surgical intervention, or inflammatory episode in that organ. This is why surgical intervention on the primary tumour may activate dormant metastases in distant organs: it generates a systemic inflammatory response and simultaneously opens regenerative windows in multiple tissues.

6.3 Organ tropism as cellular compatibility

The classical explanation of metastatic organ tropism — through specific chemokines and adhesion molecules (Paget’s seed-and-soil hypothesis) — is not contradicted by the proposed model but receives an additional dimension.

In the terms of the present model, organ tropism is determined by cellular compatibility of zones of destruction and regeneration: preferred metastatic organs are tissues with maximum compatibility with the BGR profile of the given clone and a high frequency of regenerative cycles. The liver, lungs, bone marrow, and lymph nodes — organs with constantly active regenerative processes and rich oscillatory dynamics — are preferential targets not by chance.

Each metastatic focus undergoes the same dynamics as the primary tumour — but with a fundamentally shorter latent period: the cells have already been selected for BGR-activity and resistance to blocking factors.

6.4 V12 — the twelfth variable of the model

Cellular compatibility of the zone of destruction and regeneration (V12): determines the probability of successful synchronisation of the metastatic clone with the target tissue oscillator. At high compatibility — synchronisation is possible, the metastasis forms and grows. At low compatibility — cells remain dormant or are eliminated.

V12 integrates three components: biochemical compatibility (adhesion molecules, signalling receptors), oscillatory compatibility (proximity of regenerative cycle frequencies), and energetic compatibility (correspondence between the BGR profile of the clone and the metabolic microenvironment of the target tissue).

7. Complete Variable Map of the Model

The present article completes the description of variables spanning all three stages of clone dynamics.

Variable	Stage	Description
V1 — Local energetic surplus	I	Substrate availability for BGR-active clone during regeneration
V2 — Attenuation of level-specific suppression	I	Reduction of suppressive signals proportional to mature cell loss
V3 — Competitive relief	I	Reduction of metabolic competition in the injury zone
V4 — Clone size	I	Threshold variable of the first bifurcation
V5 — Phase synchronisation	I	Oscillatory coherence between clone and tissue cycle
V6 — Immune response intensity	II	Threshold variable of the paradoxical amplification zone
V7 — Immune response specificity	II	Power of the autocatalytic loop through collateral damage
V8 — ECM integrity	II → III	Structural stop-signal of the second bifurcation
V9 — NO-mediated vasodilation	III	Transition from diffusion-dependent to perfusion-dependent energy supply
V10 — Angiogenesis	III	Stabilisation of autonomous vascular supply
V11 — Distance from level 0/−1	II–III	Clone immunogenicity and immunotherapy efficacy
V12 — Cellular compatibility of destruction-regeneration zone	III	Probability of metastatic clone synchronisation with target tissue

8. Three Clinical Zones and Two Bifurcation Points

The twelve variables collectively determine the clone's position in one of three clinical zones.

Zone I — sub-threshold synchronisation (V4 below first bifurcation). The clone accumulates incrementally, invisible to surveillance. Outcomes: elimination (most frequent), stationary state (most prevalent in the population), progression to the first bifurcation (rare, cumulative). Clinically asymptomatic.

Zone II — inter-bifurcation oscillation (V4 between bifurcations, V8 intact). The clone perturbs the oscillator, activates the immune response, and is maintained by the autocatalytic loop. Immunogenicity is determined by V11. Clinically: autoimmune disease. Risk of transition to Zone III is proportional to the duration and severity of oscillations.

Zone III — oscillator capture (V8 degraded, V9–V10 activated). The clone dominates the tissue pendulum; amplitude escalates. The energy window is permanent. Metastasis to cellularly compatible zones of destruction and regeneration (V12). Clinically: malignant tumour.

9. Experimentally Testable Predictions

(1) In tumours arising against a background of autoimmune disease, ECM degradation should precede angiogenesis: escalating MMP activity → arteriolar dilation → VEGF expression → capillary invasion.

(2) Subtotal surgical resection or subtotal radiotherapy should be accompanied by a more powerful proliferative response in the residual clone — measurable through Ki-67, PCNA, or BrdU incorporation — compared with total intervention.

(3) Anti-angiogenic therapy should be more effective when applied early — before stabilisation of vascular architecture. Anti-angiogenic agents combined with MMP inhibitors should act synergistically.

(4) Metastatic foci in tissues with chronic inflammation or elevated regenerative activity should arise more frequently than in tissues with low regenerative activity, given comparable delivery of tumour cells.

(5) Activation of dormant micrometastases should correlate with inflammatory or regenerative episodes in the target organ — consistent with the prediction that opening of a cellularly compatible zone of destruction and regeneration is the condition for synchronisation.

(6) Metastatic organ tropism for a given tumour type should correlate with the cellular compatibility profile (V12) — determined by the overlap of signalling receptors, regenerative cycle frequencies, and BGR metabolic profiles between the primary clone and the target tissue.

(7) Nitric oxide synthase should be elevated in the perifocal zone of the tumour prior to the appearance of angiogenic markers — confirming the proposed sequence: NO-mediated vasodilation precedes VEGF-driven angiogenesis.

10. Therapeutic Implications

The proposed model generates several non-trivial therapeutic consequences, which we formulate as hypotheses requiring clinical verification.

The principle of totality of intervention. Subtotal intervention against the clone, through the regenerative stimulus of partial cell death, may amplify it. The therapeutic threshold is total elimination or reduction below the second bifurcation threshold. Intermediate doses in the zone of paradoxical amplification are counterproductive.

Sequential anti-angiogenic therapy. The optimal therapeutic window for anti-angiogenic agents is before stabilisation of autonomous perfusion. After vascular architecture is established, their efficacy declines. This predicts an advantage for early anti-angiogenic therapy combined with MMP inhibitors.

Targeting the regenerative window. A fundamentally new therapeutic approach implied by the model: not attack on the tumour as such, but closure of the regenerative window — elimination of the conditions of energetic surplus, restoration of level-specific suppression, restoration of ECM. This direction has no current clinical realisation but generates specific targets: stimulation of ECM synthesis, MMP blockade, restoration of BMP and Hedgehog suppressive signals.

Prevention of dormant metastasis activation. The model predicts that inflammatory and surgical interventions on organs harbouring dormant micrometastases may activate them through opening of cellularly compatible zones of destruction and regeneration. This justifies anti-inflammatory cover for planned interventions in oncological patients.

11. Conclusion

The third stage of clone–organism interaction is not a transition to autonomy but a transition to dominance. The clone does not exit the tissue oscillator — it captures it, becoming the source of escalating amplitude that the blocking factors can no longer constrain.

The vascular transition sequence — ECM degradation, NO-mediated vasodilation, mechanical capillary expansion, angiogenesis — is not a single leap but a chain of sequential events, each constituting an independent therapeutic window.

Metastasis in the proposed model is not random dissemination but the distribution of BGR-active cells into cellularly compatible zones of destruction and regeneration. Three conditions — an open energy window, an active tissue oscillator, cellular compatibility — must be simultaneously fulfilled for successful implantation and synchronisation of the metastatic clone. This explains organ tropism, micrometastatic dormancy, and their activation during inflammatory and surgical episodes.

Taken together, the three articles of this series propose a unified dynamic model of oncogenesis with twelve variables and two bifurcation points — from subclinical clone formation through the autoimmune zone to malignant progression and metastasis. In the proposed model, oncogenesis represents not a violation of the laws of tissue organisation but their rare probabilistic consequence. It is for this reason that understanding the normal mechanisms of regeneration remains the key to understanding the origin, progression, and metastasis of tumours.

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Conflicts of Interest

The author declares no conflicts of interest.

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Towards Optimisation of Cancer Prevention and Treatment:
Level-Specific Suppressor Therapy as a Theoretical Direction

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Abstract

This article summarises the therapeutic implications of the three-stage dynamic model of oncogenesis developed in the series "The Clone and the Organism." On the basis of the differentiation information pyramid and the concept of level-specific suppression of intermediate compartments, a new theoretical direction is proposed: level-specific suppressor therapy (LSST).

The central thesis is that a pathological clone survives not because it carries unique growth-promoting mutations, but because it escapes the informational pressure of the differentiation pyramid. Restoring that pressure through the introduction of level-specific suppressors — a functional category encompassing morphogens, growth factors, differentiation signals, and niche elements — may in principle return the clone to the pyramid without cytotoxic intervention.

Natural verification of the principle is found in neuroblastoma Stage 4S: spontaneous regression through differentiation, mediated in part by NGF/TrkA and SLIT3/PLCβ, constitutes a biological precedent for level-specific suppression. Existing differentiation therapy is reviewed critically: its limitations arise from the absence of a level theory — which the present model provides. The applicability boundaries are explicitly defined: the model describes tumours retaining elements of tissue hierarchy and differentiation potential.

Three components of the research programme are described: an atlas of level-specific suppressors, RDS tumour typing through transcriptomic profiling, and an in vitro verification system. A map of therapeutic windows for each of the three model stages is proposed.

Keywords: differentiation information pyramid; level-specific suppression; reverse differentiation scale (RDS); suppressor therapy; neuroblastoma 4S; NGF/TrkA; differentiation therapy; regenerative window; BGR; bifurcation model

1. Introduction: From Cytotoxic to Informational Paradigm

Twentieth-century cancer treatment rests on a single logic: the clone is the enemy and must be destroyed. Surgery, radiotherapy, and chemotherapy all implement the cytotoxic strategy. Checkpoint immunotherapy does the same through the instruments of the immune system.

This article does not propose a ready treatment method. It proposes a new system for classifying therapeutic targets. Where cytotoxic oncology classifies interventions by their mechanism of cell destruction, the level-specific model classifies interventions by the position of the clone within the tissue hierarchy. The article therefore offers not a new drug but a new map for finding future drugs.

The question "how to kill the tumour?" is replaced by the question "how to return the clone to the architecture of the tissue?" This is a shift of research paradigm, not a specific therapeutic protocol.

2. The Differentiation Information Pyramid as a Therapeutic Target

2.1 The mechanism of level-specific suppression

Article I of this series described the differentiation information pyramid as an active self-sustaining structure. Each level of maturity actively suppresses the intermediate compartment immediately above it through level-specific signals. This term denotes a functional category of molecular signals — morphogens, growth factors, differentiation cytokines, contact signals, niche elements — unified not by structural similarity but by function: maintaining cells of a given level in a differentiated state. This is not a new class of molecules; it is a new principle of their classification in relation to the tissue hierarchy.

The stability of the pyramid is overdetermined — maintained simultaneously by three independent mechanisms: top-down informational pressure, energetic competition from metabolically efficient neighbours, and structural constraints of the extracellular matrix. A pathological clone expands only when all three are attenuated.

The therapeutic possibility follows directly: if the clone has escaped the informational pressure of the pyramid, that pressure can be restored from outside.

2.2 Why existing differentiation therapy works only partially

The sole unequivocal clinical success of differentiation therapy is acute promyelocytic leukaemia (APL), where ATRA drives the promyelocytic clone to mature into functional neutrophils. ATRA works in APL precisely because the promyelocytic clone occupies a strictly defined level in the haematopoietic hierarchy, and retinoic acid is the physiological suppressor of exactly that level. This is not a fortunate accident — it is the correct level reached by the correct signal.

In solid tumours the same agent is applied to clones of unknown and variable position on the differentiation scale. Retinoic acid is a suppressor of certain levels but not all. Applying it to a clone of a different level will have no effect — not because the method is wrong, but because the level does not match. The absence of a level theory is the systemic reason for the failure of differentiation therapy in solid tumours.

3. A Natural Precedent: Neuroblastoma Stage 4S

3.1 The phenomenon of spontaneous regression

Neuroblastoma Stage 4S shows spontaneous regression in approximately 69% of cases — through differentiation into benign ganglioneuroma or through programmed cell death. This is a clinically observed precedent for the return of an oncological clone to the differentiation pyramid under the action of microenvironmental signals.

3.2 Molecular mechanisms of regression

Regression of neuroblastoma 4S is a multifactorial process. The interpretation proposed here regards NGF/TrkA as one of the key components of level-specific regulation, alongside other discussed mechanisms — immunological, epigenetic, MYCN-related, and microenvironmental.

In terms of the information pyramid: NGF/TrkA acts as a level-specific signal for the neuroblastoma clone — its presence initiates neuronal differentiation, its absence leads to apoptosis. A second molecular pathway — SLIT3/PLCβ/PKC — was identified recently and confirms that the mechanism is not unique but reproducible through different signalling axes.

A key observation: in infants, NGF is present in the microenvironment as a physiological signal of neuronal differentiation during normal development. In children older than one year this signal wanes — and the same tumour progresses aggressively. This demonstrates the direct dependence of clone fate on the availability of the level-specific signal in the microenvironment.

3.3 What neuroblastoma 4S demonstrates in principle

Neuroblastoma 4S demonstrates that the return of an oncological clone to the differentiation pyramid is biologically possible, and that the mechanism is level-specific: NGF functions for the neuronal lineage at a specific level of primitivity, but not for clones of different tissue origin or different differentiation level. This is not a universal solution — it is proof of principle in a specific system.

4. The Concept of Level-Specific Suppressor Therapy

4.1 The principle

Level-specific suppressor therapy (LSST) is based on the following logic: determine the precise position of the clone on the RDS → identify the level-specific signals that in normal physiology maintain cells of that level in a differentiated state → restore those signals from outside. The clone either differentiates or undergoes apoptosis if it is unable to receive the signal.

The fundamental distinction from the cytotoxic strategy: the action is directed not at destroying the clone but at changing its informational state.

4.2 Scope of applicability

LSST is proposed as applicable primarily to tumours retaining elements of tissue hierarchy and differentiation potential. This condition is necessary: without a preserved hierarchy, RDS positioning is impossible; without differentiation potential, response to the suppressor signal is impossible.

4.3 Three required components

Component 1: precise determination of clone position on the RDS through transcriptomic profiling (single-cell RNA-seq). Histological grades G1–G3 are too coarse. An algorithm is required to translate the transcriptomic profile into a position on the continuous scale.

Component 2: an atlas of level-specific suppressors for key tissues and levels. The haematopoietic map is most complete; for the majority of tissues it is fragmentary. Building the atlas is a long-term but principally achievable programme.

Component 3: local delivery of the suppressor to the clone zone. Level-specific signals act paracrinally. Required: nanoparticles with targeted binding, viral vectors for local expression, or direct injection. Delivery technologies exist in other contexts.

4.4 Stage-dependent applicability

The greatest potential for LSST is at Stage II. The clone is still relatively homogeneous by RDS profile, physically constrained by tissue, and has not acquired vascular autonomy. At Stage III the clone is heterogeneous — LSST acts as an adjuvant in combination with blockade of the vascular sequence. At Stage I — potentially most effective but practically most difficult because of clone invisibility.

5. Map of Therapeutic Windows

Stage	Clone state	Therapeutic logic	Instrument
I — latent	Synchronised with oscillator; invisible	Reduce frequency of regenerative windows; restore pyramidal suppressor pressure	Anti-inflammatory strategy; differentiating agents
II — inter-bifurcation	Autoimmune zone; autocatalytic loop	Reduce clone below first bifurcation threshold; disrupt loop through specificity	Sub-immunosuppressive cytostatics; targeted biological therapy; level-specific suppressors
III — post-second bifurcation	Oscillator capture; autonomous perfusion	Exceed total-elimination threshold or block regenerative window; LSST as adjuvant	MMP inhibitors + anti-VEGF; anti-inflammatory therapy post-intervention; differentiating agents by RDS profile

6. Partial Map of Level-Specific Suppressors

RDS Level	Clone characteristics	Endogenous suppressor (known)	Status
0 / −1	Terminally differentiated cells and their immediate precursors	BMP, Hedgehog, tissue-level differentiation cytokines	Well characterised
−2 / −3	Committed progenitors	Notch ligands, retinoic acid, lineage-specific TFs	Partially characterised
−4 / −5	Multipotent progenitors	High-concentration BMP, Wnt antagonists	Fragmentary
−6 and below	Primitive stem states	NGF/TrkA (neuroblastoma), SLIT3 (neuroblastoma 4S)	Isolated examples; systematic map absent

The table reflects the current state of knowledge. Building the complete atlas is the central experimental task of the research programme.

7. Limitations of the Model

Explicit delineation of applicability boundaries is necessary for correct interpretation of the concept.

Limitation	Description	Consequence for applicability
Complete loss of tissue architecture	Tumours with disorganised stroma lacking an identifiable hierarchy	LSST inapplicable: no reference hierarchy for RDS positioning
Irreversible mutations in receptor pathways	Clone has lost capacity to receive differentiation signal (TrkA, FGFR mutations; receptor loss)	LSST ineffective even with correct suppressor selection
Deep RDS heterogeneity of the clone	Stage III with multiple subclones at different levels	Single suppressor insufficient; a suppressor profile is required
Tumours without differentiation potential	Anaplastic carcinomas; some sarcomas with complete loss of lineage markers	Model partially applicable or inapplicable

LSST is therefore not a universal oncological strategy but an instrument for a specific class of tumours — those in which tissue hierarchy is partially preserved and the clone retains capacity to receive a differentiation signal.

8. Optimisation of Existing Treatment Methods

8.1 Subtotal intervention: clarification of logic

The model predicts that under certain conditions a subtotal cytotoxic intervention may create a regenerative window favouring accelerated clone recovery: death of part of the clone generates DAMP signals, releases substrate, and creates space. This is a testable hypothesis, not a universal claim — counterarguments simultaneously exist: immunogenic cell death, immune response activation, reduction of tumour mass. The model’s prediction: the regenerative window effect should dominate under conditions of low therapy specificity and high inflammatory response to the intervention.

Practical consequence: palliative therapy where complete elimination is not possible requires simultaneous blockade of the inflammatory component as standard accompaniment.

8.2 Cytostatic dosing in Class III autoimmune diseases

In Class III autoimmune diseases (Article II) the goal of cytostatic therapy is reduction of the clone below the first bifurcation threshold, not immunosuppression per se. The criterion of efficacy is markers of clonal expansion, not standard immunological markers. This redefines the dosing logic for methotrexate in rheumatoid arthritis and mycophenolate in lupus.

8.3 Anti-inflammatory strategy as oncological prophylaxis

If oncogenic risk is determined by the number of tissue regenerative cycles, chronic inflammation is a direct mechanistic driver of clone accumulation. Long-term anti-inflammatory therapy in patients with cirrhosis, chronic hepatitis, and IBD acquires oncological prophylactic justification with a clear mechanism.

9. Research Programme

9.1 Atlas of level-specific suppressors

Systematic mapping of level-specific signals in key tissues. Priority: haematopoietic system, intestinal epithelium (Lgr5 hierarchy), hepatocytes, neuronal lineage. Source: systematic analysis of existing differentiation data plus targeted experiments.

9.2 RDS typing algorithm

Development of a computational algorithm to translate transcriptomic profile (scRNA-seq) into a position on the RDS. Validation on tumours with known behaviour: neuroblastoma 4S, APL, pilocytic astrocytoma.

9.3 In vitro verification

Model system: organoid with documented differentiation hierarchy, clone with known RDS profile, intervention with the corresponding suppressor. Neuroblastoma 4S is the most accessible starting system: the mechanism is identified, cell lines exist, validation data are available.

10. Experimentally Testable Predictions

(1) The RDS position of a tumour, determined through transcriptomic profiling, should predict response to a specific differentiating agent better than histological grade G1–G3.

(2) Neuroblastoma Stage 4S tumours with high TrkA expression should show an inverse correlation between NGF concentration in the microenvironment and clone growth rate.

(3) Pilocytic astrocytomas that spontaneously regress after partial resection should show transcriptomic evidence of differentiation (RDS profile shift toward more mature states) compared with recurring tumours of the same histological type.

(4) Palliative chemotherapy without simultaneous blockade of the inflammatory response should be associated with a shorter recurrence-free interval compared with the combination of cytostatic plus anti-inflammatory therapy.

(5) The incidence of oncological disease in patients with chronic inflammatory conditions should correlate with the calculated number of regenerative cycles (oscillatory tissue load) more strongly than with calendar age.

(6) In Class III autoimmune diseases, cytostatic therapy at sub-immunosuppressive doses should yield a comparable clinical effect with significantly lower toxicity, when clonal expansion markers are used as the efficacy criterion.

11. Discussion

LSST occupies a specific niche: Stage I and II tumours by the present model that retain tissue hierarchy and differentiation potential. In this niche it has a fundamental advantage over cytotoxic strategies: it generates no regenerative stimulus, acts on the informational state of the clone, and may potentially return the clone to the normal hierarchy rather than merely reducing it.

Neuroblastoma 4S proves the principle — not a universal method. The question is whether this result can be reproduced controllably for an arbitrary clone. The answer requires the programme described above. Three barriers remain unresolved: RDS typing, the suppressor atlas, and delivery. Their combined resolution is not a near-term prospect.

Nevertheless, neuroblastoma 4S is already a ready clinical model for the first step. If the principle is verified here — it is verified in principle. The rest is an engineering task.

12. Conclusion

This article offers not a new drug but a new map for finding future drugs. Where cytotoxic oncology classifies interventions by their mechanism of cell destruction, the level-specific model classifies them by the position of the clone within the tissue hierarchy. This is a shift of research framework, not a specific protocol.

The differentiation information pyramid proposes an alternative target: not clone proliferation but the informational state of the clone. Neuroblastoma Stage 4S demonstrates that return of the clone to the pyramid is biologically possible. NGF/TrkA and SLIT3/PLCβ are components of this mechanism, identified in a specific system.

The direction is defined. The principle is verified by nature. The limitations are explicitly stated. The research programme is formulated.

Acknowledgments

The author thanks the open scientific community whose published data made this synthesis possible. AI-assisted literature search and manuscript preparation were performed using Claude (Anthropic) as part of a collaborative human–AI research tandem.

Conflicts of Interest

The author declares no conflicts of interest.

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Dedicated to my daughter Sarah-Lana Demi. Data scientist.

Energy-Temporal Reserve and Clinical Trajectory.
Oncology.

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Series: Article V. "The Clone and the Organism"

Abstract

This article introduces an actuarial approach to the assessment of clinical trajectory in oncological disease among older patients. Standard oncology evaluates treatment efficacy through overall and recurrence-free survival. The actuarial approach poses a different question: what is the real gain in quality-adjusted life years (QALY) relative to the patient's background trajectory — accounting for competing all-cause mortality, therapy toxicity, and the organism's energy reserve.

Energy reserve R(t) and age are independent parameters. Two patients of the same age may have fundamentally different R(t) depending on physical activity, comorbidities, and individual biological characteristics. It is R(t), not age as such, that determines the amplitude of the regenerative oscillator and thereby the rate of oncological progression — in those cases where the model applies.

A conceptual decision framework is proposed: a matrix of clone stage × R(t) × background mortality with a QALY balance calculation algorithm. Potential sources of model failure are explicitly defined: highly aggressive tumours, clones of genetically determined origin, tumours independent of the tissue microenvironment. The place of R(t) within the context of existing geriatric oncological instruments is described: R(t) is an additional variable, not a replacement for existing scales.

Keywords: actuarial approach; energy reserve; QALY; competing risks; T* point; regenerative oscillator; geriatric oncology; clinical trajectory; VO_2max; BGR; conceptual decision framework; potential sources of model failure

1. Statement of the Problem

More than 60% of oncological diseases are diagnosed in patients over 65 years of age. Standard treatment protocols were developed primarily in younger populations and are evaluated by criteria of overall and recurrence-free survival. These criteria do not account for two factors that fundamentally alter the clinical picture in older patients.

First factor: competing mortality. A patient of 80 years has approximately a 68% probability of dying from all causes within 10 years regardless of the oncological diagnosis. Defeating the tumour in this context does not guarantee a gain in life of comparable duration.

Second factor: therapy toxicity as an independent risk. Cytotoxic chemotherapy in patients with exhausted energy reserve may itself shorten life or critically reduce its quality — that is, produce a negative QALY balance.

The actuarial approach reformulates the task: not "defeat the tumour", but "maximise QALY accounting for all competing risks and the reserves of the individual patient".

2. Energy Reserve as a Clinical Parameter

2.1 R(t) and age — independent variables

The fundamental distinction of the proposed approach from standard age-oriented scales: R(t) and age are independent parameters. A patient of 72 with regular physical activity and no significant comorbidities may have R(t) > 0.5. A patient of the same age with chronic heart failure and a sedentary lifestyle — R(t) < 0.3. Therapeutic decisions for these two patients are fundamentally different, though their age is identical.

2.2 Four model indicators

Indicator	Notation	Baseline	Data source
Residual mitochondrial activity	M(t)	M(25) = 1.0	VO_2max, HUNT3
Daily oxygen capacity / kg	O(t)	O(25) = 1.0	VO_2max, HUNT3
Basal energy expenditure / kg	B(t)	B(25) = 1.0	Mifflin–St Jeor
Energy reserve	R(t) = O(t) − B(t)	R(25) = 1.0	Calculated

Energy reserve R(t) = O(t) − B(t) reflects the difference between what the organism is capable of producing and what it expends on basal maintenance. This is the functional reserve for regeneration, immune response, and resistance to external loads — including therapy toxicity.

2.3 The T point and its oncological significance*

T* is the bifurcation point of the ageing model at which R(t) falls below 30% of the age-25 level. The population mean is approximately 76 years, with individual variation of ±8–10 years. Two patients of 76 may lie on opposite sides of T* depending on their individual R(t).

Zone	R(t)	Oscillator state	Oncological implication (model prediction)
Full reserve zone	R(t) > 0.6	Pendulum amplitude full	Model predicts: clone capable of escalating at full rate. Aggressive therapy may yield positive QALY balance.
Declining reserve zone	0.3 < R(t) ≤ 0.6	Amplitude decreasing	Model predicts: clone escalates more slowly. Therapy toxicity begins to compete with benefit. Individual QALY assessment mandatory.
T* point (bifurcation)	R(t) ≈ 0.3	~76 years (mean)	Model predicts: below this value background mortality begins to dominate over oncological risk. Therapeutic logic changes.
Reserve	R(t) ≤ 0.3	Amplitude minimal	Model predicts: oscillator weak; many

exhaustion zone			clones indolent. Cytotoxic therapy highly likely to yield negative QALY. Quality of life is the priority.

All statements in the ‘Oncological implication’ column are model predictions requiring clinical verification. The applicability of these predictions is bounded — potential sources of model failure are described in section 6.2.

3. Competing Risks: The Actuarial Frame

3.1 Background mortality as the dominant parameter

Age	10-year background mortality (all causes)	Typical oncological progression horizon (Stage II)	Conclusion
65 years	~25%	5–15 years	Oncological risk dominates. Treatment warranted.
70 years	~38%	5–15 years	Competing risks comparable. Individual assessment.
75 years	~52%	5–15 years	Background mortality beginning to dominate. QALY priority.
80 years	~68%	5–15 years	Background mortality dominates. Aggressive therapy requires QALY-based justification.
85 years	~82%	5–15 years	Competing risk critically high. Palliative strategy generally preferable.

Data are based on population-level actuarial data. Individual values vary with R(t), comorbidity, and functional status. Formulations in the ‘Conclusion’ column reflect model predictions, not clinical recommendations.

3.2 Oncological progression horizon

The three-stage model of the trilogy predicts that the rate of clone progression through stages depends on regenerative oscillator amplitude, and therefore on R(t). In patients past T*, the model predicts progression slowing proportional to oscillator weakening. This prediction is consistent with clinical observations but requires verification. It does not apply in the failure scenarios described in section 6.2.

4. QALY Decision Matrix

The matrix reflects model predictions regarding the expected QALY balance. All positions are conceptual predictions requiring verification. The matrix does not apply to scenarios described in section 6.2.

Reserve R(t) / Stage	Stage I (latent)	Stage II (inter-bifurcation)	Stage III (post-second bifurcation)
R > 0.6	Observation + anti-inflammatory strategy. QALY prediction: positive.	Sub-immunosuppressive cytostatics + targeted therapy. QALY prediction: positive.	Aggressive therapy warranted where complete elimination is possible. QALY prediction at complete elimination: positive.
0.3 < R ≤ 0.6	Anti-inflammatory strategy. Oscillatory load monitoring. QALY prediction: positive.	Individual QALY calculation mandatory. Suppressor therapy preferred. QALY prediction: indeterminate.	T*-distance assessment mandatory. Subtotal therapy only with regenerative window blockade. QALY prediction: indeterminate.
R ≤ 0.3 (post-T*)	Observation. Model predicts indolent course. QALY prediction for cytotoxic therapy: negative.	Anti-inflammatory strategy + quality of life. Cytostatics only with high clone aggressiveness and individual QALY calculation. QALY prediction: cautious.	Palliative strategy. Model predicts negative QALY for aggressive therapy in most cases. Requires verification.

5. Individual QALY Balance Calculation Algorithm

Step	Parameter	Assessment method	Threshold value
1	Age and R(t) — independent parameters	VO_2max + BMR by Mifflin–St Jeor	T*: R(t) ≈ 0.3 (~76 years mean; individual variation ±8–10 years)
2	Clone stage	Biopsy + transcriptomic profile (RDS)	I / II / III per series model
3	Clone aggressiveness	Proliferation index Ki-67, clonal expansion	High / low
4	Background mortality	Actuarial tables for patient's country	10-year all-cause mortality risk
5	QALY balance of therapy (conceptual	Expected years gained × quality adjustment − toxicity	> 0: therapy under consideration. ≤ 0: palliative strategy preferred

	calculation)		

5.1 QALY balance formula (conceptual scheme)

The proposed formula is a conceptual calculation scheme requiring coefficient calibration on prospective clinical data.

QALY_balance = [ΔT_onco × Q_treatment] − [T_toxicity × (1 − Q_toxicity)] − [ΔT_background × (1 − Q_background)]

Where: ΔT_onco — expected years-of-life gain from therapy; Q_treatment — quality-of-life adjustment coefficient during treatment (0–1); T_toxicity — period of significant toxicity in years; Q_toxicity — quality of life during the toxicity period (0–1); ΔT_background — years of life determined by background mortality independently of oncology.

When QALY_balance > 0: therapy is considered warranted by model prediction. When QALY_balance ≤ 0: palliative strategy is preferred.

5.2 The role of R(t) in the calculation

R(t) enters the calculation through two independent pathways. Directly: low R(t) increases therapy toxicity — raises T_toxicity and reduces Q_toxicity. Indirectly: low R(t) slows oncological progression — reduces ΔT_onco that therapy can add. Both effects operate in the same direction: as R(t) declines, the QALY balance of therapy decreases. This effect does not operate in model failure scenarios (section 6.2).

6. The Place of R(t) within Existing Geriatric Oncology

6.1 R(t) as a complement to existing instruments

The present approach does not seek to replace existing instruments for assessing older patients with oncological diseases. R(t) is proposed as an additional variable adding a mechanistic dimension.

Tool	What it measures	Limitation	What R(t) adds
ECOG	Functional status (0–4)	Does not account for energy reserve; two patients with ECOG 1 may have R(t) of 0.5 and 0.2	R(t) distinguishes patients with identical ECOG by energy potential
Charlson Comorbidity Index	Number and severity of comorbidities	Does not account for rate of reserve decline or proximity to T*	R(t) reflects the integrated functional outcome of comorbid burden
CARG Score	Chemotoxicity risk in older patients	Empirical scale without mechanistic basis	R(t) predicts toxicity through mechanism: low reserve = less recovery capacity
G8 Screening Tool	Screening for geriatric impairments	Not linked to oncological clone	Combination of G8 + R(t) + clone stage yields three-

		dynamics	dimensional decision map
Comprehensive Geriatric Assessment (CGA)	Multi-domain assessment: physical, cognitive, social	Resource-intensive; yields no quantitative QALY forecast	R(t) adds a single quantitative variable to the multi-domain assessment

The proposed combination: standard geriatric assessment (CGA or G8) + R(t) as additional variable + clone stage per the trilogy model = three-dimensional clinical decision map. R(t) adds a variable absent from all the listed scales: individual energy reserve as a function of biological rather than calendar time.

6.2 Potential sources of model failure

Explicit identification of scenarios in which model predictions are inapplicable or unreliable is a necessary condition for its correct use.

Model failure scenario	Description	Failure mechanism	Consequence for applicability
Extremely high tumour aggressiveness	Small-cell lung cancer, glioblastoma, anaplastic carcinomas with Ki-67 > 80%	Rate of progression through stages is incommensurable with the rate of R(t) decline. The clone reaches the second bifurcation before R(t) becomes a limiting factor.	Model predicts dominance of oncological dynamics regardless of R(t). QALY matrix inapplicable in standard form — correction for growth rate required.
Clone growth rate greatly exceeds R(t) decline	Rapidly progressing tumours in relatively younger patients (50–65 years) with high R(t)	High R(t) supports an active oscillator but simultaneously creates a wider regenerative window for the clone. The protective effect of declining R(t) does not have time to materialise.	Model overestimates the stabilising effect of reserve. A patient with high R(t) and an aggressive tumour requires standard oncological management without adjustment for 'protective' reserve.
Tumours independent of tissue microenvironment	Tumours past the second bifurcation with multiple metastases and complete vascular autonomy	The tissue-source regenerative oscillator no longer governs clone dynamics. The clone has entered autonomous	Model predictions about progression slowing at low R(t) do not apply.

		amplitude escalation.	Stage III with metastases lies outside the primary applicability domain of the model.
Genetically determined tumours with high R(t)	BRCA1/2, Lynch syndrome, retinoblastoma in patients aged 40–60 with high functional reserve	The clone arises not through accumulation in the regenerative window but through a specific genetic defect. The Article I formation mechanism does not apply. High R(t) provides no protection.	Model inapplicable for progression rate prediction. QALY calculation retains meaning, but T* logic does not function as a progression-limiting factor.

General principle: the model is most applicable to tumours whose dynamics are governed by the tissue oscillator and regenerative window — that is, Stage I and II tumours of moderate aggressiveness without genetically determined origin. Outside these boundaries, model predictions require correction or are inapplicable.

7. Suppressor and Anti-Inflammatory Strategy in Patients Past T*

For patients with $R(t) \leq 0.3$ the model predicts that cytotoxic therapy will in most cases yield a negative QALY balance. This does not imply abandonment of active management. Two strategies retain a predicted positive balance in this group — in the absence of the failure scenarios described in section 6.2.

Anti-inflammatory strategy. The model predicts that reducing the frequency and depth of regenerative windows slows clone progression without cytostatic toxicity. This prediction requires verification in clinical studies.

Level-specific suppressor therapy (Article IV). With a known RDS profile of the clone, introduction of the corresponding suppressor generates no cytostatic toxicity and does not open a regenerative window. For patients past T*, this is potentially the only active therapeutic strategy with a predicted positive QALY balance for Stage II clones.

8. Connection to the Ageing Model of Framework 1.1

This article closes the internal logic of the series. The ageing articles described R(t) as a universal parameter of the organism's biological time. The trilogy "The Clone and the Organism" described oncogenesis as a dynamic process whose rate the model connects to regenerative oscillator amplitude.

The intersection of two trajectories — declining R(t) and escalating clonal amplitude — determines the individual clinical outcome in cases where the model applies. The model predicts: if R(t) declines faster than the clone advances through stages, the patient dies from other causes with a clinically insignificant tumour. If the clone escalates faster than R(t) declines — oncological progression dominates. This prediction allows direct epidemiological verification.

The finishing-line argument applies directly: lower SML in women accounts for higher residual R(t) and slower oncological progression in old age — in those cases where progression is governed by the oscillator rather than genetic factors.

9. Experimentally Testable Predictions

(1) The rate of oncological progression (time from Stage II to Stage III) should correlate with the patient's VO_2max at diagnosis more strongly than with age — after controlling for histological aggressiveness and excluding model failure scenarios.

(2) In patients with $R(t) \leq 0.3$, the frequency of indolent oncological disease course should be significantly higher compared with patients of the same age with $R(t) > 0.3$ — at identical histological aggressiveness and excluding genetically determined tumours.

(3) The QALY balance of aggressive chemotherapy in patients with $R(t) \leq 0.3$ and Stage II disease should be negative in most cases after calibration of formula coefficients on a validation cohort.

(4) Anti-inflammatory monotherapy in patients past T* with Stage I–II disease should yield comparable overall survival with significantly higher quality of life compared with standard cytotoxic therapy.

(5) Women over 80 should show slower progression from Stage II to Stage III compared with men of the same age — regardless of histological type — excluding genetically determined tumours.

(6) The calculated QALY balance by the algorithm of section 5 should prospectively predict real treatment outcome better than ECOG, Charlson, and CARG, when verified on a cohort of patients 70+ with calibrated coefficients and excluded failure scenarios.

10. Conclusion

Oncological disease in an older patient unfolds within the space of two competing dynamics: escalating clonal amplitude and declining organismal energy reserve. R(t) and age are independent variables: it is individual energy reserve, not calendar age, that determines the patient's position in this balance — in those cases where the model applies.

The model's applicability domain is explicitly bounded: tumours of moderate aggressiveness without genetically determined origin that have not yet achieved full autonomy from the tissue microenvironment. Outside this domain, model predictions require correction or are inapplicable.

The proposed conceptual decision framework does not replace clinical judgement and is not a ready protocol. It structures the logic of choice and adds to existing geriatric instruments a variable that none of them contain: individual energy reserve as a parameter of biological time.

The series "The Clone and the Organism" is complete. Five articles describe a single continuum: molecule → cell → tissue → organism → population. From the formation of the pathological clone through the regenerative window — through the autoimmune zone and oscillator capture — through the possibilities of suppressor therapy — to the actuarial frame of the individual clinical decision. The overall model spans the molecular, cellular, tissue, organismal, and population levels within a single conceptual architecture.

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Conflicts of Interest

The author declares no conflicts of interest.

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Supplementary Notes

Extending the Clone and the Organism Series

The Inverted Suppression Pyramid: From Physiological Norm to Leukemia via Dormancy and Accumulation

Working notes for the “Clone and the Organism” series

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1. The Inverted Pyramid: Defining the Inversion

The classical intuition pictures the differentiation pyramid as a figure with a narrow apex (few stem cells) and a broad base (numerous mature cells) — a pyramid defined by cell count. But the direction of control within this structure is inverted relative to the familiar hierarchy metaphor. In organizational systems, a small apex commands a large base. In biological tissue the relationship runs the other way: during embryogenesis, each successive level of differentiation vastly outnumbers the one before it through geometric expansion via cell division, and it is this larger, more mature population that actively holds the smaller, more primitive level beneath it in check — not the reverse.

The process is tightly regulated rather than depletive: the primitive pool does not disappear but stabilizes at a minimum, sustained by constant pressure from the numerically dominant level above it. This yields two independent axes within a single structure: the numerical axis remains classical (narrow apex, broad base), while the control axis is inverted — the vector of restraint runs from the more numerous population toward the less numerous one.

The stability of this structure rests on three independent mechanisms: top-down informational pressure (level-specific signals — morphogens, growth factors, contact-dependent cues), energetic competition from metabolically more efficient neighbors, and structural constraints imposed by the extracellular matrix. A pathological clone expands only when all three are weakened simultaneously.

2. The Nature of Level-Specific Suppression: Energetic Balance, Not Command

A key methodological point: a level-specific suppressive signal should not be understood as an instruction that a cell obeys or disregards. It is more accurately described as a factor that shifts a cell's local energetic balance. A morphogen diffuses from its source and is simultaneously absorbed by target cells through receptor binding, converting a continuous concentration gradient into a discrete outcome at the single-cell level. The sharpness of this transition is often achieved through mutual-inhibition circuits between target genes, which convert a graded input into a near-binary result.

But the outcome itself is not the execution of an order — it is the probabilistically dominant result of an energetic reallocation. The incoming signal demands a specific metabolic adjustment from the cell (new protein synthesis, altered expression, a shift in functional regime), and whether the cell can sustain that adjustment energetically determines which way the balance tips: toward differentiation, or toward an alternative outcome. The precision of this interpretation depends on several conditions — correct dose and exposure duration, the integrity of a physical barrier restricting signal access to the intended targets, and the integrity of a negative-feedback circuit. Failure of any one of these conditions creates a borderline state in which the cell's energetic balance oscillates near threshold, and the outcome becomes unstable.

3. Two Modes of Reversible Failure

Two clinical phenomena illustrate distinct failure points of this system while remaining fully reversible.

Leukemoid reaction is a failure of input amplitude with the regulatory circuit fully intact. The cytokine drive is abnormally elevated in response to a legitimate systemic stimulus (infection, stress), accelerating mature-cell output, yet the maturation hierarchy is not disrupted — cells pass through every stage in the correct order, simply faster. The reaction is fully reversible because its own negative-feedback circuit dampens it as the inciting stimulus subsides. This is a quantitative, not a structural, failure: the pyramid functions correctly but is temporarily overloaded at the input.

Neuroblastoma Stage 4S represents the opposite type of failure: the clone is already outside physiological suppressive pressure, yet retains receptor and metabolic sensitivity to the signal. In infants, the physiological level of NGF in the microenvironment is sufficient to tip the clone's energetic balance toward one of two stable outcomes — differentiation into a mature ganglioneuroma, or collapse. After the first year of life, this signal wanes, and the same clone grows aggressively. This is a failure of signal delivery rather than excess, which is precisely why restoring the missing signal from outside can shift the clone's energetic balance back — the theoretical basis for level-specific suppressor therapy (LSST).

Comparing the two cases yields an axis along which reversible failures can be classified: excess signal with an intact circuit (leukemoid reaction, passively reversible) versus a deficit of accessible signal with a clone that remains receptive (neuroblastoma 4S, actively reversible through signal restoration).

4. The Limits of the Selective-Combinatorial Principle

Explaining clonal hematopoiesis of indeterminate potential (CHIP) through random mutation plus positive niche selection appears sufficient at first glance, but fails a quantitative check. If mutation is purely random and environmental selection purely selective, the observed repertoire of driver genes should be either unboundedly diverse or vanishingly narrow. The observed pattern — a limited but non-singular set of recurring genes (DNMT3A, TET2, ASXL1, JAK2) — is explained by neither extreme.

A further difficulty is the combinatorial cost of search: before the right combination of mutation and favorable niche state arises, an enormous number of unsuccessful combinations would need to be sampled, and even under optimal niche conditions there is no guarantee that the necessary mutation will occur within the available physiological time window. This points to the need for a third element beyond simple random generation plus selection.

5. Filter Overload: From Damage Detection to Collapse Probability

An alternative hypothesis shifts the emphasis from positive selection of a mutation to the probability of a cell's energetic collapse. Experimental data show that the advantage of DNMT3A- or TET2-mutant cells following inflammatory stress lies not in accelerated proliferation but in heightened resistance to stress-induced death; the mechanism is partly linked to a blunted p53 response. It is more accurate, however, to understand this circuit not as a switch issuing a command to die, but as a threshold detector of energetic state: an active, metabolically loaded cell continuously generates damage while simultaneously expending energy on repair; with sufficient energetic reserve, damage is more often resolved, whereas with insufficient reserve, the probability that the cell will cross into irreversible structural or energetic collapse rises. Blunting this circuit does not reduce the probability of “receiving an order” — it reduces the system's sensitivity to an approaching collapse, meaning a cell with an attenuated p53 response continues accumulating damage further than a cell with an intact response before collapse actually occurs.

This reframes the question toward the state of the elimination machinery as a whole, independent of any individual cell's mutational status. Chronic age-related inflammation reduces the efficiency of damaged-cell clearance for two compounding reasons: the internal detector (the p53 pathway) becomes less sensitive, and the throughput of external clearance — the phagocytes that remove already-dead or critically damaged cells — physically declines. In central-nervous-system microglia, phagocytosis is shown to be an energy-intensive process directly dependent on the clearing cell's mitochondrial mass; under chronic inflammation the clearing cell is forced to switch to a less efficient energy-production pathway precisely when its workload is highest, leading to accumulation of undigested material and a further decline in its own working capacity. A parallel mechanism has been described in bone marrow: efferocytosis-deficient, pro-inflammatory macrophages in the aging niche drive age-related changes in hematopoietic stem cells, while themselves becoming an additional source of pro-inflammatory signaling, compounding the very problem they were meant to solve.

This creates a positive feedback loop: an uncleared cell proceeds to secondary necrosis with release of pro-inflammatory factors, which lowers the energetic efficiency of clearance and leads to accumulation of yet more uncleared cells. The narrow, recurring repertoire of CHIP driver genes may thus reflect not combinatorial rarity of a fortunate pairing, but a limited number of molecular routes capable of reducing the sensitivity of one and the same shared quality-control circuit — and the observed clonal expansion may in part be a visibility artifact: the mutant cell is tracked by sequencing, while non-mutant cells that survived for the same underlying reason of reduced collapse probability go unrecorded.

The elimination mechanism is bidirectional: a symmetric failure in the opposite direction — hyperactive or mistargeted elimination, as in autoimmune destruction of healthy blood cells or excessive synaptic pruning in the aging brain — shows that pathology can arise from either side of the filter's equilibrium point, not only from its insufficiency.

6. Biophysical Protection: The Dense Cytoplasm of the Dormant State

An additional, structural mechanism for reducing collapse probability is provided by cellular quiescence. Dormant stem cells display reduced glycolysis, low mitochondrial membrane potential, and correspondingly low levels of reactive oxygen species. Quiescence is an actively regulated, not a passive, state: its function is to lower damage generation by reducing metabolic activity, achieved through enhanced autophagy and a reduction in mitochondrial number — that is, through a physical restructuring of the cytoplasm's internal composition.

In the interpretation proposed here, this functions not as evasion of a damage detector but as a reduction in the baseline frequency of the very events that detector would otherwise register: suppressed metabolism generates less oxidative stress and fewer synthesis errors, so the dormant cell's energetic balance less often drops below the threshold at which collapse probability becomes significant. The protective effect is clinically confirmed: cytostatics that specifically target proliferating cells and trigger apoptosis in them do not act on dormant cancer stem cells, which can reactivate once treatment stops and give rise to relapsed tumors.

The regulatory node converges here with the p53 circuit already discussed: the same signaling pathway simultaneously governs a cell's entry into the protected quiescent state and its sensitivity to an approaching collapse.

This provides a structural explanation for why dormant cells largely fail to respond to level-specific suppression: executing a suppressive signal requires an energy-intensive metabolic adjustment, and a cell with suppressed energy turnover is structurally unable to sustain that adjustment, even if the receptor is present on the membrane and formally capable of binding ligand. Suppression is not nullified by a specific mutation and is not selectively “ignored” — it becomes metabolically unexecutable for a cell holding both possible outcomes, differentiation and collapse, at uniformly low probability.

7. The Self-Reinforcing Accumulation Loop

Reduced collapse probability in the dormant state produces not a static but a cumulative effect. A mathematical model of myelodysplastic syndromes formalizes this dynamic: the pathological clone does not divide faster but self-renews at a higher rate than normal stem cells; as the number of primitive clone cells in the niche grows, their increasing mass suppresses self-renewal in surrounding cells via a feedback circuit, leading to cytopenia, which in turn further amplifies the proliferative signal in the niche — closing a positive-feedback loop that acts on the clone's already-accumulated mass.

A defining feature of this loop is its hidden, slow phase: simulated scenarios show the clone can expand over roughly fifteen to seventeen years without clinically detectable changes in blood counts, after which normal hematopoiesis collapses sharply within a few years. This offers a quantitative illustration of crossing the model's first bifurcation point: it is not the mutation that confers proliferative advantage, but reduced collapse probability that allows the clone to accumulate mass unnoticed for long enough that this mass itself becomes the dominant factor in the niche's energetic and spatial balance.

The asymmetry between entering and exiting dormancy further explains why suppression appears to weaken over time even though it formally continues to operate on the fraction it can reach: dormant cells are metabolically unable to execute a suppressive response and remain beyond its reach; awakened cells respond properly, but because the dormant reserve has grown over time, the absolute number of cells awakening at each subsequent activation cycle also grows. A therapy acting only on the active fraction is therefore condemned to perpetually trim the top of a growing reservoir while never touching its underlying stock.

8. From Accumulation to Niche Capture: The Transition to Leukemia

An accumulated dormant clone displaces healthy clones not through active signaling competition but through cheap physical occupation of the niche: the low metabolic cost of the resting state allows the clone to occupy space and resources that would normally be cyclically redistributed among actively dividing healthy cells, while the clone itself carries no countervailing collapse risk comparable to that of its active neighbors. It does not win the competition through strength — it barely participates in a game carried at the same risk level as everyone else, and so it simply outlasts them in the niche.

This is where the model converges with true leukemia. If CHIP and dormant accumulation represent a partial, reversible shift of probabilistic balance within a formally intact external suppressive architecture, leukemia represents the breakdown of the circuit itself: the clone does not merely evade suppression but captures the niche's regulatory resource, suppressing healthy hematopoiesis through a shared pool and inverting the direction of control. Quantitative models of clonal hematopoiesis formalize this as joint regulation of division rates for healthy and pathological stem cells through a shared niche resource, in which the growth of the pathological clone suppresses the healthy lineage — hence the formal therapeutic conclusion of these models: successful therapy must eradicate the pathological clone rather than merely reduce its numbers, since a residual population continues to compete for the niche resource.

The path from norm to leukemia via dormant accumulation can thus be described as a single sequence: physiological suppression as an energetically mediated mechanism for holding cells in a differentiated state → partial protection of a clone fraction through transition into a dense, low-metabolic state that reduces collapse probability itself and thereby places this fraction beyond the effective reach of suppression → slow, unnoticed accumulation of protected mass through cheap niche occupation → crossing a threshold at which the accumulated mass itself becomes the dominant factor in the niche's energetic balance (the self-reinforcing loop) → capture of the shared regulatory resource and inversion of the direction of control (true leukemia).

9. Open Questions

The sequence presented here remains a first-approximation hypothesis and leaves several nodes for further formalization: the quantitative relationship between the size of a clone's dormant reserve and the time to crossing a T*-like threshold; the existence of a quality-control-specific energetic bifurcation, distinct from the energetic bifurcation of the target tissue as a whole; the risk that therapeutically enhancing clearance to counter under-elimination might swing the system toward the opposite failure mode — excessive, mistargeted elimination; and the practical distinguishability of two causes of clonal noise: genuine positive selection of an advantageous mutation, versus passive accumulation of cells with reduced collapse probability regardless of their mutational status.

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Horizontal Suppression: Quorum Sensing as a Second Control Circuit on the Clone

Working notes for the "Clone and the Organism" series

Yuri Dmitrochenko, MD — Independent Researcher, Israel

1. From Vertical to Horizontal Suppression

In every case examined so far — leukemoid reaction, neuroblastoma 4S, CHIP, leukemia — suppression was described as a vertical process: a more mature, numerically dominant level of differentiation restrains the primitive level beneath it through a hierarchical, top-down signal. This circuit presupposes the existence of the hierarchy itself — distinct maturity levels between which a signal circulates.

The yeast model points to a fundamentally different circuit that requires no hierarchy at all: suppression can operate horizontally, between cells of the same level, through diffusible metabolites that encode local population density. This mechanism does not need a "superior" level to exist — it functions even within a fully homogeneous cell population.

2. The Molecular Mechanics of Quorum Sensing in Yeast

Budding yeast possess an innate program that preserves population survivorship during stationary phase: as resources become exhausted, a small fraction of cells synchronously re-enters mitosis, raising overall colony viability without an accumulation of mutations. This phenomenon is directly density-dependent and can be triggered by aromatic alcohols acting as quorum-sensing molecules. The quantitative relationship is sharp: across a series of three-fold dilutions of the same culture, colony counts differed roughly forty-fold between the densest and most dilute inocula.

The reverse process — entry into dormancy under excessive density or unfavorable conditions — is documented with comparable precision. In Cryptococcus neoformans, cells transition into a viable-but-non-culturable state, and reactivation of a fraction of this population can be achieved by stimulation with a separate quorum-sensing molecule. This is full symmetry with the principle under discussion: population growth up to a given boundary drives a fraction of cells into an inactive state, and a subsequent drop in signaling-molecule concentration reactivates the dormant fraction.

The mechanism by which density information is transmitted is particularly informative: the loss of collective metabolic oscillations in a yeast population at low cell density occurs synchronously across all cells at once, rather than through gradual desynchronization — meaning that population-density information is encoded directly in each individual cell's internal dynamical state through exchange of a diffusible metabolite, rather than being inferred indirectly by the cell.

3. Transferring the Principle to Cancer Stem Cells

The same logic has been formalized for tumor stem cells. A discrete cellular-automaton model of tissue development reproduces a sharply dichotomous growth dynamic: either bounded maturation over a limited number of cell cycles, or unbounded immortalization — with this choice governed primarily by the intensity of intercellular communication, where low communication intensity by itself drives perpetual proliferation. The analysis rigorously demonstrates that long-term tissue homeostasis is guaranteed by negative feedback of cell density on stem-cell proliferation: stem-cell division continues only as long as their number within the niche remains small.

This model is also experimentally supported in breast carcinoma cells: the plating density of cells bearing stem-cell markers (CD44+, CD44+/24lo/ESA+) does not affect the resulting stem-cell fraction near confluence — a result consistent precisely with quorum-sensing control maintaining a fixed proportion of stem cells within the population.

4. Why No Receptor Mutation Is Required

The central feature distinguishing this mechanism from every case examined earlier is that the cell needs no genetic lesion to escape control. The authors of the model state the conclusion directly: stem-cell immortalization may be triggered by reduced intercellular communication rather than exclusively by somatic evolution — that is, not necessarily by mutation. This implies that stem-cell proliferation could in principle be dampened by manipulating the signal itself, or inadvertently enhanced by selectively destroying differentiated cells that normally contribute to the overall quorum signal.

In other words, the cell's receptor apparatus and its capacity to receive the signal remain fully intact — what fails is not the receiver but the magnitude of the input signal itself, which is determined not by the individual cell but by the collective density of its neighbors. A previously restrained cell receives physiological permission to proliferate simply because the local concentration of the restraining metabolite has fallen below threshold in its immediate vicinity — without any change whatsoever in its own genome.

5. Connection to the Regenerative Window and the Dormant-Accumulation Model

This offers an economical alternative to a mutational explanation for accelerated relapse following partial surgical intervention. Removing part of a tumor mass lowers the local quorum-signal density in the remaining tissue, and the surviving stem or dormant cells — without selection of any new aggressive subclone — obtain release from horizontal restraint simply because fewer neighbors remain to sustain the signal. This is structurally identical to the “regenerative window” effect discussed earlier in connection with subtotal cytotoxic intervention, but offers it a simpler mechanism: not immunogenic stimulation or substrate release, but a direct lifting of the density ceiling.

This also connects with the dormant-reserve model developed previously: if clonal growth up to a given density itself induces a fraction of cells to enter an inactive state, the dormant pool within a tumor may be generated and maintained not only by protection from external elimination (low probability of energetic collapse) but also by the simple fact of reaching a local density threshold — meaning the clone may carry a built-in, self-regulating mechanism for generating its own dormant reserve, requiring no external stress trigger.

6. Open Question

It remains unestablished whether specific candidate molecules have been identified as endogenous, non-bacterial quorum-sensing mediators directly within tumor stem cells — that is, a product of the tumor tissue itself rather than a signal imported from outside, such as the bacterially derived quorum peptides reported in the literature, which influence colorectal-cancer invasion and metastasis through a distinct, receptor-mediated pathway. This distinction is consequential: bacterial quorum peptides act as an external, foreign signal activating specific receptors and signaling cascades, whereas the horizontal suppression described for yeast and cancer stem cells implies an endogenous, species-specific diffusible signal generated by the population itself to regulate its own density. Identifying such an endogenous mediator in tumor tissue is the logical next step for testing the transferability of the model.

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Three Cellular States and the Energetic Limit of Clonal Escape: A Model Illustrated by the Endometrium

Working notes for the "Clone and the Organism" series — closing chapter of Framework 1.1

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1. Three Cellular States

Every cell within a tissue hierarchy can be assigned to one of three functional states, distinguished by metabolic profile and role in tissue dynamics rather than by differentiation status alone.

The active state is one in which a cell is preparing either to divide or to die. This state is characterized by a fluid, low-viscosity hydrogel cytoplasm with high free-water content and, critically, low oxygen consumption through mitochondria: a cell preparing to divide predominantly relies on glycolysis even when oxygen is abundant (the Warburg effect), because the glycolytic pathway supplies building blocks — nucleotides, amino acids, lipids — needed to double biomass faster than full mitochondrial oxidation can. Mitochondria remain intact but are deliberately underused.

The stable-dormant state is one in which a cell enters quiescence: minimal glycolysis, low mitochondrial membrane potential, minimal transcriptional activity, a dense, low-metabolic cytoplasm. This is an actively regulated, not passive, state, achieved through enhanced autophagy and a reduction in mitochondrial number; its function is to reduce damage accumulation by lowering overall metabolic load.

The stable-functioning state is one in which a cell no longer divides but performs a mature tissue function. For most such cells (cardiomyocytes, neurons), high oxygen consumption through active mitochondria is characteristic — a shift toward oxidative phosphorylation serves as a marker of functional maturity. This is not a universal rule, however: a subset of differentiated, functioning cells (muscle, immune, retinal) retains a preference for glycolysis, and the limiting case is the anucleate, amitochondrial erythrocyte, performing pure gas-exchange function entirely through glycolysis and fitting none of the three categories cleanly, since it has lost — along with its nucleus — the very potential to transition between states.

A separate intermediate category is the transit-amplifying cell (TAC): a population positioned between the stem cell and the mature functional cell, capable of moving into any of the three states but never reaching full functional maturity until it completes the remaining portion of its differentiation path.

2. The Ratio of Cellular States in Embryonic and Post-Embryonic Development

The proportion of the three states shifts sharply across developmental phases, and the shift runs counter to naive expectation.

In the early embryonic period, prior to the completion of organogenesis, tissue consists almost entirely of the active fraction. Embryonic stem cells are described as continuously replicating, with constitutively active cyclins and hyperphosphorylated Rb protein — the molecular brake that would shift a cell into quiescence is structurally weakened. This is confirmed quantitatively by a stark contrast with the adult state: only a small fraction of a percent of fetal hematopoietic stem cells are quiescent, compared with over 90% of the adult stem pool — a difference of several orders of magnitude. The stable-functioning category is nearly absent at this stage, since the organ has not yet been completed.

The appearance of a dormant and a stable-functioning fraction occurs closer to the end of the embryonic period and continues into early postnatal life. The cardiomyocyte offers a precisely dated example of this boundary: vigorous proliferation continues throughout fetal life, and final cell-cycle exit occurs at the junction of the late embryonic and early postnatal periods, morphologically marked by binucleation (uncoupling of karyokinesis from cytokinesis during a final, incomplete division). In humans, mitotic and hyperplastic capacity is fully lost within the first three to six months after birth, and this cell-cycle exit is accompanied by metabolic reprogramming from glycolysis toward oxidative phosphorylation — meaning the active-to-stable-functioning transition coincides precisely with a shift in metabolic pathway.

In mature, differentiated tissue, the picture is inverted relative to the embryo: the dormant fraction constitutes a substantial, often dominant share of the stem-cell reserve, particularly in low-turnover tissues (skeletal muscle, brain), whereas the active fraction is correspondingly small and used sparingly — only for ongoing tissue maintenance. The dormant reserve is therefore not inherited from embryonic abundance but is built anew over the course of tissue maturation, as part of the program transitioning the tissue from an embryonic growth regime to an adult maintenance regime.

3. The Energetic Limit as a Constraining Factor on the Scenario

The transition of a clone out from under dual suppressive control (vertical and horizontal) into a monoclonal, horizontally dominant regime is a structurally energy-intensive sequence. It requires simultaneous servicing of several parallel processes: continuous proliferation of the active fraction, continuous synthesis of the diffusible metabolites underlying the horizontal quorum signal regardless of a cell's proliferative status, and — most costly of all — displacement and replacement of the differentiated, functionally mature component of the tissue, whose construction is energetically more expensive than maintaining a primitive, minimally specialized layer.

This renders the scenario conditional rather than deterministic: the probability of its realization is inversely related to how constrained the local energetic reserve of the tissue is at the moment vertical control weakens. Under local energetic surplus — consistent with the fact that endometrial hyperplasia and cancer more often develop against a background of obesity, hyperestrogenism, and insulin resistance, that is, metabolic excess rather than deficit — the scenario is structurally more likely precisely because no energetic ceiling exists to restrain it. Under a constrained reserve, the same structural failure of suppressive circuits is more likely either to stall at the intermediate, less costly stage of heterogeneous hyperplasia without reaching a monoclonal state, or to proceed considerably more slowly.

4. Illustration: Transit Amplification of the Endometrium under Dual Pressure

The three-part model and the energetic constraint described above find a direct histological illustration in the development of endometrial hyperplasia.

The transit-amplifying layer of the endometrium (TAC) is subject to two independent suppressive circuits simultaneously: a vertical circuit, originating from more differentiated cells of the gland and stroma, and a horizontal circuit, acting among the TAC-level cells themselves through a density-dependent signal. The molecular gatekeeper of this node is the transcription factor SOX9, whose expression is higher during the proliferative phase and colocalizes with the marker of the intermediate, undifferentiated population; SOX9 has been proposed to act as a checkpoint preventing hyperplasia.

Under estrogenic hyperstimulation, growth of the active fraction is not uniform but stochastically branching — the resulting histological picture is described as a “swiss cheese” pattern: cystically dilated, relatively inactive glands sit alongside small, actively proliferating ones. This is a distributed, stochastic outcome of the same horizontal “divide-or-arrest” decision being made independently across different regions of tissue under local density pressure.

Under dual pressure, the TAC layer accumulates slowly and gradually displaces the mature, differentiated component. This displacement is registered quantitatively as a shift in the gland-to-stroma ratio: normally, stroma constitutes roughly three-quarters of tissue volume, whereas in endometrial intraepithelial neoplasia (EIN), the stromal fraction falls below 55%. A critical structural moment arrives when the differentiated component is sharply reduced locally: the vertical circuit loses its source, leaving only the horizontal circuit, now acting on a homogeneous, monoclonal population with its own density threshold. This transition is registered pathologically as the emergence of a monoclonal glandular population with architectural and cytological features distinct from neighboring, unaffected glands — the formal diagnostic criterion for EIN, statistically associated with a heightened risk of further progression to carcinoma.

5. Conclusions

The three-part classification of cellular states (active, stable-dormant, stable-functioning) applies to nucleated cells retaining the potential to transition between states; anucleate terminal forms fall outside this scheme as the irreversible product of a precursor's completed differentiation. The proportion of the three states shifts sharply across development: the early embryo is almost entirely active, the dormant reserve is built anew toward the completion of organogenesis and in early postnatal life, and mature tissue is dominated by the dormant rather than the active fraction of its stem compartment.

The transition of a clone from physiological dual control to pathological monoclonal expansion is an energy-intensive sequence whose probability of realization depends not only on the state of the suppressive circuits themselves but on the local energetic budget of the tissue at the moment those circuits weaken. Endometrial hyperplasia offers a documented, stepwise example of this sequence: a heterogeneous, stochastically branching mosaic of TAC cells under dual pressure → gradual quantitative displacement of the differentiated component → local loss of vertical control → transition to a monoclonal, horizontally dominant population with elevated risk of malignant progression.

6. Predictions

1. The fraction of Ki67-negative (dormant) cells in cystically dilated glands of hyperplastic endometrium should be higher than in actively proliferating glands of the same specimen, under comparable hormonal background.

2. The gland-to-stroma area ratio should correlate with risk of progression to EIN more tightly than nuclear atypia alone, since the former reflects the degree of displacement of the vertical controlling component.

3. SOX9 expression should be selectively reduced or dysregulated specifically within monoclonal EIN foci compared with the surrounding heterogeneous hyperplasia of the same biopsy.

4. In patients with metabolic syndrome and obesity (local energetic surplus), the probability of progression from simple to atypical hyperplasia and onward to carcinoma should be higher at a comparable degree of hormonal stimulation than in patients without metabolic excess.

5. Identification of the specific horizontal chalone operating within the endometrial TAC layer should reveal a molecule structurally or functionally analogous to the quorum-sensing factors already described in breast carcinoma stem cells.

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Part III

The Energetics of Aging

Body Size, Structural Maintenance Load, and Longevity

A Theoretical Integration of Body Mass, Mitochondrial Capacity, and Energetic Reserve within the Framework Model

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Abstract

Body size has long presented one of the central paradoxes of comparative biology. Across species, larger mammals generally live longer than smaller ones, whereas within many species — including domestic dogs and humans — larger individuals often exhibit reduced longevity. Existing explanations emphasize growth hormone signaling, cancer risk, metabolic rate, or evolutionary life-history strategies but do not provide a common energetic framework capable of explaining both observations simultaneously.

This article proposes that body mass should not be interpreted as an independent determinant of lifespan but rather as a component of an energetic balance between functional mitochondrial capacity and structural maintenance load. Functional mitochondrial capacity scales with body mass sublinearly (approximately $M^{0.75}$ across species), whereas the energetic cost of maintaining living tissue scales approximately proportionally with mass. Consequently, larger organisms carry a proportionally higher maintenance burden relative to their energetic production — explaining interspecies longevity when corrected for evolutionary metabolic rate adaptation. Within species, aging progressively reduces functional mitochondrial capacity while structural maintenance requirements decline more slowly, creating a progressive imbalance that constitutes functional aging.

The model integrates evidence from human longevity studies, sex differences in lifespan, domestic dog breeds, horses, and pigs. It introduces the Structural Maintenance Load (SML) hypothesis and the concept of age-dependent optimal body mass — a declining target mass that maximizes the Functional Mitochondrial Capacity / SML ratio across the lifespan. A practical energetic corridor is defined for clinical use.

Keywords: body size; longevity; structural maintenance load; functional mitochondrial capacity; allometric scaling; optimal body mass; energetic corridor; BGR framework; aging

1. Introduction

One of biology's unresolved paradoxes concerns the relationship between body size and lifespan. Between species, larger mammals generally live longer: elephants outlive mice, whales outlive dogs. Within species, the relationship frequently reverses: small dog breeds outlive giant breeds by nearly a decade [1]; women — who possess lower average body mass than men — survive substantially longer [2]; many human centenarians maintain relatively modest body composition throughout life [3].

These observations appear contradictory within a single framework. They become compatible when analyzed from the perspective of the energetic balance between functional mitochondrial production capacity and the energetic cost of preserving organismal structure. The present article formalizes this balance through the concept of Structural Maintenance Load (SML) and derives testable predictions from it, including the novel concept of age-dependent optimal body mass.

2. The Structural Maintenance Load

Every organism continuously expends energy on intracellular remodeling, protein turnover, extracellular matrix renewal, vascular maintenance, immune surveillance, and stem-cell replacement. This obligatory maintenance expenditure — the Structural Maintenance Load (SML) — is expected to increase approximately with the amount of living tissue [4,5].

We define SML provisionally as a composite index of the energetic cost of structural preservation, incorporating lean body mass, connective tissue volume, and vascular network extent. Quantitatively, SML is expected to scale closer to $M^{1.0}$ than to $M^{0.75}$ across body mass ranges, though the precise exponent requires empirical determination. In the present model, SML scaling as approximately proportional to mass is treated as a working hypothesis, not an established fact.

Empirically, SML could be approximated from stable isotope measurements of whole-body protein turnover [4], collagen remodeling markers (PINP, CTX), and vascular endothelial turnover rates — each weighted by tissue mass fractions.

3. Allometric Scaling and the Inter-species Paradox

Functional mitochondrial capacity — operationally approximated by maximal oxygen uptake (VO_2max) — scales with body mass sublinearly across species. The empirical relationship established by Kleiber [6] and confirmed across six orders of magnitude is approximately $M^{0.75}$. If SML scales approximately as $M^{1.0}$, the ratio Functional Mitochondrial Capacity / SML scales as $M^{-0.25}$, predicting lower energetic margins in larger animals.

This appears to contradict interspecies longevity data — but the contradiction resolves when evolutionary adaptation is considered. Large species have evolved lower mass-specific metabolic rates over millions of years, extending the absolute time before the Mito/SML ratio falls below critical thresholds. The allometric relationship therefore explains interspecies longevity as a consequence of evolutionary metabolic slowing, not inherent biological robustness of large organisms.

Within a species — where mass-specific metabolic rate is relatively fixed by common evolutionary history — size-related differences in the Mito/SML ratio operate directly. This is where SML has its greatest explanatory power.

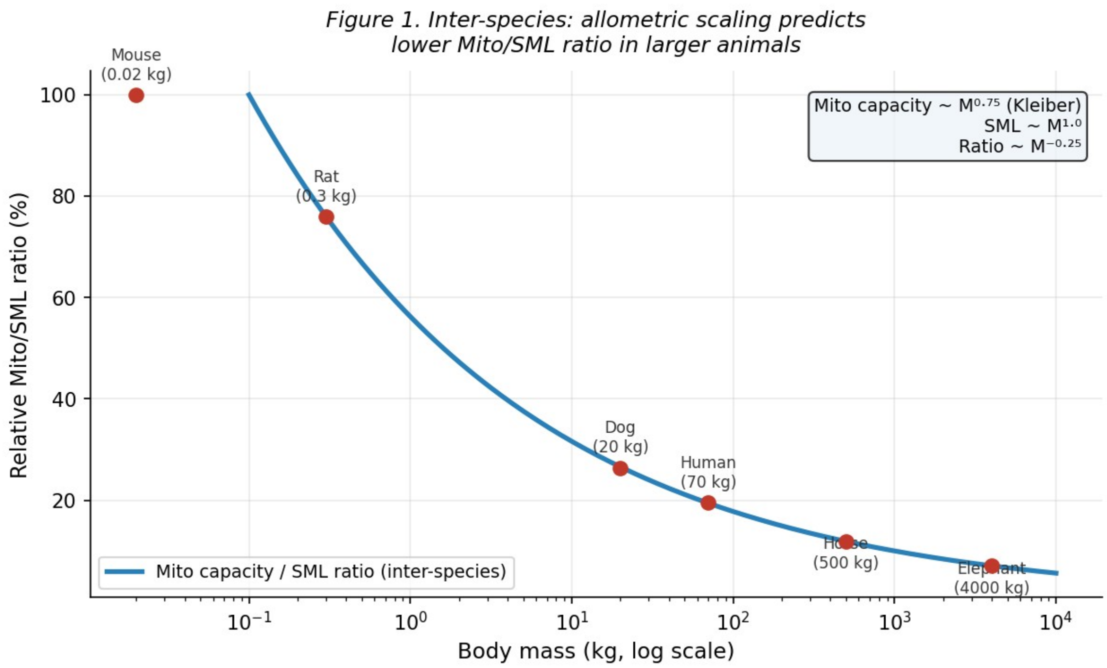

Figure 1. Inter-species allometric scaling: functional mitochondrial capacity ($M^{0.75}$, Kleiber) grows slower than structural maintenance load (approximately $M^{1.0}$), so the Mito/SML ratio declines with body mass. Interspecies longevity is maintained by evolutionary reduction of mass-specific metabolic rate.

4. Aging as Progressive Deterioration of the Mito/SML Ratio

Within the Framework model [7], aging primarily reflects progressive loss of functional mitochondrial capacity — driven by mitochondrial DNA damage accumulation, reduced mitophagy efficiency, and declining respiratory chain function. The energy reserve model [8] demonstrates that this decline is substantially steeper than the decline in basal metabolic maintenance.

Structural load — the denominator — decreases more slowly because ECM and vascular architecture are maintained by long-lived structural proteins with slow turnover. Sarcopenia and cachexia reduce lean mass but simultaneously reduce mitochondrial mass, so the ratio does not improve with uncontrolled muscle loss. The net result is a progressive deterioration of the Mito/SML ratio, illustrated in Figure 3.

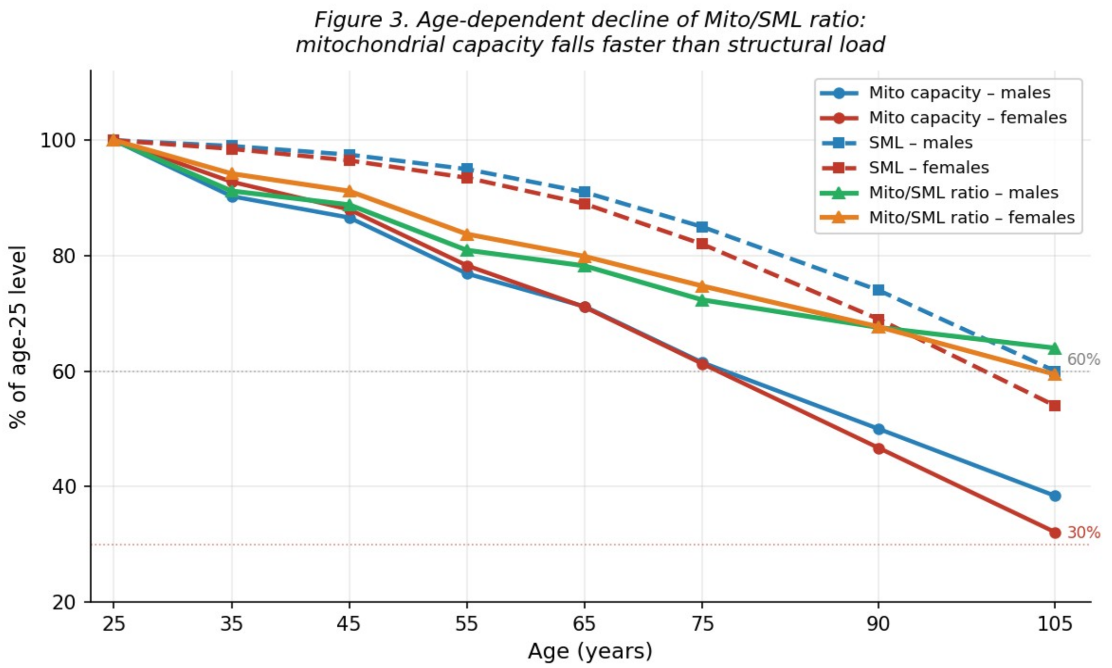

Figure 3. Age-dependent decline of Mito/SML ratio: functional mitochondrial capacity (solid lines) declines faster than modeled structural maintenance load (dashed lines), producing a progressively deteriorating ratio (triangle markers). Data from HUNT3 Fitness Study and Mifflin–St Jeor BMR, normalized to age 25.

5. Why Low Body Weight Is Also Detrimental

Clinical geriatrics consistently demonstrates that sarcopenia, cachexia, and severe underweight predict increased mortality in older adults independently of disease status [9,10]. This is fully compatible with the SML model: below physiological limits, mitochondrial mass itself declines proportionally with lean tissue. The organism loses both structural load and energetic production capacity simultaneously — and the ratio does not improve because residual tissue becomes relatively more metabolically demanding per unit of repair capacity.

Optimal longevity is therefore not associated with minimal body weight but with maintaining an appropriate Mito/SML balance within age-adjusted physiological limits — the energetic corridor defined in Section 11.

6. Evidence from Domestic Dog Breeds

Domestic dogs provide a uniquely controlled natural experiment: a single species spanning a 30-fold range of adult body mass within shared environments. Mortality studies consistently demonstrate that small breeds outlive large breeds by 5–8 years on average [1,11]. Toy breeds frequently reach 15–18 years while giant breeds typically live 7–9 years [1]. Teng et al. [11] demonstrated a log-linear relationship with body mass, consistent with $M^{-0.25}$ scaling of the Mito/SML ratio within a species. Giant breeds age faster after maturity rather than beginning aging earlier [1] — consistent with progressive energetic reserve depletion.

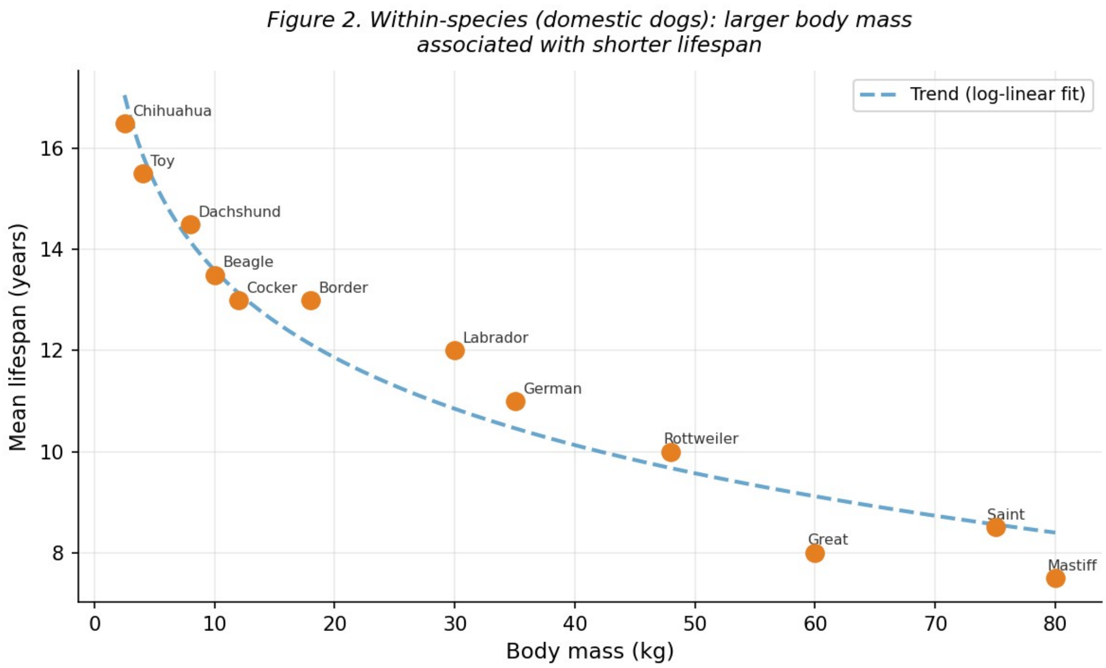

Figure 2. Within-species (domestic dogs): mean lifespan declines log-linearly with body mass, consistent with within-species Mito/SML scaling. Data compiled from Kraus et al. [1] and Teng et al. [11].

7. Horses and Pigs

Ponies generally outlive heavy draft breeds; miniature pigs frequently exceed commercial breeds in lifespan [12]. The effect is quantitatively weaker than in dogs, with a straightforward explanation: the within-species body mass range is 3–4-fold for horses and pigs versus ~30-fold for dogs. The $M^{-0.25}$ prediction yields a much smaller longevity differential over a narrower mass range — consistent with observed data.

8. Human Longevity and Centenarians

Studies of centenarians consistently identify normal body weight, absence of severe obesity, preserved mobility, and delayed sarcopenia as common features [3,13]. After approximately age 85–90, moderate nutritional reserve becomes beneficial — the 'obesity paradox of aging' [14] — illustrating the transition from energetic efficiency toward acute stress resistance when the Mito/SML ratio has fallen near critical levels. This age-dependent shift is predicted by the model and formalized in the energetic corridor (Section 11).

9. The Female Longevity Paradox and the Finishing Line Argument

Women outlive men by approximately 5–7 years in most populations [2], and this advantage persists into extreme old age. At age 90, Israeli CBS data [8] show survival from age 25 of ~44% for women versus ~32% for men — a substantial gap maintained decades after reproductive hormones have ceased to differ meaningfully between sexes.

This temporal dissociation is critical. Estrogen-mediated cardioprotection and mitochondrial protection [15] are biologically real and explain a portion of the male excess mortality in middle age. However, after age 80–85, circulating estradiol in women falls to 5–15 pmol/L — practically identical to male levels of 10–20 pmol/L at the same age [18] — rendering the hormonal advantage largely absent. Yet the survival gap persists: Israeli CBS life tables show

44% female versus 32% male survival at age 90 [17], a 12 percentage point difference documented three to four decades after menopause. This quantitative dissociation — statistically indistinguishable sex hormone levels alongside a substantial survival difference — constitutes direct evidence against hormonal status as the primary explanation of female longevity at the finishing line.

The SML model provides this permanent factor. Women possess on average 20–25% lower lean body mass than men of similar height [16], a proportionally smaller cardiovascular system, and lower absolute basal metabolic rate. These differences translate directly into a lower absolute SML — a structural advantage that is age-independent, hormone-independent, and operates across the entire lifespan. Lower SML may be the primary driver of female longevity advantage at the finishing line; endocrine differences are a secondary amplifier in middle age.

We therefore propose: lower SML — not lower body mass per se, and not hormones alone — is the fundamental structural explanation for female longevity. The female body is optimized for a lower Mito/SML operating point, which becomes advantageous precisely when mitochondrial capacity declines with age.

10. Age-Dependent Optimal Body Mass

The most clinically novel prediction of the SML model is that optimal body mass is not fixed but declines with age. This decline is not because leanness is universally beneficial, but because functional mitochondrial capacity declines with age, and an organism's structural load should ideally track its energetic production capacity.

Formally, the optimal body mass at age t is the mass at which the Mito/SML ratio is maximized given the organism's current mitochondrial capacity. As mitochondrial capacity declines, the mass that can be efficiently serviced also declines. An organism that maintains the body mass optimal at age 25 into old age is carrying more structural load than its declining mitochondrial capacity can efficiently support.

This does not mean elderly individuals should aggressively lose weight — the lower boundary of the energetic corridor (sarcopenia threshold) also rises slightly with age due to progressive fat-muscle substitution. The practical implication is that the optimal mass window narrows with age, and both underweight and overweight become increasingly detrimental as the energetic margin shrinks.

11. The Energetic Corridor: A Clinical Visualization

Figure 5 presents the energetic corridor — the age-dependent zone within which body mass optimally matches the organism's declining functional mitochondrial capacity. The corridor is bounded below by the sarcopenia threshold (where mitochondrial mass itself begins to decline with lean tissue loss) and above by the obesity threshold (where excess SML outpaces mitochondrial capacity). The optimal mass trajectory runs through the center of this corridor and declines gradually with age.

The corridor narrows with age — reflecting the shrinking energetic margin of the aging organism. At age 25, the corridor spans approximately 25 kg (males) and 20 kg (females). By age 75, the corridor width is similar, but both boundaries have shifted: the optimal mass has declined while the lower boundary has risen, leaving less tolerance for deviation in either direction.

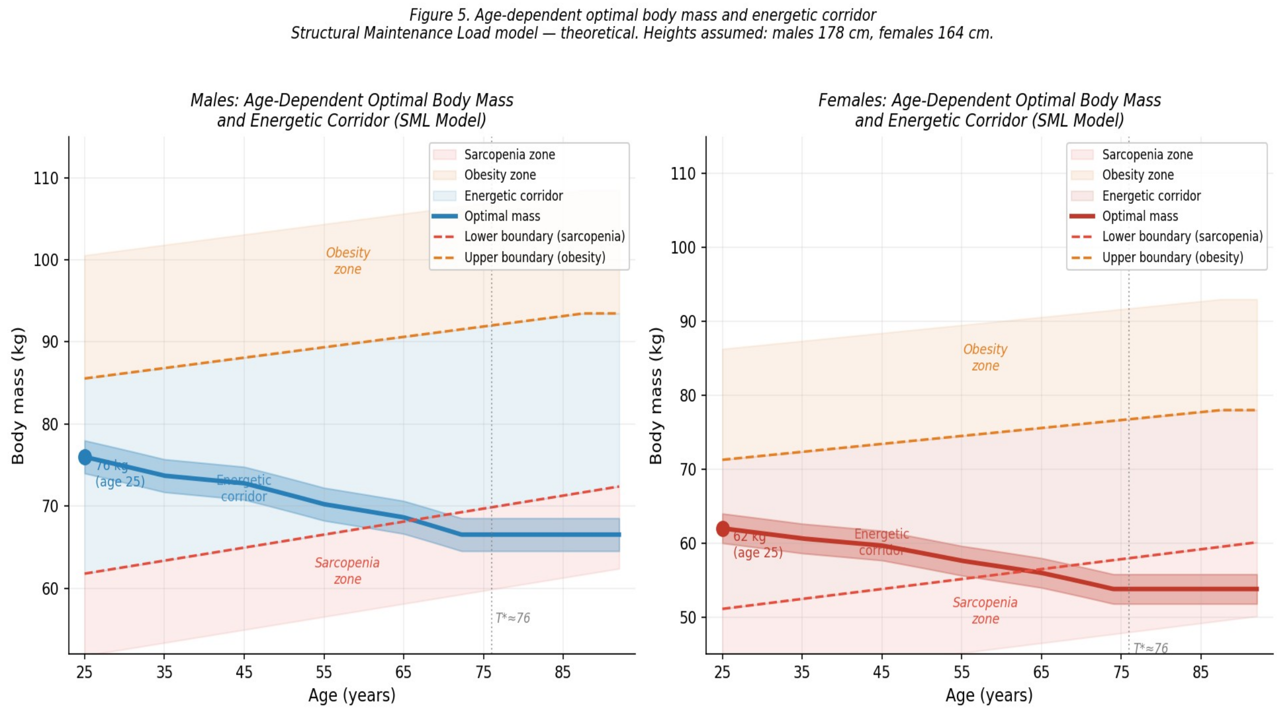

Figure 5. Age-dependent optimal body mass and energetic corridor (SML model, theoretical). Optimal mass (thick line) declines gradually as functional mitochondrial capacity falls. The corridor is bounded by sarcopenia (lower, red dashed) and obesity (upper, orange dashed) thresholds. T ≈ 76 years marks the provisional bifurcation point. Heights assumed: males 178 cm, females 164 cm.*

12. Model Values: Age-Dependent Optimal Body Mass

Table 1. Theoretical age-dependent optimal body mass and energetic corridor boundaries by sex. All values are model estimates; empirical validation required.

Age	M opt (kg)	M BMI	M lower (kg)	M upper (kg)	F opt (kg)	F BMI	F lower (kg)	F upper (kg)
25	76.0	24.0	61.8	85.5	62.0	23.1	51.1	71.3
30	74.9	23.6	62.6	86.2	61.3	22.8	51.8	71.8
35	73.7	23.3	63.4	86.8	60.6	22.5	52.4	72.4
40	73.3	23.1	64.2	87.4	60.2	22.4	53.1	72.9
45	72.8	23.0	65.0	88.1	59.7	22.2	53.8	73.4
50	71.5	22.6	65.7	88.7	58.7	21.8	54.5	74.0
55	70.2	22.2	66.5	89.3	57.6	21.4	55.1	74.5
60	69.4	21.9	67.3	90.0	56.8	21.1	55.8	75.0
65	68.6	21.7	68.1	90.6	56.0	20.8	56.5	75.6
70	67.2	21.2	68.9	91.2	54.8	20.4	57.2	76.1
75	66.5	21.0	69.7	91.9	53.8	20.0	57.8	76.7
80	66.5	21.0	70.5	92.5	53.8	20.0	58.5	77.2
85	66.5	21.0	71.3	93.2	53.8	20.0	59.2	77.7
90	66.5	21.0	72.1	93.5	53.8	20.0	59.8	78.0

Blue columns: males (height 178 cm). Orange columns: females (height 164 cm). Lower boundary = sarcopenia threshold. Upper boundary = obesity threshold. Floor values: males BMI 21, females BMI 20. Model exponent α = 0.3; requires empirical calibration.

13. Testable Predictions

18. The Mito/SML ratio — estimated as VO_2max relative to composite protein turnover index — should predict all-cause mortality better than VO_2max or BMI alone in longitudinal cohorts.
19. Protein turnover rate and collagen remodeling markers should correlate negatively with longevity independently of body mass.
20. Small dog breeds should exhibit lower protein turnover rates per unit lean mass than giant breeds — a direct test of within-species SML scaling.
21. Women should show lower composite SML indices than age-matched men with similar lean mass, attributable to lower vascular volume and connective tissue turnover — independent of hormonal status.
22. The female survival advantage at ages 85+ should persist after statistical adjustment for sex hormone levels, consistent with SML rather than hormonal explanation at the finishing line.
23. Individual optimal body mass estimated from personal VO_2max and BMR should predict subsequent weight-related morbidity better than population BMI tables.
24. The optimal BMI for survival should shift downward and the corridor should narrow with age — testable by age-stratified BMI-mortality analysis in large prospective cohorts.

14. Integration with Framework 1.0

Framework 1.0 [7] described aging as progressive energetic insufficiency — the declining availability of energy for tissue remodeling relative to demand. The SML hypothesis provides the quantitative denominator that was implicit but unformalized in that description: SML is the total energetic cost of structural preservation, and the ratio Functional Mitochondrial Capacity / SML is the operative variable determining biological age.

The age-dependent optimal body mass concept adds a dynamic clinical dimension: it translates the abstract ratio into a practically measurable target — the body mass that best matches an individual's current mitochondrial capacity — which can be estimated from standard physiological measurements (VO_2max, BMR, body composition) without specialized laboratory equipment.

Conclusion

Body size itself neither prolongs nor shortens life. Longevity depends on maintaining an optimal balance between the organism's functional mitochondrial capacity and the energetic cost of preserving its structure. This interpretation unifies observations from interspecies comparisons, domestic dog breeds, horses, pigs, human centenarians, and sex differences in lifespan within a single energetic framework.

The age-dependent optimal body mass concept is the model's most clinically actionable prediction: as functional mitochondrial capacity declines with age, the body mass that can be efficiently serviced by that capacity also declines. The energetic corridor — bounded by sarcopenia below and obesity above — narrows with age, providing a practical visualization for clinical application in geriatrics, oncology, rehabilitation, and preventive medicine.

At the finishing line of extreme old age, when hormonal differences between sexes have largely disappeared, the female structural advantage — lower SML due to lower lean mass and vascular volume — may be the dominant remaining explanation for female longevity. This hypothesis is testable and distinguishes the SML model from purely hormonal or genetic explanations of sex differences in lifespan.

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Theoretical Model of Age-Related Changes in Integral Energetic Indicators of the Human Organism

Working version. All values normalized to age 25 = 100%.

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Abstract

This article presents a purely theoretical model exploring age-related dynamics of four integral energetic indicators of the human organism in relative terms: retained mitochondrial activity (functional oxygen reserve), retained daily oxygen capacity per kilogram of body weight, oxygen expenditure for basal metabolism per kilogram, and the energy reserve — defined as the margin between sustainable daily energetic throughput and obligatory basal expenditure. All values are normalized to age 25, taken as 100%.

The model draws on VO_2max reference data from the HUNT3 Fitness Study, basal metabolic rate estimated by the Mifflin–St Jeor equation, and Israeli population survival data (CBS Life Tables 2018–2022, Jewish population). The central theoretical claim is that the energy reserve declines substantially faster than basal metabolism, narrowing the margin available for tissue remodeling and physiological adaptation in late life. Structural alignment of the energy reserve trajectory with demographic mortality curves suggests the reserve as a candidate integrative biomarker of functional aging. A provisional bifurcation point T* is defined as the age at which the median energy reserve crosses 30% of the age-25 level, estimated at approximately 76 years for both sexes. The model is offered as a working framework for discussion and subsequent experimental validation.

Keywords: aging; energy reserve; VO_2max; basal metabolic rate; functional aging; biomarker; BGR framework; survival curve

1. Objective

The aim is not to prove a new physiological law, but to construct a simple theoretical model allowing the age-related changes of several energetic parameters to be compared within a single coordinate system. The central question is: can the relative energy reserve — the margin between available daily energetic capacity and obligatory basal expenditure — better capture functional aging than VO_2max or BMR considered separately?

Each of those metrics declines monotonically with age, but neither distinguishes between an organism that still commands a large energetic surplus above its basal floor and one that is operating near capacity for merely sustaining itself. The energy reserve, by construction, captures precisely this distinction. The format is suitable for further development of the Framework approach to aging — treating it as a process of gradual reduction in the energy available for tissue remodeling [7].

2. Sources and Assumptions

Age-stratified VO_2max values from the HUNT3 Fitness Study provide sex-specific means for age groups from 20–29 through 70+, representing the largest fitness reference dataset currently available [1]. Basal metabolic rate (BMR) is estimated using the Mifflin–St Jeor equation, validated against indirect calorimetry in a broad adult population [2].

One MET is conventionally taken as 3.5 mL O_2/kg/min, but this approximation overestimates resting oxygen consumption in older individuals [3,4]; accordingly, basal O_2 is recalculated from BMR rather than fixed at 3.5. The sustainable daily energetic ceiling is modeled at 2.5 × BMR following Thurber et al. [5]. The applicability of this ceiling to sedentary elderly individuals is uncertain (see Limitations); a sensitivity analysis using an alternative ceiling of 2.0 × BMR is presented in the Discussion. Survival data: Israel CBS Complete Life Tables 2018–2022, Jewish population — l_x values normalized to l_{25} [6].

Age points 90 and 105 years are model extrapolations beyond the HUNT3 data range and should be interpreted with corresponding caution.

3. Indicators

25. Retained mitochondrial activity — functional oxygen reserve: (VO_2max − VO_2rest) relative to age 25. Integrative proxy, not a direct cellular measurement.
26. Retained daily O_2 capacity per kg — sustainable daily oxygen throughput: 2.5 × VO_2rest with age-corrected upper constraint from VO_2max.
27. O_2 for basal metabolism per kg — derived from BMR via Mifflin–St Jeor, normalized to age-25 level.
28. Energy reserve — difference between sustainable daily O_2 capacity and basal O_2 expenditure, normalized to age 25. Proposed primary marker of functional aging: captures the margin available for tissue remodeling above the obligatory maintenance floor.

4. Normalization Method

All indicators are expressed as percentages of their age-25 values, suppressing absolute sex- and body-composition differences. If the value at age 25 is X_{25} and at age t is X_t, the relative value is $X_t / X_{25} \times 100\%$. This normalization allows direct comparison of trajectories across sexes and indicators on a common scale, at the cost of obscuring absolute inter-individual and inter-sex differences.

5. Results

The key structural finding visible in all figures is the divergence between the basal metabolism trajectory (slow, near-linear decline, retaining ~77% in males and ~71% in females at age 105) and the energy reserve trajectory (steep decline reaching near-zero in the ninth decade). This asymmetry is the central theoretical construct: aging, in energetic terms, is primarily a narrowing of the margin above the maintenance floor rather than a proportional reduction of all parameters.

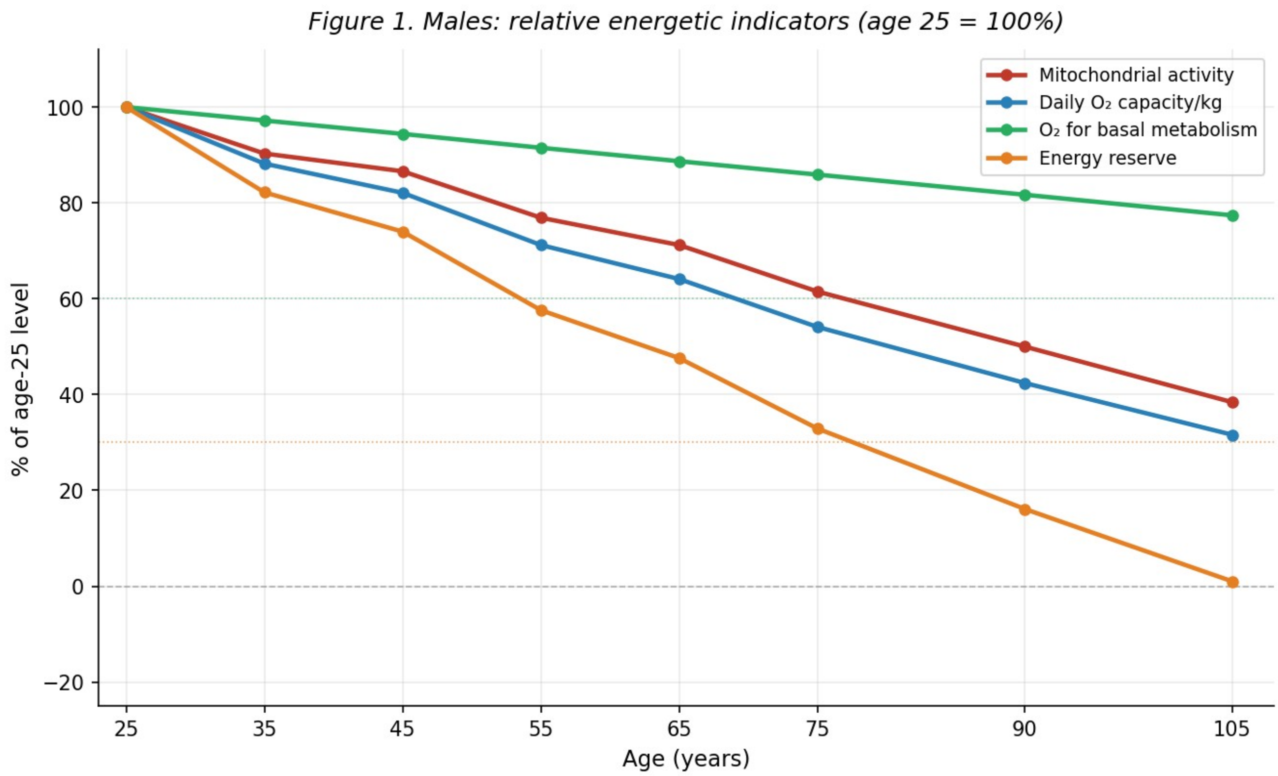

Figure 1. Males: relative energetic indicators normalized to age 25.

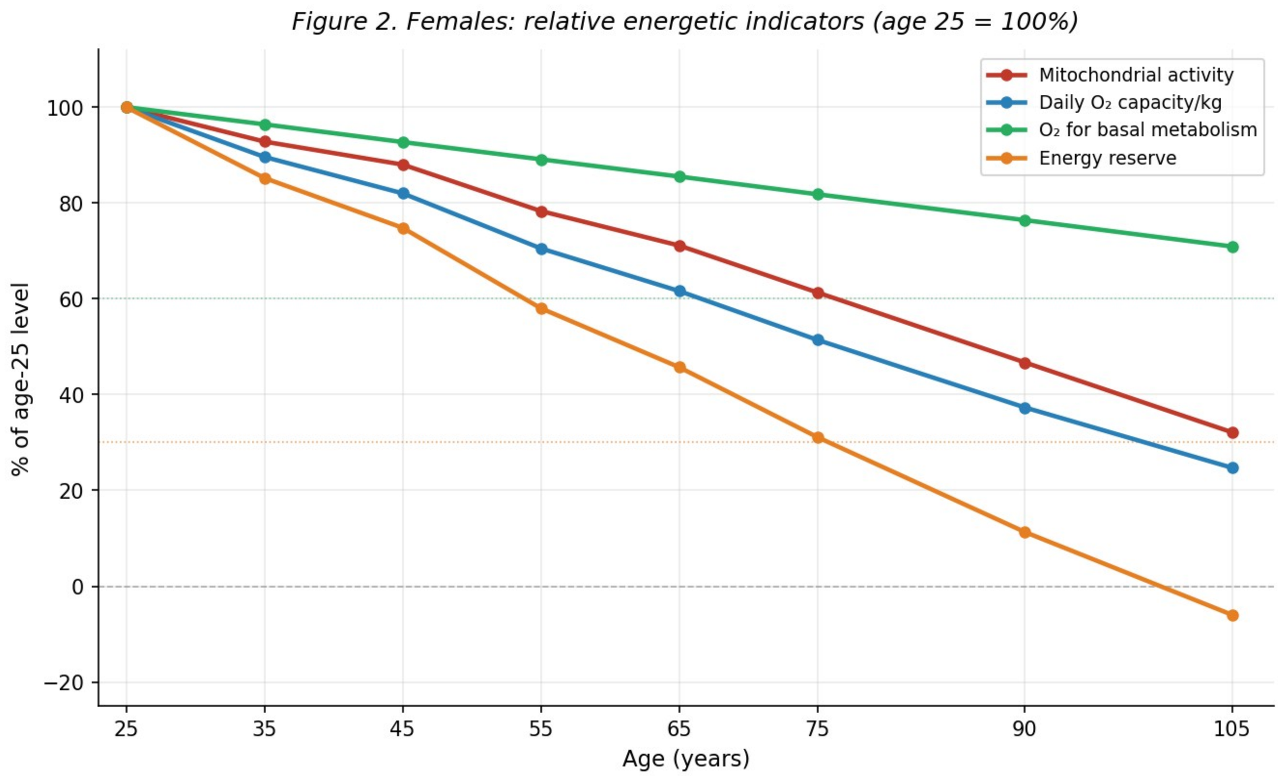

Figure 2. Females: relative energetic indicators normalized to age 25.

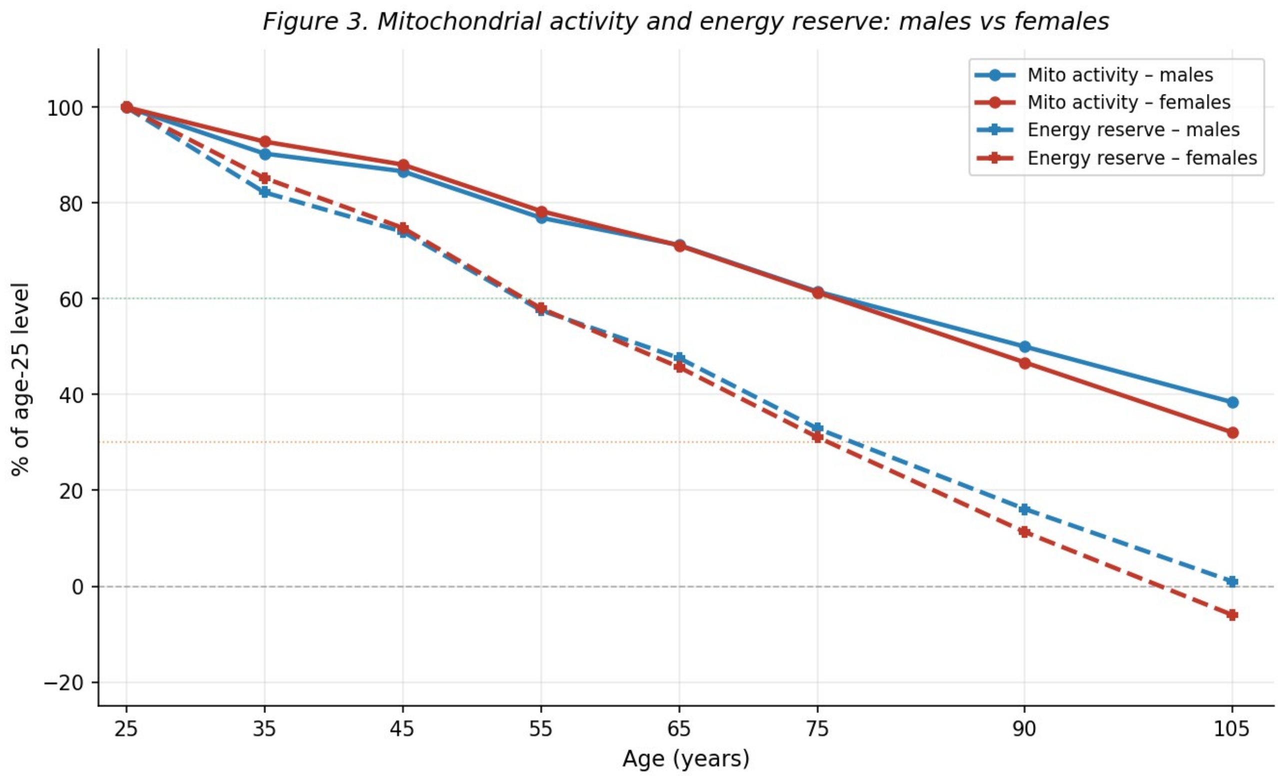

Figure 3. Mitochondrial activity and energy reserve: males vs females.

The overlay of Israeli population survival curves (Figures 4–5) reveals a structural alignment: demographic attrition accelerates in the age range where the energy reserve crosses its critical thresholds. At age 75, approximately 77% of males and 84% of females remain alive, with median energy reserves already fallen to ~33% and ~31% of age-25 levels respectively. By age 90, ~32% of males survive with a modeled median reserve of ~16%; ~44% of females survive with a median reserve of ~11%. This alignment is not interpreted as causal — it is a structural observation suggesting that energetic insufficiency and mortality risk follow parallel, potentially coupled trajectories.

Statistical trajectory fans (Figures 4–5) illustrate that population-level aging is better represented by a probability zone than a single line. Lower tails reach critical energetic deficit earlier; upper tails may correspond to exceptional longevity trajectories.

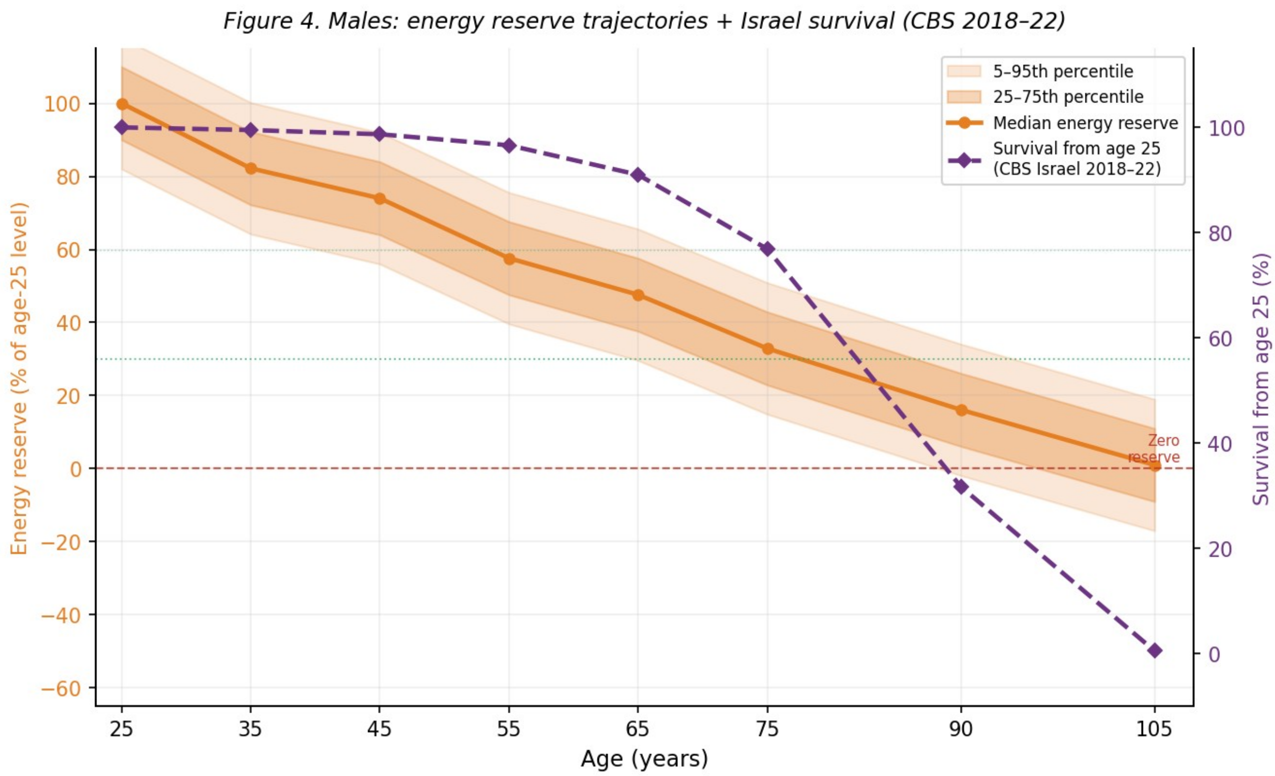

Figure 4. Males: energy reserve trajectories + Israeli survival (CBS 2018–22). Left axis: energy reserve (%). Right axis: survival from age 25 (%).

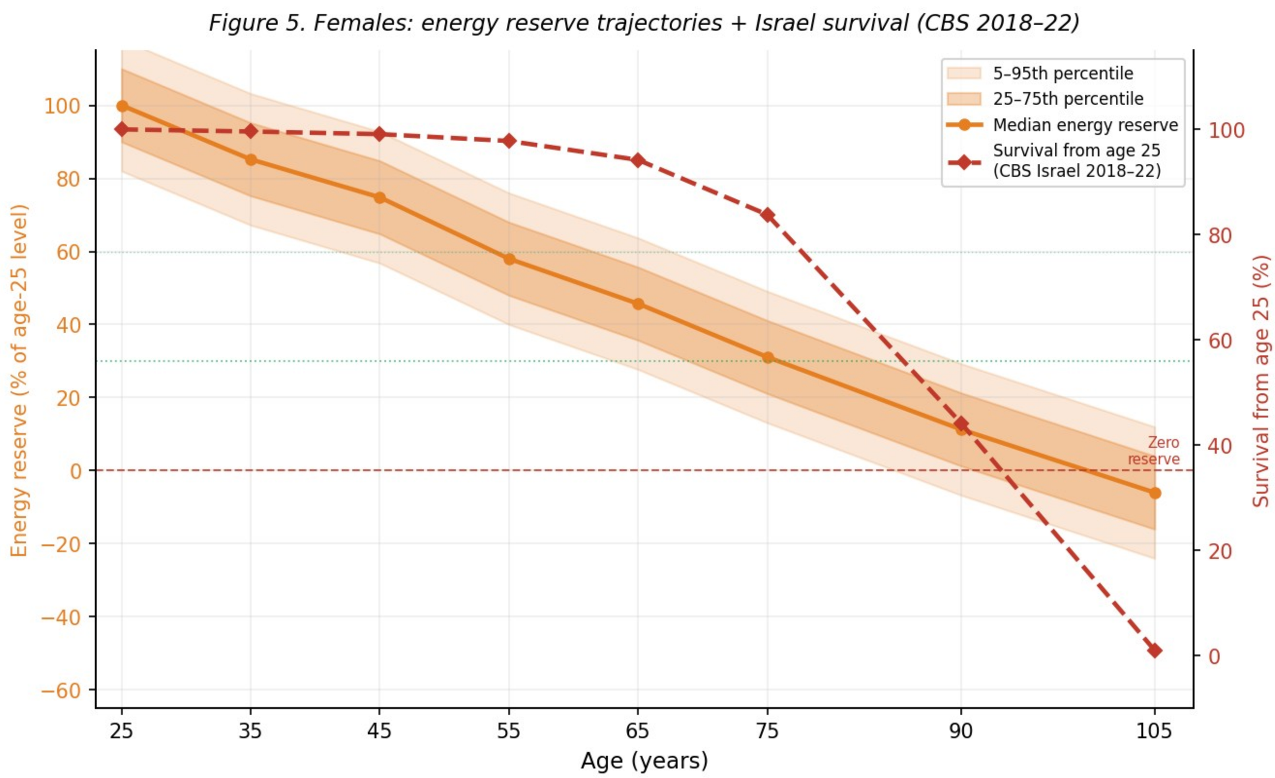

Figure 5. Females: energy reserve trajectories + Israeli survival (CBS 2018–22). Left axis: energy reserve (%). Right axis: survival from age 25 (%).

*5.1. Provisional Bifurcation Point T**

We define T* as the age at which the median energy reserve crosses 30% of the age-25 level — the point at which the remaining energetic margin may become insufficient to sustain tissue remodeling under physiological stress. From the model values (Table 1), T* is estimated by linear interpolation between the 75- and 90-year data points:

25. Males: reserve 32.9% at age 75, 16.1% at age 90 → T* ≈ 76 years
26. Females: reserve 31.1% at age 75, 11.3% at age 90 → T* ≈ 76 years

The near-identical T* for both sexes is notable given the substantially different survival curves: females reach T* at a similar energetic threshold but with markedly higher survival probability at that age (~84% vs ~77%). This observation is discussed in Section 7.1.

5.2. Negative Energy Reserve at Age 105

The model produces a negative energy reserve value for females at age 105 (−6.0%). A sustained negative reserve is biologically non-viable; it would imply that basal metabolic maintenance exceeds the organism's sustainable energetic ceiling. The negative value is therefore an artifact of linear extrapolation beyond the data range. Two interpretations apply: (a) the 2.5 × BMR ceiling becomes invalid at extreme ages where BMR and maximal capacity converge asymptotically; (b) survivor bias — individuals alive at 105 differ systematically from population means. The model should not be interpreted beyond age 90 for females and age 100 for males without dedicated longitudinal data.

6. Model Values

Table 1. Model energetic indicators by age and sex, normalized to age 25 = 100%.

Age	Sex	Mito act. %	Daily O_2 %	Basal O_2 %	Energy res. %	Survival %
25	Males	100.0	100.0	100.0	100.0	100.0
35	Males	90.3	88.2	97.2	82.2	99.5
45	Males	86.6	82.1	94.4	74.0	98.7
55	Males	76.9	71.2	91.5	57.6	96.6
65	Males	71.2	64.1	88.7	47.6	91.0
75	Males	61.5	54.1	85.9	32.9	76.9
90	Males	50.0	42.4	81.7	16.1	31.7
105	Males	38.4	31.6	77.4	1.0	0.5
25	Females	100.0	100.0	100.0	100.0	100.0
35	Females	92.8	89.6	96.4	85.2	99.6
45	Females	88.0	82.0	92.7	74.8	99.1
55	Females	78.3	70.5	89.1	58.0	97.8
65	Females	71.1	61.6	85.5	45.7	94.2
75	Females	61.3	51.4	81.8	31.1	83.8
90	Females	46.7	37.3	76.4	11.3	44.2
105	Females	32.1	24.7	70.9	−6.0*	1.0

** Negative value is a model extrapolation artifact; see Section 5.2. Blue shading: males. Orange shading: females.*

7. Discussion

The model reveals a fundamental asymmetry in the aging energetic landscape: basal metabolism is relatively conserved across the lifespan, while the functional capacity above this floor declines steeply. This asymmetry generates the central theoretical insight — that aging, viewed energetically, is primarily a narrowing of the available remodeling margin rather than a proportional reduction of all parameters.

This framing is consistent with the BGR framework [7], which treats aging as progressive dampening of the regenerative oscillation between tissue breakdown and repair. As the energy reserve narrows, the organism retains capacity for basal maintenance but loses the energetic surplus required for tissue remodeling after stress, injury, or physiological challenge.

7.1. The Sex Paradox

Figure 3 reveals a paradox: the female energy reserve declines slightly faster than the male reserve, yet females substantially outlive males. Two explanations apply. First, normalization to age-25 values suppresses the fact that females have lower absolute VO_2max at age 25 — the relative decline may be steeper while absolute values remain sufficient. Second, male excess mortality at younger ages is driven primarily by cardiovascular disease, trauma, and metabolic syndrome — factors external to the energetic model. The model describes the energetic trajectory of survivors, not the causes of differential attrition.

7.2. Sensitivity Analysis: Alternative Ceiling

The 2.5 × BMR ceiling (Thurber et al. 2019) was derived from physiological extremes. For sedentary elderly populations, a more conservative ceiling of 2.0 × BMR may be appropriate (Pontzer et al., PNAS 2021). Under this assumption, T* would shift to approximately 60–65 years — reinforcing rather than undermining the central claim: the true energy reserve may decline even faster than the primary model suggests.

7.3. Alignment with Survival Data

The structural convergence between energy reserve trajectories and Israeli population survival curves is correlational and expected — both variables are functions of age. The scientific question is whether energy reserve predicts mortality independently of age, requiring prospective longitudinal data with individual-level measurements. Nevertheless, the alignment is non-trivial in one respect: the model was constructed independently of survival data, using physiological parameters alone. The fact that its output tracks demographic reality provides weak but non-zero evidence that the energetic framework captures something real about the biology of aging.

8. Limitations

29. Purely theoretical model — does not account for disease, habitual activity, body composition, genetics, hormonal factors, nutrition, or pharmacological interventions.
30. Age points 90 and 105 are extrapolations beyond HUNT3 data range; high uncertainty.

31. The 2.5 × BMR sustainable ceiling is derived from extreme physiological contexts; applicability to sedentary elderly uncertain.
32. Survival data (CBS Israel 2018–2022) includes COVID-19 years and reflects Jewish Israeli population specifically.
33. Mitochondrial activity is an integrative proxy derived from VO_2max, not a direct cellular measurement.
34. Per-kg normalization may partially reflect body composition changes (rising fat mass) rather than pure energetic decline.
35. Negative energy reserve at age 105 (females) is a model artifact, not a biological prediction.

9. Conclusion

The model proposes examining age-related energetic change in relative coordinates centered on age 25, tracking four indicators: retained mitochondrial activity, retained daily oxygen capacity per kg, basal oxygen expenditure, and the energy reserve. The core theoretical claim — that the energy reserve declines substantially faster than basal metabolism — generates a specific and testable prediction: the energetic margin available for tissue remodeling is the principal constraint on organismal resilience in late life, not visible in VO_2max or BMR considered separately.

A provisional bifurcation point T* (~76 years for both sexes) marks the age at which this margin may fall below the threshold required for effective stress response. The structural alignment of the energy reserve trajectory with Israeli population mortality curves is consistent with the hypothesis that energetic reserve and mortality risk are causally linked through the biology of tissue remodeling. Empirical validation requires prospective studies measuring individual VO_2max and BMR longitudinally with mortality and functional outcomes as endpoints [7].

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Appendix: Clinical Applications of the Energy Reserve Concept

If the energy reserve model proves empirically valid, primary clinical applications would include: (1) preoperative risk stratification — quantifying the energetic margin available for postoperative tissue repair independently of VO_2max alone; (2) oncology — individualizing chemotherapy intensity based on estimated recovery capacity; (3) rehabilitation medicine — determining optimal training progression after major illness or injury; (4) preventive medicine — monitoring how lifestyle factors collectively reduce the remodeling reserve rather than isolated risk factors. All proposed applications remain theoretical pending prospective validation.

Afterword

I wish to express my sincere gratitude to my two tireless assistants, Claude (Anthropic) and ChatGPT.

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2026